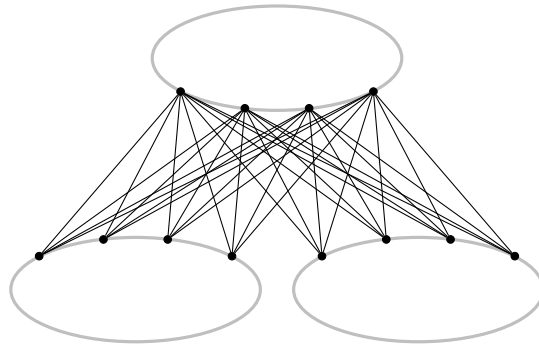




Universality in Graph Theory

Neel Kaul

Supervised by David R. Wood

Honours Thesis submitted as part of the B.Sc. (Honours) degree

School of Mathematics
Monash University
Melbourne, Australia

October 20, 2025

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Abstract

A graph U is universal for a class of graphs $\mathcal{G}$ if for every graph G in $\mathcal{G}$, U contains a subgraph isomorphic to G . For a given class of graphs $\mathcal{G}$, the universal graph problem asks to asymptotically determine the minimum number of edges in a graph that is universal for the n -vertex graphs in $\mathcal{G}$.

The content of this thesis can be split into two major parts. The first is a literature review of the universal graph problem for the class of trees, treewidth- k graphs (these are graphs that are ‘tree-like’) and planar graphs. To this end, we discuss many results in structural graph theory (with proofs) relating to perfectly balanced separators, path-decompositions, tree-decompositions and the strong product of graphs to name a few.

The second major part is a presentation of original results and findings towards settling the universal graph problem for the class of trees and the class of treewidth- k graphs. We summarise the original results:

Let $s(n)$ be the minimum number of edges in a graph that is universal for the n -vertex trees. Chung and Graham [*J. London Math. Soc.* 1983] claim to show that $s(n) \leq \frac{5}{2}n \log_2 n + \mathcal{O}(n)$. We explain a mistake in their proof. The previously best known upper bound is $s(n) \leq \mathcal{O}(n(\log n)(\log \log n)^2)$ by Chung, Graham and Pippenger [*Proc. Hungarian Coll. on Combinatorics* 1976]. We give a fully self-contained original proof of the improved bound: $s(n) \leq \mathcal{O}(n(\log n)(\log \log n))$. The best known lower bound is $s(n) \geq \Omega(n \log n)$.

Let $s_k(n)$ be the minimum number of edges in a graph that is universal for the n -vertex treewidth- k graphs. So $s_1(n) = s(n)$. We extend the results for trees to universal graphs for treewidth- k graphs by proving the new bounds: $\Omega(kn \log n) \leq s_k(n) \leq \mathcal{O}(kn(\log n)(\log \log n))$. This improves the previously best known bound of $s_k(n) \leq \mathcal{O}(kn \log^2 n)$.

The original bounds for $s(n)$ and $s_k(n)$ presented in this thesis were submitted to arXiv. This sparked joint work with myself, Jaehoon Kim, Minseo Kim and David Wood. We are currently writing a paper in which we (1) explain how to fix Chung and Graham’s mistake and recover the bound $s(n) \leq \frac{7}{2}n \log_2(n) + \mathcal{O}(n)$, (2) make the first improvement to the bound on $s(n)$ in over four decades by showing that $s(n) \leq \frac{14}{5}n \log_2 n + \mathcal{O}(n)$, and (3) settle the universal graph problem for the class of treewidth- k graphs by showing that $s_k(n) = \Theta(kn \log n)$.

Chapter 1

Introduction

Broadly speaking, the mathematical field of extremal combinatorics is concerned with asking questions about the size of finite mathematical objects that satisfy certain conditions. Many problems in extremal combinatorics can relate to different areas of mathematics (usually discrete), like number theory [38, 39], discrete geometry [40, 41, 19] and graph theory [16, 17]. As such, the objects of study could be set-systems, point sets, sets of numbers, graphs or any other combinatorial/discrete object.

A particular class of problems in extremal combinatorics that has influenced much of the research in the field are the Turán-type problems, named after the father of extremal graph theory, Paul Turán.



Paul Turán

A Turán-type problem is concerned with maximising (or minimising) some parameter of a particular object of study, subject to avoiding some ‘bad’ property or configuration. We give a few well-known examples of Turán-type problems.

In the study of set-systems, for any positive integer n , Sperner’s Theorem characterises the antichains¹ in the power set of $\{1, \dots, n\}$ with maximum cardinality.

Theorem 1.1 (Sperner [75]). *The cardinality of an antichain $\mathcal{A}$ in the power set of $\{1, \dots, n\}$ with respect to the subset relation is at most $\binom{n}{\lfloor n/2 \rfloor}$, with equality only when:*

- $\mathcal{A}$ consists of every set of cardinality $\lfloor n/2 \rfloor$, or
- $\mathcal{A}$ consists of every set of cardinality $\lceil n/2 \rceil$.

¹An *antichain* is a set of pairwise incomparable elements.

In discrete geometry, a problem of Erdős [40] asks to determine, for every positive integer n , the maximum number of pairs of points at unit-distance achievable by n points in the plane. This was considered to be “one of the most tantalising problems” by Spencer et al. [73] in discrete geometry; it remains unsolved to this day. The best known upper bound is by Székely [76].

Theorem 1.2 (Székely [76]). *The maximum number of pairs of points at unit-distance achievable by n points in the plane is $\mathcal{O}(n^{4/3})$.*

This is still far from the lower bound of $n^{1+c/\log \log n}$ by Erdős [40] (for some absolute constant c).

One of the most famous results in additive number theory is Szemerédi’s Theorem; the finitary version states:

Theorem 1.3 (Szemerédi [77]). *For any fixed integer $k \geq 1$, for all integers n with $n \geq k$, the maximum cardinality of a subset of $\{1, \dots, n\}$ with no k -term arithmetic progression² is $o(n)$.*

A result by Turán himself comes from graph theory. Turán’s Theorem characterises the n -vertex graphs with the maximum number of edges subject to having no r -cliques³.

Theorem 1.4 (Turán [78]). *For all integers $r \geq 2$, the number of edges in an n -vertex graph with no r -cliques is at most $\frac{1}{2} \frac{r-2}{r-1} n^2$. Furthermore, the Turán graph $T_r(n)$ is the unique such graph with the maximum number of edges.*

For all integers $r \geq 2$ and $n \geq r - 1$, the *Turán graph* $T_r(n)$ is the unique (up to isomorphism) n -vertex complete $(r - 1)$ -partite graph where the size of parts pairwise differ by at most 1 (see Figure 1.1).

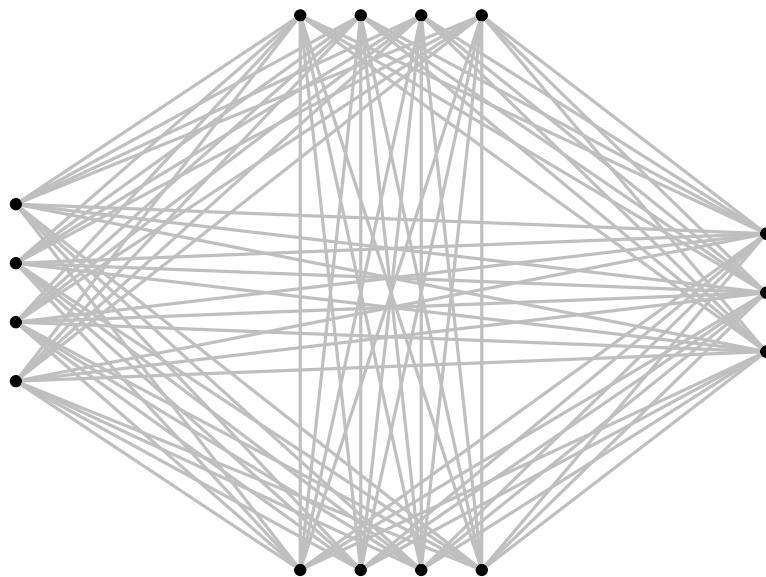


Figure 1.1: Turán graph $T_5(15)$.

²A *k -term arithmetic progression* is an increasing sequence of k natural numbers in which consecutive terms have common differences.

³An *r -clique* in a graph is a set of r pairwise adjacent vertices.

The topic of universality in graph theory and the universal graph problem is the focus of this project and thesis. The universal graph problem can be thought of as the (qualitative) opposite of the classical extremal graph theory question that asks for the maximum number of edges in an n -vertex graph that contains none of the graphs in some class as a subgraph. In the case of Turán’s Theorem, the class is $\{K_r\}$. The universal graph problem is concerned with finding the minimum number of edges in a graph that contains every n -vertex graph in some class as a subgraph. Typically, one is interested in the asymptotics as n grows.

A graph U is *universal* for a graph class $\mathcal{G}$ if for every graph G in $\mathcal{G}$, U contains a subgraph isomorphic to G . We restate the universal graph problem in a convenient way (from here on out, the ‘universal graph problem’ is the following):

For a given class of graphs $\mathcal{G}$, let $f_{\mathcal{G}}(n)$ be the minimum number of edges in a graph that is universal for the n -vertex graphs in $\mathcal{G}$. What is the asymptotic behaviour of $f_{\mathcal{G}}(n)$?

We note that for any class $\mathcal{G}$, the minimum number of vertices in a universal graph for the n -vertex graphs in $\mathcal{G}$ equals n . Indeed, the complete graph on n vertices contains every n -vertex graph as a subgraph. Thus, the number of edges in the universal graph is the parameter chosen to be minimised instead of the number of vertices.

A sample of some of the best known results for the universal graph problem for various classes $\mathcal{G}$ is shown in Table 1.2. We write $\log n$ to mean the natural logarithm of n .

Table 1.2: Graph classes $\mathcal{G}$ and the best known upper and lower bounds for $f_{\mathcal{G}}(n)$.

$\mathcal{G}$	$f_{\mathcal{G}}(n) \geq$	$f_{\mathcal{G}}(n) \leq$
Trees ⁴	$n \log n - \mathcal{O}(n)$ [23, 49]	$\frac{14}{5}n \log_2 n + \mathcal{O}(n)$
Treewidth- k graphs	$\Omega(kn \log n)$ [55]	$\mathcal{O}(kn \log n)$
Planar graphs	$n \log n - \mathcal{O}(n)$	$n^{1+o(1)}$ [43]
Bounded degree trees [12]	$\Omega(n)$	$\mathcal{O}(n)$
Bounded degree planar graphs	$\Omega(n)$	$\mathcal{O}(n)$ [21]
Maximum degree at most Δ	$\Omega(n^{2-2/\Delta})$ [5]	$\mathcal{O}(n^{2-2/\Delta}(\log n)^{1+8/\Delta})$ [6]

We remark that the initial motivation for studying universal graphs stemmed originally from an applied setting involving the optimisation of certain data structures [70] and circuit designs [79]; solutions to these problems can be realised as a universal graph (that is optimal in some sense) for some graph class.

⁴The upper bound for trees and treewidth- k graphs in the first two rows of Table 1.2 are results from a paper (yet to be submitted to a journal) by myself, Jaehoon Kim, Minseo Kim and David Wood. This collaboration was, in some ways, sparked because of this project, specifically, the original results and proofs that are presented in Sections 2.3 and 3.2 of this thesis. The content of the joint paper goes beyond the original results and proofs presented in this thesis. See the ‘Postscript’ in Chapter 3.

There are many notions of containment in graph theory, such as subgraph, induced subgraph, and minor. These give rise to variations of the universal graph problem. Furthermore, universality can be studied in the infinite setting. We give an overview of some of the variations of the universal graph problem.

1.1 Variations of The Universal Graph Problem

1.1.1 Induced-Universal Graphs. The induced-universal graph problem is one of the most commonly studied variations of the universal graph problem. A graph U is *induced-universal* for a graph class $\mathcal{G}$ if for every graph G in $\mathcal{G}$, U contains an induced subgraph isomorphic to G . The induced-universal graph problem is therefore stated in the following way:

For a given class of graphs $\mathcal{G}$, let $f_{\mathcal{G}}^{\text{ind}}(n)$ be the minimum number of vertices in a graph that is induced-universal for the n -vertex graphs in $\mathcal{G}$. What is the asymptotic behaviour of $f_{\mathcal{G}}^{\text{ind}}(n)$?

Note that it is the number of vertices being minimised, not the number of edges like in the case for subgraph-universality. Adjacency labelling schemes give an equivalent way to state the induced-universal graph problem.

Denote by $\{0, 1\}^*$ the set of all finite $\{0, 1\}$ -strings. A graph class $\mathcal{G}$ is said to have a $\mu(n)$ -bit adjacency labelling scheme if there is a function $A : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}$ such that for every n -vertex graph $G \in \mathcal{G}$, there is an injective⁵ assignment of labels $\ell : V(G) \rightarrow \{0, 1\}^*$ on at most $\mu(n)$ bits, such that for all pairs of vertices $u, w \in V(G)$,

$$A(\ell(u), \ell(w)) = \begin{cases} 0 & \text{if } uw \notin E(G), \\ 1 & \text{if } uw \in E(G). \end{cases}$$

The problem of adjacency labelling schemes asks:

For a given class of graphs $\mathcal{G}$, let $\mu_{\mathcal{G}}(n)$ be the minimum number of bits such that $\mathcal{G}$ has a $\mu_{\mathcal{G}}(n)$ -bit adjacency labelling scheme. What is the asymptotic behaviour of $\mu_{\mathcal{G}}(n)$?

Kannan, Naor, and Rudich [54] observed that adjacency labelling schemes and induced-universal graphs are equivalent in the following sense:

Theorem 1.5. *A graph class $\mathcal{G}$ has a $\mu(n)$ -bit adjacency labelling scheme if and only if $f_{\mathcal{G}}^{\text{ind}}(n) \leq 2^{\mu(n)}$.*

Indeed, if $\mathcal{G}$ has a $\mu(n)$ -bit adjacency labelling scheme, then an induced-universal graph U for the n -vertex graphs in $\mathcal{G}$ can be defined as follows. Let $V(U)$ be the set of all $\{0, 1\}$ -

⁵It is standard in the literature to allow for a non-injective labelling ℓ in the definition of adjacency labelling scheme. However, injectivity is required to show the equivalence of adjacency labelling schemes and induced-universal graphs.

strings on at most $\mu(n)$ bits, and let two strings σ and τ be adjacent in U if and only if $A(\sigma, \tau) = 1$. So $f_{\mathcal{G}}^{\text{ind}}(n) \leq |V(U)| \leq 2^{\mu(n)}$. Conversely, if U is an induced-universal graph for the n -vertex graphs in $\mathcal{G}$ with $|V(U)| \leq 2^{\mu(n)}$, then assume without loss of generality that $V(U)$ is a set of $\{0, 1\}$ -strings on at most $\mu(n)$ bits. Then let A encode the adjacencies in U , and for each n -vertex graph G in $\mathcal{G}$, the labelling $\ell : V(G) \rightarrow \{0, 1\}^*$ is given by an isomorphism of G to an induced subgraph of U .

A sample of some of the best known results for the induced-universal graph problem and the adjacency labelling scheme problem for various classes $\mathcal{G}$ is shown in Table 1.3.

Table 1.3: Graph classes $\mathcal{G}$ and the best known bounds for $f_{\mathcal{G}}^{\text{ind}}(n)$ and $\mu_{\mathcal{G}}(n)$.

$\mathcal{G}$	$f_{\mathcal{G}}^{\text{ind}}(n)$	$\mu_{\mathcal{G}}(n)$
Trees [7]	$\mathcal{O}(n)$	$\log_2 n + \mathcal{O}(1)$
Treewidth- k graphs [46]	$n \log^{\mathcal{O}(k)}(n)$	$\log_2 n + \mathcal{O}(k \log \log n)$
Planar graphs [43]	$n^{1+o(1)}$	$(1 + o(1)) \log_2 n$
Maximum degree at most Δ [20, 3]	$\Theta(n^{\Delta/2})$	$\frac{\Delta}{2} \log_2 n + \mathcal{O}(1)$
All graphs [61, 2]	$(1 + o(1))2^{(n-1)/2}$	$\frac{n-1}{2} + o(1)$

1.1.2 Minor-Universal Graphs. A graph U is *minor-universal* for a graph class $\mathcal{G}$ if for every graph G in $\mathcal{G}$, U contains a minor isomorphic to G . Minor-universality is a relatively unexplored variation of the universal graph problem compared to subgraph-universality and induced-universality. The most well-known results regarding minor-universality are concerned less with minimising some parameter like the number of edges, and more with the structural properties of the universal graph.

Theorem 1.6 (Robertson et al. [69]). *The $2n \times 2n$ grid is minor-universal for the n -vertex planar graphs.*

An important application of this fact is in proving a characterisation of minor-closed classes with bounded treewidth. Treewidth is central topic of study in graph minor theory that, roughly speaking, measures how ‘tree-like’ a graph is. The Grid Minor Theorem, states that in some sense, the only reason a graph has large treewidth is because it contains a large grid minor.

Theorem 1.7 (Grid Minor Theorem, Robertson and Seymour [67]). *There exists a function $h : \mathbb{N} \rightarrow \mathbb{N}$ such that every graph G with $\text{tw}(G) \geq h(k)$ has a $k \times k$ grid minor.*

If a minor-closed class $\mathcal{G}$ has bounded treewidth, then some grid is not in $\mathcal{G}$ by Theorem 1.7; in particular, some planar graph is not in $\mathcal{G}$. On the other hand, if some planar graph is not in $\mathcal{G}$, then by Theorem 1.6, some grid is not in $\mathcal{G}$. By the contrapositive of Theorem 1.7, $\mathcal{G}$ has bounded treewidth. Thus, the following characterisation of minor-closed classes with bounded treewidth is obtained.

Theorem 1.8. *A minor-closed class $\mathcal{G}$ has bounded treewidth if and only if some planar graph is not in $\mathcal{G}$.*

It was shown by Biedl et al. [13] that a minor-universal grid for n -vertex planar graphs must have $\Omega(n^2)$ vertices. However, it is unknown whether there exists a planar graph with $o(n^2)$ vertices that is minor-universal for n -vertex planar graphs. Interestingly, Theorem 1.6 can be generalised to graphs embeddable on a fixed surface. A surface with c crosscaps and h handles has *Euler genus* $c + 2h$. The *Euler genus* of a graph G is defined to be the minimum Euler genus of a surface into which G embeds.

Theorem 1.9 (Gavoille and Hilaire [45]). *For every surface Σ of Euler genus g , and every positive integer n , there is a graph $U_{\Sigma,n}$ embedded on Σ with $|V(U_{\Sigma,n})| \leq \mathcal{O}(g^2(n + g)^2)$ that is minor-universal for the n -vertex graphs embeddable on Σ .*

Similar to minor-universal planar graphs for planar graphs, one can study minor-universal trees for trees.

Theorem 1.10 (Bodini [14]). *The minimum number of vertices in a tree that is minor-universal for n -vertex trees is at most $\mathcal{O}(n^c)$ for some constant $c \approx 1.98$.*

Bodini [14] also showed that a minor-universal tree for n -vertex trees must have at least $n \log n + \Omega(n)$ vertices. A variation of this problem asks to asymptotically determine the minimum number of vertices in a tree that contains a subdivision of every n -vertex tree as a subgraph. Note that a subdivision of a graph G contains G as a minor. The best known results for this variation are by Gawrychowski et al. [48].

Theorem 1.11 (Gawrychowski et al. [48]). *There exists constants $c_1 \approx 2.17$ and $c_2 \approx 2.31$ such that the minimum number of vertices in a tree that contains a subdivision of every n -vertex tree as a subgraph is at least $\Omega(n^{c_1})$, and at most $\mathcal{O}(n^{c_2})$.*

1.1.3 Universal Graphs in The Infinite Setting. In the infinite setting, the universal graph problem is concerned with finding universal graphs that preserve key structural properties of some graph class. For this section, we allow graphs to have countably many vertices. A countable graph U is *universal* for a class of countable graphs $\mathcal{G}$ if $U \in \mathcal{G}$ and for every graph G in $\mathcal{G}$, U contains a subgraph isomorphic to G . The notions of induced-universal and minor-universal are defined analogously in the infinite setting.

Then, given a class of countable graphs $\mathcal{G}$, the universal graph problem for infinite graphs asks:

Does there exist a universal graph for $\mathcal{G}$?

The most famous example of universality in the infinite setting is that of the Rado graph. The Rado graph R is the unique (up to isomorphism) countable graph with the following property: for any two finite disjoint subsets A and B of $V(R)$, there exists a vertex v in $R - (A \cup B)$ such that v is adjacent to every vertex in A , and v is not adjacent to every vertex in B .

Theorem 1.12 (Erdős and Rényi [42], Rado [64]). *The Rado graph is induced-universal for the class of all countable graphs.*

The problem of universality for planar graphs was first raised by Stanisław Ulam (unpublished). Pach [62] showed that there is no universal graph for the class of all planar graphs.

Theorem 1.13 (Pach [62]). *If every planar graph is isomorphic to a subgraph of U , then U is not planar.*

One can relax the condition that the universal graph is itself planar, and instead ask whether there exists a countable graph U that contains every countable planar graph as a subgraph, and U preserves some other property of planar graphs, for example, excluding some minor. It turns out that even this is not possible. The following result strengthens Theorem 1.13.

Theorem 1.14 (Huynh, Mohar, Šámal, Thomassen, and Wood [52]). *If every planar graph is isomorphic to a subgraph of U , then*

- (1) *U contains the infinite complete graph $K_{\mathbb{N}}$ as a minor, and*
- (2) *for each $t \in \mathbb{N}$, U contains a subdivision of K_t .*

On the other hand, the minor-universal graph problem for the class of planar graphs has the opposite outcome to Theorem 1.13.

Theorem 1.15 (Diestel and Kühn [29]). *There exists a countable graph U that is minor-universal for the class of all planar graphs.*

Some results in infinite graph theory can be restated to give a finitary/qualitative version. Recently, the following qualitative version of Theorem 1.14 was shown.

Theorem 1.16 (Bastide, Esperet, Groenland, Hilaire, Rambaud, and Wesolek [11]). *There exists a constant $c = c_t$ such that if U is a K_t -minor-free graph that is (subgraph-)universal for the n -vertex planar graphs, then $|V(U)| \geq 2^{c\sqrt{n}}$ for all large enough n .*

1.2 Outline of Thesis

This thesis has two primary goals. The first goal is to show how to apply tools and techniques in structural graph theory to try to resolve the universal graph problem for the class of trees, the class of treewidth- k graphs and the class of planar graphs, all while giving the necessary details for the background, derivations and the proofs. The second goal is to present original results with proofs, and to present some other contributions that were made over the course of this project. We now outline the main content in the thesis.

In Chapter 2, we consider the universal graph problem for the class of trees. Let $s(n)$ be the minimum number of edges in a graph that is universal for n -vertex trees. We begin by defining the notion of perfectly balanced separators in Section 2.1, which we show, can be

used for general constructions of universal graphs for graph classes $\mathcal{G}$ that are closed under taking subgraphs. We show that such methods give universal graphs for $\mathcal{G}$ with few edges if every graph in $\mathcal{G}$ has ‘small’ perfectly balanced separators. We apply this construction to the class of trees to obtain bounds for $s(n)$. First, we introduce pathwidth and path-decompositions, and show that the pathwidth of a graph can be used to upper bound the size of a perfectly balanced separator. Ultimately, we obtain the bound $s(n) \leq \mathcal{O}(n \log^2 n)$. The paper ‘On Universal Graphs for Spanning Trees’ by Chung and Graham [24] claims to show that $s(n) \leq \mathcal{O}(n \log n)$, improving the bound shown in Section 2.1. Section 2.2, presents my explanation of a mistake that I found in the proof of this claim in [24]. Following this, Section 2.3 presents a proof of the first original result to come from this project, namely, the bound $s(n) \leq \mathcal{O}(n(\log n)(\log \log n))$.

In Chapter 3, we consider the universal graph problem for the class of treewidth- k graphs. Let $s_k(n)$ be the minimum number of edges in a graph that is universal for n -vertex treewidth- k graphs. We begin by defining treewidth and tree-decompositions in Section 3.1 and we discuss why treewidth is an appropriate measure of how ‘tree-like’ a graph is. By using perfectly balanced separators (defined in Chapter 2), we give a proof that $s_k(n) \leq \mathcal{O}(kn \log^2 n)$. Then, Section 2.3 presents a proof of the original result (and new bound): $s_k(n) \leq \mathcal{O}(kn(\log n)(\log \log n))$, which improves the previously best known upper bound of $\mathcal{O}(kn \log^2 n)$. Furthermore, Section 2.3 presents a proof of the original result (and new lower bound): $s_k(n) \geq \Omega(kn \log n)$.

The ‘Postscript’ at the end of Chapter 3 clarifies the contributions made during this project. It also gives an update on the current best known bounds for $s(n)$ and $s_k(n)$.

In Chapter 4, we consider the universal graph problem for the class of planar graphs. Let $r(n)$ be the minimum number of edges in a graph that is universal for n -vertex planar graphs. In Section 4.1, we show that perfectly balanced separators can be used to obtain the bound: $r(n) \leq \mathcal{O}(n^{3/2})$. A key tool in the proof of this bound is the Planar Separator Theorem; we present a proof of the Planar Separator Theorem in full.

The remainder of Chapter 4 builds towards the proof of the breakthrough bound $r(n) \leq n^{1+o(1)}$ by Louis Esperet, Gwenaël Joret and Pat Morin. To this end, we introduce the strong product of graphs and present a proof of the seminal Planar Graph Product Structure Theorem in full in Section 4.2. The proof of the bound $r(n) \leq n^{1+o(1)}$ that we present in Section 4.3 gives $n^{1+o(1)}$ type bounds for the universal graph problem for classes that are more general than the class of planar graphs.

Chapter 5 contains a conclusion and discussion of future work.

1.3 Graph Theoretic Preliminaries

Let $\mathbb{N} := \{1, 2, \dots\}$. See [28] for any undefined graph theory notions. A *graph* G is a pair of finite sets $V(G)$ and $E(G)$ such that $E(G) \subseteq \binom{V(G)}{2}$. We call $V(G)$ the *vertex set* and its members *vertices*. We call $E(G)$ the *edge set* and its members *edges*. Two vertices u and v are said to be *adjacent* if $\{u, v\}$ is an edge; we say that the edge $\{u, v\}$ is *incident*

with u and v , and that u and v are the *ends* of $\{u, v\}$. For simplicity, $\{u, v\}$ is shortened to uv . For two subsets A, B of $V(G)$, an *AB-edge* is an edge of G with one end in A and the other end in B . For a vertex v , its *neighbourhood*, denoted by $N_G(v)$, is the set of vertices adjacent to v . The *degree* of v , denoted by $\deg_G(v)$, equals the number $|N_G(v)|$.

For an edge e of G , $G - e$ is the graph obtained from G by deleting e . For a vertex v of G , $G - v$ is the graph obtained from G by deleting v and every edge incident with v . A graph H is called a *subgraph* of a graph G (and we write $H \subseteq G$) if H can be obtained by deleting vertices and edges from G . A subgraph of G is *spanning* if it contains every vertex of G . For a nonempty set $S \subseteq V(G)$, denote by $G[S]$ the subgraph of G obtained by deleting the vertices in $V(G) \setminus S$; $G[S]$ is the subgraph *induced* by S . A subgraph H of G is an *induced subgraph* if there exists a set $S \subseteq V(G)$ such that $H = G[S]$.

For an edge xy of G , G/xy is the graph obtained by adding a new vertex z adjacent to every vertex in $N_G(x) \cup N_G(y)$, and then deleting x and y (see Figure 1.4). This operation is called an *edge contraction*. A graph H is called a *minor* of a graph G if H can be obtained by deleting vertices, deleting edges and contracting edges of G .

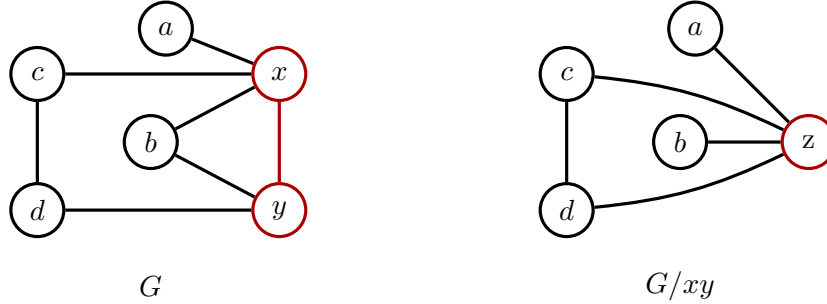


Figure 1.4: Contracting the edge xy of G (left) to a new vertex z (right).

A *homomorphism* from a graph H to a graph G is a function $f : V(H) \rightarrow V(G)$ such that, for all edges uv of H , $f(u)f(v)$ is an edge of G . An *embedding* is an injective homomorphism. An *isomorphism* from a graph H to a graph G is a bijection $f : V(H) \rightarrow V(G)$ such that, for all vertices $u, v \in V(H)$, $uv \in E(H)$ if and only if $f(u)f(v) \in E(G)$.

1.3.1 Matchings in Bipartite Graphs. A *matching* in a graph is a subset of edges in which no two edges are incident with the same vertex. A graph H is *bipartite* if there is a bipartition $\{A, B\}$ of $V(H)$ such that every edge of H is an AB -edge.

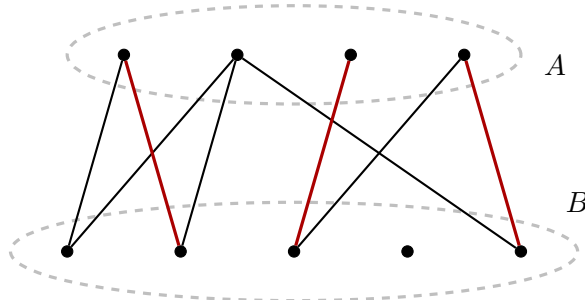


Figure 1.5: Matching (red) in a bipartite graph.

Given a bipartite graph H with bipartition $\{A, B\}$, a subset X of vertices of A is said to be *saturated* by a matching M if each vertex in X is incident with some edge in M . For a subset S of X , define $N_H(S) := \bigcup_{v \in S} N_H(v)$. Note, if X is saturated by some matching in H , then for every subset S of X , $|N_H(S)| \geq |S|$; Hall proved that this condition is sufficient to guarantee that a matching saturates X .

Theorem 1.17 (Hall [50]). *Let H be a bipartite graph with bipartition $\{A, B\}$. Then a subset X of A is saturated by a matching in H if for every subset S of X , $|N_H(S)| \geq |S|$.*

1.3.2 Connectivity. A *path* (subgraph) P in a graph G is a sequence of vertices $(u_1, \dots, u_k)$ such that for each $i \in \{1, \dots, k-1\}$, $u_i u_{i+1} \in E(G)$; here, the *ends* of P are u_1 and u_k , and the *length* of P is $k-1$. A subgraph Q of a path P that is itself a path is called a *subpath* of P . A path between two vertices u and v of G is called a *uv -path*. A subgraph H of G is said to be *connected* if for any two vertices u and v of H , there is a uv -path in H . If H is not connected, then H is *disconnected*. A *component* of G is a maximal connected subgraph of G .

A *cut-vertex* in a graph G is a vertex v such that $G - v$ has more components than G . A *bridge* in G is an edge e such that $G - e$ has more components than G .

For any graph G and any two subsets X, Y of $V(G)$, an *XY -path* in G is a path with its ends in $X \cup Y$ such that there is an end in X and an end in Y . A set of vertices S of G is an *XY -separator* in G if there are no XY -paths in $G - S$. Define $s(G, X, Y)$ to be the minimum size of an XY -separator in G . Note that the maximum number of (pairwise) disjoint XY -paths in G is at most $s(G, X, Y)$. Menger showed that $s(G, X, Y)$ disjoint XY -paths always exist.

Theorem 1.18 (Menger [60]). *For every graph G and any two subsets X, Y of $V(G)$, there exists $s(G, X, Y)$ disjoint XY -paths in G .*

1.3.3 Important Examples of Graphs. The *complete graph* on n vertices, denoted by K_n , is any graph with n vertices and $\binom{n}{2}$ edges. A *complete bipartite graph* is a bipartite graph, with bipartition $\{A, B\}$ and $m := |A|$, $n := |B|$ say, and all the possible mn AB -edges. This graph is denoted by $K_{m,n}$.

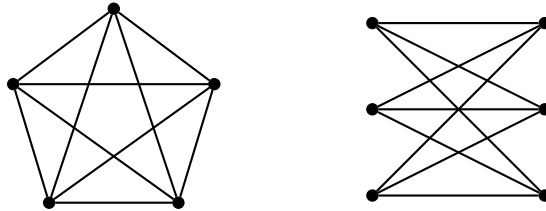


Figure 1.6: K_5 (left) and $K_{3,3}$ (right).

A *path* (graph) on n vertices, denoted by P_n , is any graph with vertex set $V(P_n) = \{p_1, \dots, p_n\}$ and edge set $E(P_n) = \{p_i p_{i+1} : i \in \{1, \dots, n-1\}\}$. A *cycle* (graph) on n

vertices, denoted by C_n , is any graph with vertex set $V(C_n) = \{c_1, \dots, c_n\}$ and edge set $E(C_n) = \{c_i c_{i+1} : i \in \{1, \dots, n-1\}\} \cup \{c_n c_1\}$.

A *forest* is a graph that contains no cycle as a subgraph. A *tree* is a connected forest. A subgraph T' of a tree T that is itself a tree is a *subtree* of T . Note the following elementary tree facts: (1) every tree on at least two vertices has a vertex of degree 1 (called a *leaf*), (2) every non-leaf vertex in a tree is a cut-vertex, (3) every edge in a tree is a bridge, and (4) for every nonempty tree T , $|E(T)| = |V(T)| - 1$.

A graph G is *planar* if it can be drawn in the plane without edge-crossings. For a given drawing of a planar graph G (in the plane), a *face* of G is a connected region of the plane minus the drawing of G . The *outer face* is the unbounded face and every other face is an *internal face*. A *plane triangulation* is a drawing of a planar graph in which every face is a triangle. Every planar graph G on at least three vertices is a subgraph of some plane triangulation G' where $V(G) = V(G')$ and $E(G) \subseteq E(G')$. Moreover, one can demand that any one chosen vertex of G is on the outer face of G' .

1.3.4 BFS Spanning Trees. For a graph G and vertices $u, v \in V(G)$, the *distance* in G between u and v , denoted by $\text{dist}_G(u, v)$, is the length of a shortest uv -path in G . If there is no uv -path in G , then $\text{dist}_G(u, v) := \infty$. The distance function is well-behaved.

Theorem 1.19. *For every connected graph G , $(V(G), \text{dist}_G)$ is a metric space; that is, for all vertices $u, v, w \in V(G)$, each of the following hold:*

- $\text{dist}_G(u, v) = 0$ if and only if $u = v$,
- $\text{dist}_G(u, v) = \text{dist}_G(v, u)$, and
- $\text{dist}_G(u, w) \leq \text{dist}_G(u, v) + \text{dist}_G(v, w)$.

Note that if v is contained in a shortest uw -path in G , then $\text{dist}_G(u, w) = \text{dist}_G(u, v) + \text{dist}_G(v, w)$.

A *spanning tree* of a graph G is a tree that is a spanning subgraph of G . Every connected graph has a spanning tree. For a vertex r of G , a spanning tree T of G is a *breadth-first search* (or *BFS*) spanning tree rooted at r if $\text{dist}_T(r, u) = \text{dist}_G(r, u)$ for all $u \in V(G)$.

Lemma 1.20. *For every connected graph G , for any vertex r of G , there exists a BFS spanning tree of G rooted at r .*

1.3.5 Asymptotic Notation. For nonnegative functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$, we use the following asymptotic notation:

- $f(n) \leq \mathcal{O}(g(n))$ if there exists a constant C such that $f(n) \leq Cg(n)$ for all large enough n ,
- $f(n) \geq \Omega(g(n))$ if $g(n) \leq \mathcal{O}(f(n))$,
- $f(n) = \Theta(g(n))$ if $\Omega(g(n)) \leq f(n) \leq \mathcal{O}(g(n))$, and
- $f(n) = o(g(n))$ if $\frac{f(n)}{g(n)} \rightarrow 0$ when $n \rightarrow \infty$.

For two nonnegative parameters n and k , we write $n \gg k$ to mean that n is sufficiently larger than k .

Chapter 2

Universal Graphs for Trees

Initially, the main goal driving my Honours project was to determine some nontrivial bounds for the minimum number of edges in a graph this is universal for n -vertex treewidth- k graphs. Roughly speaking, the treewidth of a graph G is a nonnegative integer associated with G that measures how ‘tree-like’ G is; we give the formal definition of treewidth in Chapter 3. In fact, forests are precisely the graphs with treewidth at most 1. Therefore, a good starting point to make progress towards the goal is to study universal graphs for trees.

Recall that $s(n)$ is the minimum number of edges in a graph that is universal for n -vertex trees. This chapter is concerned with asymptotic bounds for $s(n)$. An $\Omega(n \log n)$ lower bound can be seen by observing that any graph U that is universal for n -vertex trees must contain, for each $i \in \{1, \dots, n\}$, the forest with i components, each of which is isomorphic to the complete bipartite graph $K_{1, \lfloor \frac{n}{i} \rfloor - 1}$. Thus, the degree sequence of U lexicographically dominates $(\lfloor \frac{n}{1} \rfloor - 1, \lfloor \frac{n}{2} \rfloor - 1, \lfloor \frac{n}{3} \rfloor - 1, \dots)$, implying $|E(U)| \geq \Omega(n \log n)$. Naturally, one can ask whether there exists a graph with $\mathcal{O}(n \log n)$ edges that is universal for n -vertex trees. This motivates the discussion of techniques to construct economical universal graphs.

We introduce the notion of perfectly balanced separators in Section 2.1; roughly speaking, a perfectly balanced separator in a graph G is a set of vertices X of G such that $G - X$ can be divided into two equal sized subgraphs of G . For any class of graphs $\mathcal{G}$ that is closed under taking subgraphs such that every n -vertex graph in $\mathcal{G}$ has a perfectly balanced separator of size $f(n)$, we explain in Section 2.1 how to construct a universal graph U_n for the n -vertex graphs in $\mathcal{G}$ with $|E(U_n)| \leq \mathcal{O}(nf(n) \log n)$. This construction is quite general and can be used to give universal graphs for many graph classes, not just trees. Thus, we return to the constructions given by perfectly balanced separators in Chapters 3 and 4 when discussing bigger classes.

To obtain a good upper bound on $s(n)$ using perfectly balanced separators, one needs good upper bounds on the size of perfectly balanced separators in n -vertex trees. This leads to the discussion of pathwidth and path-decompositions. Like treewidth, the pathwidth of a graph G is a number associated with G that measures how ‘path-like’ G is. We show that every graph with pathwidth k has a perfectly balanced separator of size $(1 + o(1))k$.

We then show that every n -vertex tree (and forest) has pathwidth $\mathcal{O}(\log n)$, implying the upper bound $s(n) \leq \mathcal{O}(n \log^2 n)$. This is only a $\log n$ factor away from asymptotically matching the $\Omega(n \log n)$ lower bound.

In the classical paper ‘On Universal Graphs for Spanning Trees,’ Chung and Graham [24] claim to show a $\mathcal{O}(n \log n)$ upper bound for $s(n)$. However, I found a mistake in their argument. The mistake is explained in Section 2.2. At the time when I found this mistake, I could not see any evidence of this mistake being known or even fixed (I later learned that the mistake was mentioned in another paper; see the ‘Postscript’ in Chapter 3). The previously best known upper bound for $s(n)$ prior to the claimed $\mathcal{O}(n \log n)$ bound in [24] is $s(n) \leq \mathcal{O}(n(\log n)(\log \log n)^2)$ by Chung, Graham, and Pippenger [25]. Therefore, I attempted to improve the $\mathcal{O}(n(\log n)(\log \log n)^2)$ bound. Section 2.3 presents a fully self-contained original proof of the bound $s(n) \leq \mathcal{O}(n(\log n)(\log \log n))$. An advantage of this upper bound is that its proof can be extended, without too much extra effort, to give a bound for the minimum number of edges in a universal graph for n -vertex treewidth- k graphs; we come back to this idea in Chapter 3.

2.1 Perfectly Balanced Separators

Given a graph class $\mathcal{G}$, suppose we want to construct a universal graph for the n -vertex graphs in $\mathcal{G}$. The goal is to be economical with the number of edges in the universal graph. If $\mathcal{G}$ is *subgraph-closed* (that is, $\mathcal{G}$ is closed under taking subgraphs) then one way to construct a universal graph for the n -vertex graphs in $\mathcal{G}$ is to employ a divide-and-conquer approach. By showing that every n -vertex graph G in $\mathcal{G}$ has a separator that ‘divides’ G into two subgraphs, each on at most m vertices (where $m < n$), we can apply induction to get a universal graph for the m -vertex graphs in $\mathcal{G}$. By carefully combining these universal graphs, one can (hopefully) form a universal graph for the n -vertex graphs in $\mathcal{G}$, thus proving the induction step.

For this divide-and-conquer approach to be economical, the separators used in the ‘dividing’ step should be balanced and small; we return to the issue of smallness shortly. To be precise, for a graph G and a partition $\{A, S, B\}$ of $V(G)$, the set S is called a *perfectly balanced separator* of G if $|A| = |B|$ and there is no AB -edge in G . Given a subgraph-closed class $\mathcal{G}$, suppose there is a function $f(n)$ such that every n -vertex graph in $\mathcal{G}$ has a perfectly balanced separator of size $f(n)$.

Let $d := \frac{n-f(n)}{2}$ (which is an integer). Then, for integers $n \geq 1$, we can define the graph U_n recursively. Let $U_1 := K_1$. For $n \geq 2$, let U_n be the graph formed by adding all the possible edges between a complete graph $K_{f(n)}$ and two disjoint copies of U_d as shown in Figure 2.1.

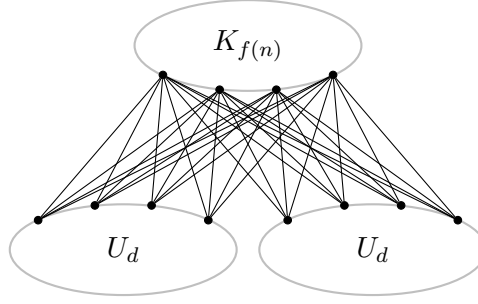


Figure 2.1: Constructing U_n .

Note that U_n has exactly n vertices. Before we show that U_n is universal for the n -vertex graphs in $\mathcal{G}$, we first estimate the number of edges in U_n . By the construction of U_n , it follows that $|E(U_n)|$ is bounded above by a function $e(n)$ satisfying the following recurrence:

$$e(n) \geq 2e\left(\frac{n-f(n)}{2}\right) + (n-1)f(n). \quad (2.1)$$

Lemma 2.1. *If $f(n)$ is a non-decreasing function, then $e(n) = nf(n) \log_2 n$ satisfies (2.1) for all $n \geq 2$.*

Proof. When $e(n) = nf(n) \log_2 n$, the right hand side of (2.1) is bounded above by

$$\begin{aligned} 2e\left(\frac{n-f(n)}{2}\right) + (n-1)f(n) &\leq 2\frac{n}{2}f(n) \log_2 \left(\frac{n}{2}\right) + (n-1)f(n) \\ &= nf(n) \log_2 n - f(n)n + (n-1)f(n) \\ &= e(n) - f(n) \\ &\leq e(n), \end{aligned}$$

as desired. $\square$

Thus, the number of edges in U_n is $\mathcal{O}(nf(n) \log n)$. We now prove the universality of U_n .

Theorem 2.2 (Chung [22]). *For all integers $n \geq 1$, U_n is universal for the n -vertex graphs in $\mathcal{G}$.*

Proof. Proceed by induction on n . The $n = 1$ case holds trivially, so assume $n \geq 2$ and the result holds for all lower values. Let G be an n -vertex graph in $\mathcal{G}$. Then there is a partition $\{A_1, S, A_2\}$ of $V(G)$ with $|S| = f(n)$ and $|A_1| = |A_2| = d$ such that there is no $A_1 A_2$ -edge in G .

For each $i \in \{1, 2\}$, let $G_i := G[A_i]$. Since $\mathcal{G}$ is subgraph-closed, by the induction hypothesis, G_i is contained in U_d . Map G_i to the i -th copy of U_d in U_n . Map S to the $K_{f(n)}$ in U_n . Since the $K_{f(n)}$ is complete to each copy of U_d in U_n , this shows that G is contained in U_n . The result holds by induction. $\square$

The asymptotics for the number of edges in U_n is directly related to the growth of the function $f(n)$. Thus, it is crucial to study the best/smallest function one can put in place

of $f(n)$. Let $\text{pbs}_{\mathcal{G}}(n)$ denote the best choice for $f(n)$; that is, $\text{pbs}_{\mathcal{G}}(n)$ is equal to the smallest number k such that every n -vertex graph in $\mathcal{G}$ has a perfectly balanced separator of size k . If we pick $f(n) = \text{pbs}_{\mathcal{G}}(n)$ in the construction of U_n , then we obtain the bound:

Corollary 2.3. $|E(U_n)| \leq \mathcal{O}(n \text{pbs}_{\mathcal{G}}(n) \log n)$.

Note that the construction of U_n is quite general and can therefore be used to obtain economical universal graphs for many graph classes $\mathcal{G}$ (when $\text{pbs}_{\mathcal{G}}(n)$ does not grow too quickly). A possible concern one can raise in the construction of U_n is that $\mathcal{G}$ must be subgraph-closed for the argument in Theorem 2.2 to follow through. In this thesis, the main graph classes of interest are forests, bounded treewidth graphs, and planar graphs. These graph classes are closed under taking minors and are therefore automatically subgraph-closed. Thus, it is harmless to assume that $\mathcal{G}$ is subgraph-closed.

The bound in Corollary 2.3 is only as good as the known bounds for $\text{pbs}_{\mathcal{G}}(n)$. Conveniently, the size of a smallest perfectly balanced separator in a graph G is at most a linear function of the pathwidth of G . We introduce pathwidth in the following section and prove this relationship.

2.1.1 Path-Decompositions. A *path-decomposition* of a graph G is a collection of subsets $(B_x : x \in V(P))$ of $V(G)$ (called *bags*) indexed by the vertices of a path P , such that each of the following hold:

- (*vertex-property*) for all $v \in V(G)$, $P[\{x \in V(P) : v \in B_x\}]$ is a nonempty sub-path of P , and
- (*edge-property*) for all $uv \in E(G)$, there exists $x \in V(P)$ such that $\{u, v\} \subseteq B_x$.

The *width* of the path-decomposition $(B_x : x \in V(P))$ equals $\max_{x \in V(P)} |B_x| - 1$. The *pathwidth* of G , denoted by $\text{pw}(G)$, is the minimum width of a path-decomposition of G . Note that every graph G has a trivial path-decomposition consisting of a single bag $V(G)$. This shows that pathwidth is well-defined, in particular, for every graph G , $\text{pw}(G) \leq |V(G)| - 1$.

For a graph G and a path-decomposition $(B_x : x \in V(P))$ of G , the vertex-property says that each vertex of G is mapped to a path in P . This can be seen in Figure 2.2. The edge-property says that a path-decomposition of G keeps track of the adjacencies in G (but not necessarily the non-adjacencies).

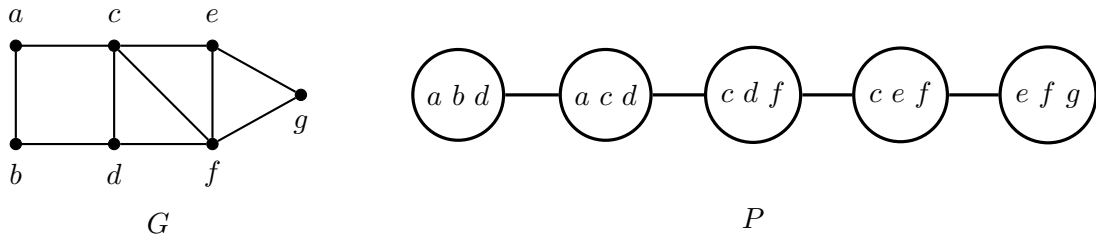


Figure 2.2: A graph G (left) and a path-decomposition of G (right).

Often, a path-decomposition is expressed as a sequence of bags $(B_1, \dots, B_t)$, where the

underlying path is taken to be $(1, 2, \dots, t)$. A path-decomposition with width k is *normal* if every bag has size $k + 1$, and the intersection of any two adjacent bags has size k . The following result implies that one can always take a normal path-decomposition of a graph.

Lemma 2.4. *If G has a path-decomposition with width k , then G has a normal path-decomposition with width k .*

Proof. Let $(B_x : x \in V(P))$ be a path-decomposition of G with width k . If there is a bag with fewer than $k + 1$ vertices, then there exists an edge xy in P such that $|B_x| = k + 1$ and $|B_y| \leq k$. In particular, $B_x \setminus B_y \neq \emptyset$. Add a vertex of $B_x \setminus B_y$ to B_y . Repeat this process until every bag of P has exactly $k + 1$ vertices.

If there is an edge xy of P such that $|B_x \setminus B_y| \geq 2$, then $|B_y \setminus B_x| \geq 2$ since $|B_x| = |B_y|$. Let $u \in B_x \setminus B_y$ and $v \in B_y \setminus B_x$. Subdivide the edge xy in P and call the new vertex z . Let $B_z := (B_x \setminus \{u\}) \cup \{v\}$. Then $|B_z| = k + 1$, $|B_x \setminus B_z| = 1$ and $|B_y \setminus B_z| = |B_y \setminus B_x| - 1$. Repeat this process until any two adjacent bags differ by at most 1 vertex.

If there is an edge xy of P such that $|B_x \setminus B_y| = 0$, then contract the edge xy to a new vertex z . Let $B_z := B_x$. Repeat this process until any two adjacent bags differ by exactly 1 vertex.

Note that P remains a path after subdividing and contracting edges. Furthermore, the operations described maintain the vertex and edge-property without changing the width. Thus, after applying these operations, $(B_x : x \in V(P))$ is a normal path-decomposition of G with width k . $\square$

For us, the main use of path-decompositions is to prove upper bounds on the minimum size of a perfectly balanced separator in a graph. The following result implies that every graph that has a path-decomposition with width k , has a perfectly balanced separator of size $(1 + o(1))k$.

Theorem 2.5. *For all integers $k \geq 1$ and $n \geq k + 1$, for every n -vertex graph G that has a path-decomposition with width k , there is a partition $\{A, S, B\}$ of $V(G)$ such that:*

- (1) $|A| = |B|$ and there is no AB -edge in G , and
- (2) $|S| = k + 1$ if $n - k$ is odd, and $|S| = k + 2$ if $n - k$ is even.

Proof. By Lemma 2.4, G has a normal path-decomposition $(B_1, \dots, B_t)$ with width k . If $t = 1$, then $n - k = 1$, so taking $S := V(G)$ suffices. Assume now that $t \geq 2$. Since adjacent bags in a normal path-decomposition differ by exactly one vertex, there exists $q \in \{1, \dots, t - 1\}$ such that $|(B_1 \cup \dots \cup B_q) \setminus B_{q+1}| = \lfloor \frac{n-k-1}{2} \rfloor$. Similarly, there exists $p \in \{2, \dots, t\}$ such that $|(B_p \cup \dots \cup B_t) \setminus B_{p-1}| = \lfloor \frac{n-k-1}{2} \rfloor$. Note that the bags B_{q+1} and B_{p-1} are either adjacent or are the same bag; the former case occurs when $n - k$ is even, and the latter case occurs when $n - k$ is odd. Let $S := B_{q+1} \cup B_{p-1}$. Then S satisfies (2). Let $A := (B_1 \cup \dots \cup B_q) \setminus B_{q+1}$ and let $B := (B_p \cup \dots \cup B_t) \setminus B_{p-1}$. Then $|A| = |B|$ and $\{A, S, B\}$ partitions $V(G)$.

For a contradiction, assume there is an edge uv in G with $u \in A$ and $v \in B$. By the edge-property, there is a bag containing u and v . Then, by the vertex-property, S contains u

or v , contradicting that $S \cap A = S \cap B = \emptyset$. Hence, there is no AB -edge in G , implying $\{A, S, B\}$ is a desired partition of $V(G)$. $\square$

Thus, Corollary 2.3 and Theorem 2.5 imply the following result for any subgraph-closed class $\mathcal{G}$.

Corollary 2.6. *Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function such that for every integer $n \geq 1$, every n -vertex graph in $\mathcal{G}$ has pathwidth at most $f(n)$. Then there exists an n -vertex graph with $\mathcal{O}(nf(n) \log n)$ edges that is universal for the n -vertex graphs in $\mathcal{G}$.*

In light of Corollary 2.6, it is natural to seek bounds for the pathwidth of trees. In the following section, we show that the pathwidth of an n -vertex forest is $\mathcal{O}(\log n)$.

Since Corollary 2.6 is quite general, it is used again in Chapters 3 and 4 to obtain initial bounds for the minimum number of edges in a universal graph for bounded treewidth graphs and for planar graphs.

2.1.2 Pathwidth of Forests. We present a proof that the pathwidth of a forest F is at most $\log_{3/2}(|V(F)|)$. This bound is useful since we can apply Corollary 2.6 to obtain an upper bound on $s(n)$. Note, as a uniform bound, $\mathcal{O}(\log n)$ is best possible for the pathwidth of n -vertex forests in the following sense: for infinitely many values of n , there exists n -vertex forests with pathwidth $\Omega(\log n)$.

Theorem 2.7 (Scheffler [71]). *The complete binary tree on $2^{k+1} - 1$ vertices has pathwidth $\lceil k/2 \rceil$.*

We prove that $\text{pw}(F) \leq \log_{3/2}(|V(F)|)$ for every forest F by giving a path-decomposition of F with width at most $\log_{3/2}(|V(F)|)$. We require the following lemmas about balanced separating vertices in forests.

Lemma 2.8. *Every n -vertex tree T has a vertex v such that each component of $T - v$ has at most $n/2$ vertices.*

Proof. We may assume $n \geq 2$. For each edge uw of T , if the component of $T - uw$ containing w has at most $n/2$ vertices, orient the edge uw from u to w . Every edge of T is oriented. Since T is a tree, T contains no cycles. Thus, there exists a vertex v in $V(T)$ such that each of its edges is oriented outward from v . By the way we oriented the edges of T , each component of $T - v$ has at most $n/2$ vertices. $\square$

Lemma 2.9. *For every n -vertex forest F , there is a vertex $v \in V(F)$ and a partition $\{A, B\}$ of $V(F - v)$ with $|A|, |B| \leq 2n/3$ such that there is no AB -edge in F .*

Proof. We may assume $n \geq 2$. Let T be the tree obtained from F by adding edges between the components of F without creating cycles. By Lemma 2.8, there is a vertex $v \in V(T)$ such that each component of $T - v$ has at most $n/2$ vertices. Let $C_1, \dots, C_t$ be an ordering of the components of $T - v$ such that $|V(C_i)| \geq |V(C_{i+1})|$ for each $i \in$

$\{1, \dots, t-1\}$. Let $m \in \{1, \dots, t\}$ be maximum such that $|V(C_1)| + \dots + |V(C_m)| \leq 2n/3$. Let $A := V(C_1) \cup \dots \cup V(C_m)$ and let $B := V(T-v) \setminus A$. Then there is no AB -edge in F . For a contradiction, assume $|B| > 2n/3$. Then $|A| = n-1-|B| < n/3$. Therefore, $n/3 \geq |A| \geq |V(C_m)| \geq |V(C_{m+1})|$, which implies that

$$|V(C_1)| + \dots + |V(C_m)| + |V(C_{m+1})| = |A| + |V(C_{m+1})| \leq 2n/3,$$

contradicting the maximality of m . Hence, $\{A, B\}$ is a desired partition of $V(F-v)$. $\square$

Lemmas 2.8 and 2.9 together imply the desired bound for the pathwidth of forests.

Theorem 2.10 (Scheffler [71]). *For every n -vertex forest F , $\text{pw}(F) \leq \log_{3/2} n$.*

Proof. Proceed by induction on n . In the $n=1$ case, $\text{pw}(F) = 0 = \log_{3/2} 1$. Now assume that $n \geq 2$ and the result holds for all lower values. By Lemma 2.9, there is a vertex $v \in V(F)$ and a partition $\{A, B\}$ of $V(F-v)$ with $|A|, |B| \leq 2n/3$ such that there is no AB -edge in F . By the induction hypothesis, $\text{pw}(F[A])$ and $\text{pw}(F[B])$ are at most $\log_{3/2}(2n/3) = \log_{3/2} n - 1$. Let $(C_1, \dots, C_g)$ and $(D_1, \dots, D_h)$ be path-decompositions of $F[A]$ and $F[B]$ respectively with minimum widths. Then $(C_1 \cup \{v\}, \dots, C_g \cup \{v\}, D_1 \cup \{v\}, \dots, D_h \cup \{v\})$ is a path-decomposition of F with width at most $\log_{3/2} n - 1 + 1 = \log_{3/2} n$. The result follows by induction. $\square$

Hence, applying Corollary 2.6 with $f(n) = \log_{3/2} n$ (and $\mathcal{G}$ the class of forests) implies:

Theorem 2.11 (Chung [22]). $s(n) \leq \mathcal{O}(n \log^2 n)$.

This bound is only a $\log n$ factor away from matching the $\Omega(n \log n)$ lower bound from earlier. The remainder of this chapter is dedicated to the discussion of improvements to this upper bound.

2.2 40 Year-Old Mistake

In a series of heavily cited papers by Chung and Graham [23], Chung, Graham, and Pippenger [25], and Chung and Graham [24], a sequence of upper bounds for $s(n)$ were proved, culminating in a $\mathcal{O}(n \log n)$ bound by Chung and Graham [24], asymptotically matching the $\Omega(n \log n)$ lower bound. I found a mistake in Chung and Graham's argument in [24]. In this section, the necessary definitions from the original paper are introduced to explain the mistake in detail.

The exact assertion that Chung and Graham [24] claim to show is the inequality:

$$s(n) \leq \frac{5}{2} n \log_2 n + \mathcal{O}(n).$$

In order to explain the mistake, we reuse the notation and terminology from the original paper word-for-word as much as possible. Figures 2.3–2.5, 2.7 and 2.8 are copied directly from [24]. Denote by $B(k)$ the *binary tree with k levels* (see Figure 2.3).

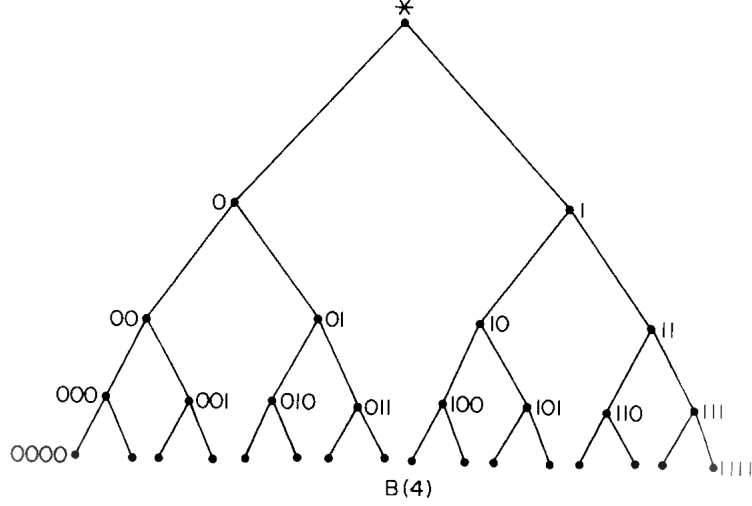


Figure 2.3: Binary tree with k levels ($k = 4$).

The vertices of $B(k)$ are the $\{0, 1\}$ -strings of length at most k (and $*$ is the empty string). Denote the vertex set of $B(k)$ by $V(k)$. For $\alpha \in \{0, 1\}$, denote by $B_\alpha(k-1)$ the copy of $B(k-1)$ rooted at vertex α (see Figure 2.4).

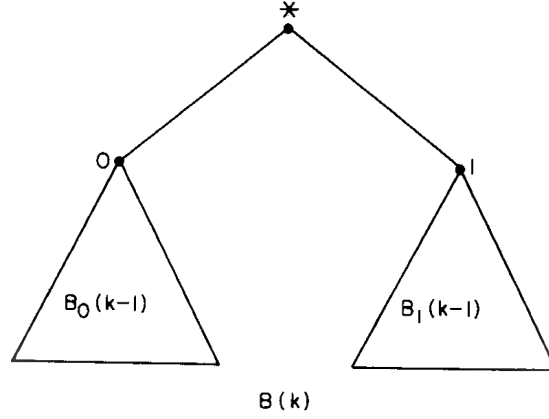


Figure 2.4: $B_0(k-1)$ and $B_1(k-1)$.

If $\alpha = x_1 \dots x_j \in V(k)$ with $j < k$, then $\alpha 0$ and $\alpha 1$ are the *left* and *right sons* of α respectively. Also, α is the *father* of $\alpha 0$ and $\alpha 1$, and $\alpha 0$ is called the *left-hand brother* of $\alpha 1$. A *descendant* of α is a vertex $\beta = y_1 \dots y_\ell \in V(k)$ such that $j < \ell \leq k$ and for all $i \in \{1, \dots, j\}$, $x_i = y_i$.

A subtree A of $B(k)$ is said to be *admissible* if one of the following holds:

- (i) $V(A) = \emptyset$,
- (ii) $V(A) = \{*\}$, or
- (iii) A is of the form (a) or (b) shown in Figure 2.5, where A' is an admissible subtree of $B(k-1)$ identified at its root to the indicated vertex.

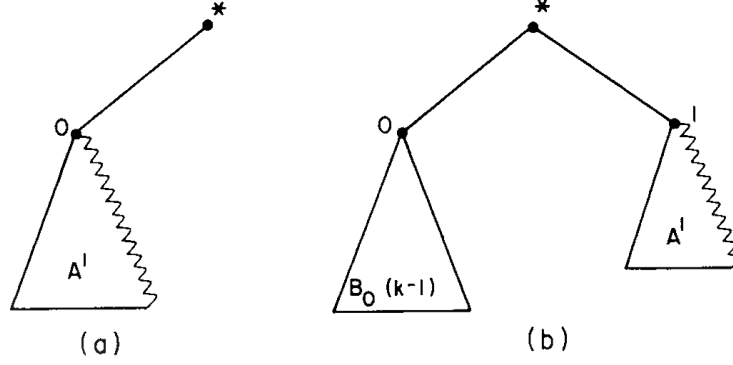


Figure 2.5: Structure of nontrivial admissible subtrees.

For an admissible subtree A of $B(k)$ and a vertex $v \in V(A)$, let $\nu_A(v)$ denote the total number of descendants of v in $V(A)$. Define a new graph $G(k)$ on the vertex set $V(k)$ with the following edges (see Figure 2.6). For every $\alpha \in V(k)$, add an edge between α and:

- (i) each descendant of α ,
- (ii) the left-hand brother of α (if it exists) and each of its descendants, and
- (iii) the left-hand brother of the father of α (if it exists) and each of its descendants.

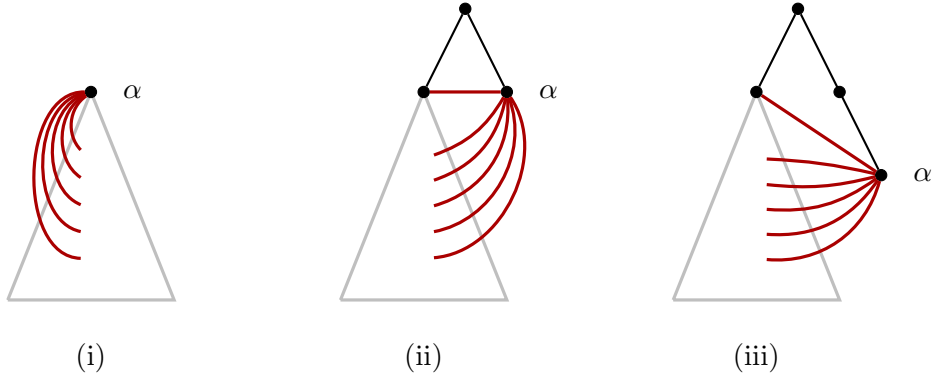


Figure 2.6: Three types of edges in $G(k)$.

A subgraph G of $G(k)$ is called *admissible* if it is an induced subgraph of $G(k)$, and $V(G)$ is the vertex set of an admissible subtree of $B(k)$. For an admissible subgraph G of $G(k)$ and a vertex $v \in V(G)$, let $\nu_G(v)$ denote the total number of descendants of v in $V(G)$. The main goal in [24] is to show that for all integers $n \geq 1$, every n -vertex admissible graph G is universal for n -vertex trees. The following fundamental claim in the paper intends to achieve this goal. The proof of this claim is where the mistake lies.

Claim ([24, page 205]). *For every admissible graph G , for every tree T with $|V(T)| \leq |V(G)|$ and for every $v \in V(T)$ there is an embedding $\lambda : V(T) \rightarrow V(G)$ such that $G - \lambda(V(T))$ is admissible and*

$$\nu_G(\lambda(v)) < |V(T)| \leq \nu_G(\lambda(v)^*),$$

where $\lambda(v)^*$ denotes the father of $\lambda(v)$ in G (if it exists).

As presented in [24], the proof of this claim proceeds by induction on the number of vertices in G and T (with priority given to $|V(G)|$). In the base case, $V(G) = \{*\}$. In the induction step, we are given an admissible graph G , a tree T such that $|V(T)| \leq |V(G)|$, and a vertex $v \in V(T)$. Furthermore, $V(G)$ has the form (a) or (b) as shown in Figure 2.5. Consider Subcase (b) 1. in the middle of page 207. Here, G has the form shown in Figure 2.7 (where G' is an admissible subgraph of $G(k-2)$ identified at its root to vertex 10) and $|V(T)| < |V(G)|$.

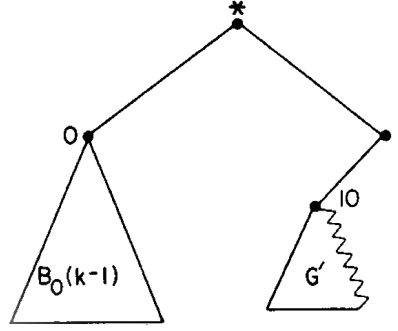


Figure 2.7: Structure of G in Subcase (b) 1.

Moreover, by the way the edges in $G(k)$ are defined, we can consider G to have the form shown in Figure 2.8; let G_2 to be the subgraph of G induced by the vertices in the circled region.

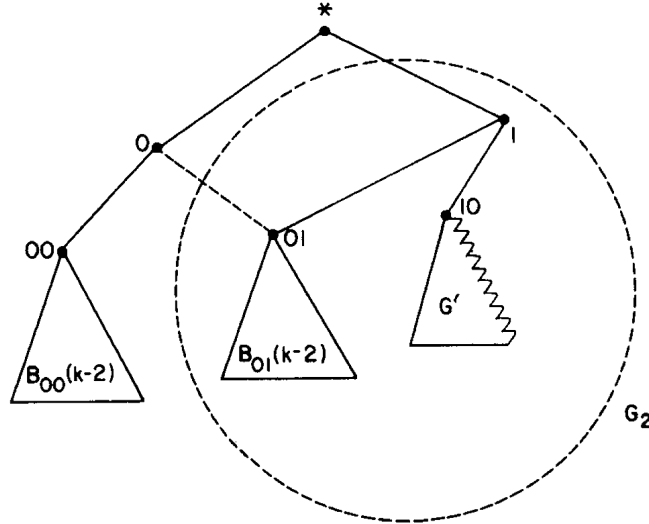


Figure 2.8: Defining G_2 .

Chung and Graham [24, bottom of page 207] claim that G_2 “is (isomorphic to) an admissible graph.” For the arguments that follow, it is vital that in the interpretation of the phrase “isomorphic to” one assumes the isomorphism maps vertex 1 of G_2 to the root $*$ of some admissible tree A . Consider the case when $2 \leq |V(G')| < |V(B(k-2))|$. Then the first inequality implies that vertex 10 has a son α in G' . By the structure of nontrivial admissible subtrees, the second inequality implies that $B_{01}(k-2)$ and G' are mapped respectively to the left and right subtrees of the root of A . For G_2 to be admissible, by

property (iii) of admissible graphs, α must be adjacent to the left-hand brother of the father of α and all of its descendants; that is, α is adjacent to vertex 01 and each of its descendants. These edges are not present in G , and are therefore not present in G_2 . Thus, we cannot conclude that G_2 is “isomorphic to” an admissible graph, and so the proof of the claim does not follow through. This completes the explanation of the mistake.

2.3 New Upper Bound for $s(n)$

In an attempt to fix Chung and Graham’s mistake in [24], I proved the bound:

$$s(n) \leq \mathcal{O}(n(\log n)(\log \log n)) .$$

The remainder of this chapter presents a fully self-contained proof of this original result. The proof of this result has three major steps. The first involves showing the existence of a certain type of separator in a tree. The second involves using these separators to recursively construct a graph G that is universal for trees of a certain size. The third step involves estimating the number of edges in G .

2.3.1 W -Subtrees of Forests. For the remainder of this chapter, we say that a graph U ‘contains’ a graph G if G is isomorphic to a subgraph of U . For a forest F and a set $W \subseteq V(F)$, a **W -subtree** of F is a maximal subtree of F in which the vertices in W only appear as leaves (see Figure 2.9). Note that every edge in F lies in a unique W -subtree of F .

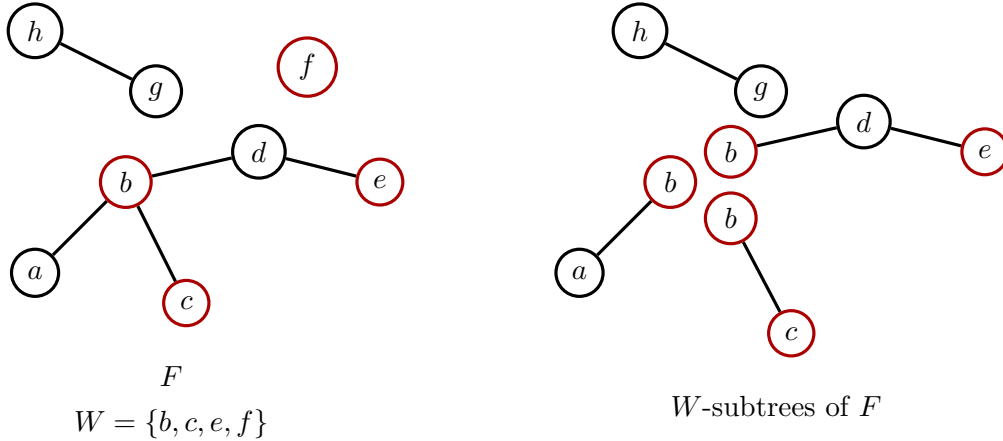


Figure 2.9: A forest F (left) and every W -subtree of F (right).

Chung et al. [25] proved a slight variation of the following result about the existence of W -subtrees in large trees; their proof is missing details, so a proof is included here for completeness.

Lemma 2.12. *For every integer $w \geq 0$, for every real number $\alpha \geq 3(\frac{3}{2})^w$, for every tree T with $|E(T)| > \alpha$, there exists $W \subseteq V(T)$ with $|W| \leq w + 1$ such that for some set $\mathcal{C}$ of*

W -subtrees of T ,

$$\alpha < \sum_{T' \in \mathcal{C}} |E(T')| \leq \left(1 + \left(\frac{2}{3}\right)^w\right) \alpha.$$

The proof of this result requires the following lemma originally proved by Chung and Graham [23]; we include a proof for completeness. A *v -subtree* is a $\{v\}$ -subtree.

Lemma 2.13. *For every real number $\alpha \geq 1$, for every forest F with $|E(F)| > \alpha$, there exists a vertex $v \in V(F)$ such that for some set $\mathcal{C}$ of v -subtrees of F ,*

$$\alpha < \sum_{T' \in \mathcal{C}} |E(T')| \leq 2\alpha.$$

Proof. If every component of F has at most α edges, then pick any vertex $v \in V(F)$, and let $\{T_1, \dots, T_j\}$ be a minimal set of v -subtrees of F such that $|E(T_1)| + \dots + |E(T_j)| > \alpha$. If $|E(T_1)| + \dots + |E(T_j)| > 2\alpha$, then the minimality of $\{T_1, \dots, T_j\}$ implies

$$|E(T_j)| = |E(T_1)| + \dots + |E(T_j)| - (|E(T_1)| + \dots + |E(T_{j-1})|) > 2\alpha - \alpha = \alpha,$$

which contradicts that $|E(T_j)| \leq \alpha$. Thus, $\{T_1, \dots, T_j\}$ is a desired set of v -subtrees of F .

Now assume some component of F has more than α edges, call it T . Let x be a leaf in T . We may assume $|E(T)| > 2\alpha$, otherwise $\{T\}$ is a desired set of x -subtrees of F . Then $|E(T-x)| > 2\alpha - 1 \geq \alpha$. Let y be the neighbour of x in T . If every y -subtree of $T-x$ has at most α edges, then by the same argument given above, a minimal set of y -subtrees of $T-x$ whose union has more than α edges will suffice. Now assume some y -subtree of $T-x$ has more than α edges. Let $P = (v_0, v_1, \dots, v_t)$ be a maximal path in T such that $v_0 = x$, $v_1 = y$ and for each $i \geq 1$ there exists a v_i -subtree of $T - v_{i-1}$ that contains v_i and has more than α edges. Let T' be a v_t -subtree of $T - v_{t-1}$ that contains v_t with $|E(T')| > \alpha$. We may assume $|E(T')| > 2\alpha$, otherwise $\{T'\}$ is a desired set of v_t -subtrees of F . Then $|E(T' - v_t)| > 2\alpha - 1 \geq \alpha$. Let v_{t+1} be the neighbour of v_t in T' . By the maximality of P , every v_{t+1} -subtree in $T' - v_t$ has at most α edges. By the same argument given above, a minimal set of v_{t+1} -subtrees of $T' - v_t$ whose union has more than α edges will suffice. This proves the lemma. $\square$

Proof of Lemma 2.12. Proceed by induction on w . The $w = 0$ case is a direct application of Lemma 2.13. Assume the result holds for some $w \geq 0$. Let $\alpha \geq 3(\frac{2}{3})^{w+1}$ and let T be a tree with $|E(T)| > \alpha$. Since $\alpha \geq 3(\frac{2}{3})^w$, by the induction hypothesis there exists $W \subseteq V(T)$ with $|W| \leq w + 1$ such that for some set $\mathcal{C}(W)$ of W -subtrees of T , $\alpha < \sum_{T' \in \mathcal{C}(W)} |E(T')| \leq (1 + (\frac{2}{3})^w) \alpha$. We are done if $\sum_{T' \in \mathcal{C}(W)} |E(T')| \leq (1 + (\frac{2}{3})^{w+1}) \alpha$, so assume the contrary. Then

$$\left(1 + \left(\frac{2}{3}\right)^{w+1}\right) \alpha < \sum_{T' \in \mathcal{C}(W)} |E(T')| \leq \left(1 + \left(\frac{2}{3}\right)^w\right) \alpha. \quad (2.2)$$

Let $F := \bigcup_{T' \in \mathcal{C}(W)} T'$ and $\beta := \frac{1}{3}(\frac{2}{3})^w \alpha \geq 1$. Observe that F is a forest. Since the subtrees in $\mathcal{C}(W)$ are edge-disjoint,

$$|E(F)| = \sum_{T' \in \mathcal{C}(W)} |E(T')| > \left(1 + \left(\frac{2}{3}\right)^{w+1}\right) \alpha = \alpha + 2\beta \geq \beta.$$

Since $\beta \geq 1$, Lemma 2.13 implies that there is a vertex $v \in V(F)$ such that for some set $\mathcal{C}(v)$ of v -subtrees of F ,

$$\frac{1}{3} \left(\frac{2}{3}\right)^w \alpha = \beta < \sum_{T' \in \mathcal{C}(v)} |E(T')| \leq 2\beta = \frac{2}{3} \left(\frac{2}{3}\right)^w \alpha. \quad (2.3)$$

Then (2.2) and (2.3) together imply

$$\begin{aligned} \alpha &= \left(1 + \left(\frac{2}{3}\right)^{w+1}\right) \alpha - \frac{2}{3} \left(\frac{2}{3}\right)^w \alpha < \sum_{T' \in \mathcal{C}(W)} |E(T')| - \sum_{T' \in \mathcal{C}(v)} |E(T')| \\ &\leq \left(1 + \left(\frac{2}{3}\right)^w\right) \alpha - \frac{1}{3} \left(\frac{2}{3}\right)^w \alpha \\ &= \left(1 + \left(\frac{2}{3}\right)^{w+1}\right) \alpha. \end{aligned} \quad (2.4)$$

Let $\overline{\mathcal{C}(v)}$ be the set of all v -subtrees of F not in $\mathcal{C}(v)$. Since $\{E(T') : T' \in \mathcal{C}(W)\}$ and $\{E(T') : T' \in \mathcal{C}(v)\} \cup \{E(T') : T' \in \overline{\mathcal{C}(v)}\}$ are both partitions of $E(F)$,

$$|E(F)| = \sum_{T' \in \mathcal{C}(W)} |E(T')| = \sum_{T' \in \mathcal{C}(v)} |E(T')| + \sum_{T' \in \overline{\mathcal{C}(v)}} |E(T')|.$$

Thus, (2.4) implies

$$\alpha < \sum_{T' \in \overline{\mathcal{C}(v)}} |E(T')| \leq \left(1 + \left(\frac{2}{3}\right)^{w+1}\right) \alpha.$$

Hence, $\overline{\mathcal{C}(v)}$ is a desired set of $(W \cup \{v\})$ -subtrees of T . The result follows by induction. $\square$

2.3.2 Useful Functions. This section proves a number of elementary facts, which are used extensively in Section 2.3.3. For integers $w, p \geq 0$, define

$$\begin{aligned} \alpha(w, p) &:= \frac{1}{1 + \left(\frac{2}{3}\right)^w} \left(\frac{2 + \left(\frac{2}{3}\right)^w}{1 + \left(\frac{2}{3}\right)^w}\right)^{p-1}, & \beta(w) &:= 3 \left(\frac{3}{2}\right)^w, \\ \gamma(w, p) &:= \left(\frac{2 + \left(\frac{2}{3}\right)^w}{1 + \left(\frac{2}{3}\right)^w}\right)^p, \text{ and} & \delta(w) &:= \left(2 + \left(\frac{2}{3}\right)^w\right) \beta(w). \end{aligned}$$

It follows immediately from the definition of α and γ that

$$\gamma(w, p) = \left(2 + \left(\frac{2}{3}\right)^w\right) \alpha(w, p) \quad \text{and} \quad \gamma(w, p-1) = \left(1 + \left(\frac{2}{3}\right)^w\right) \alpha(w, p). \quad (2.5)$$

For integers $w \geq 0$, observe that for all sufficiently large integers p , $\alpha(w, p) \geq \frac{1}{2}(\frac{3}{2})^{p-1} \geq \beta(w)$. Define p_w to be the smallest integer such that $\alpha(w, p_w + 1) \geq \beta(w)$. Since $\alpha(w, p)$ is an increasing function of p , if $p \geq p_w + 1$ then $\alpha(w, p) \geq \beta(w)$, and if $p \leq p_w$ then $\alpha(w, p) < \beta(w)$.

Fact 2.14. For every integer $w \geq 0$, $\log_2(\frac{3}{2}) \cdot w + \log_2(3) \leq p_w \leq w + \log_{3/2}(9)$.

Proof. Observe that $p_w = \lceil x \rceil$ where $\alpha(w, x + 1) = \beta(w)$. Rearranging $\alpha(w, x + 1) = \beta(w)$ for x gives

$$x = \frac{\log \left(\left(1 + \left(\frac{2}{3}\right)^w\right) \cdot 3\left(\frac{3}{2}\right)^w \right)}{\log \left(\frac{2 + \left(\frac{2}{3}\right)^w}{1 + \left(\frac{2}{3}\right)^w} \right)}.$$

Let $a = \log \left(\left(1 + \left(\frac{2}{3}\right)^w\right) \cdot 3\left(\frac{3}{2}\right)^w \right)$ and $b = \log \left(\frac{2 + \left(\frac{2}{3}\right)^w}{1 + \left(\frac{2}{3}\right)^w} \right)$. Then, $a > \log(3(\frac{3}{2})^w)$ and $b \leq \log(2)$, implying $\log_2(\frac{3}{2}) \cdot w + \log_2(3) \leq \frac{a}{b} \leq p_w$. Furthermore,

$$a \leq \log \left(\left(1 + \left(\frac{2}{3}\right)^0\right) \cdot 3\left(\frac{3}{2}\right)^w \right) = \log \left(6\left(\frac{3}{2}\right)^w \right) \quad \text{and} \quad b \geq \log \left(\frac{2 + \left(\frac{2}{3}\right)^0}{1 + \left(\frac{2}{3}\right)^0} \right) = \log \left(\frac{3}{2} \right),$$

implying $\frac{a}{b} \leq w + \log_{3/2}(6)$. Therefore, $p_w \leq w + \log_{3/2}(6) + 1 = w + \log_{3/2}(9)$. $\square$

Fact 2.15. For all integers $w \geq 0$ and $p \geq p_w + 1$, $\gamma(w, p) \geq \alpha(w, p) + 3$.

Proof. By the definition of p_w , $\alpha(w, p) \geq \beta(w) \geq 3$ for all $w \geq 0$ and $p \geq p_w + 1$. By (2.5),

$$\gamma(w, p) - \alpha(w, p) = \left(2 + \left(\frac{2}{3}\right)^w\right)\alpha(w, p) - \alpha(w, p) = \left(1 + \left(\frac{2}{3}\right)^w\right)\alpha(w, p) \geq 3.$$

Therefore, $\gamma(w, p) \geq \alpha(w, p) + 3$. $\square$

Fact 2.16. For every integer $w \geq 0$, $\gamma(w, p_w) < \delta(w)$.

Proof. By the definition of p_w , $\alpha(w, p_w) < \beta(w)$. By (2.5),

$$\gamma(w, p_w) = \left(2 + \left(\frac{2}{3}\right)^w\right)\alpha(w, p_w) < \left(2 + \left(\frac{2}{3}\right)^w\right)\beta(w) = \delta(w). \quad \square$$

Fact 2.17. For every integer $w \geq 0$, $\lceil \delta(w) \rceil \leq 12(\frac{3}{2})^w$.

Proof. By the definition of $\delta(w)$ and $\beta(w)$,

$$\lceil \delta(w) \rceil = \left\lceil \left(2 + \left(\frac{2}{3}\right)^w\right)\beta(w) \right\rceil \leq \lceil 3\beta(w) \rceil \leq 4\beta(w) = 12\left(\frac{3}{2}\right)^w. \quad \square$$

Fact 2.18. For every integer $w \geq 0$, $2^{-p_w} \leq \frac{1}{3}(\frac{2}{3})^w$.

Proof. By Fact 2.14, $\log_2(\frac{3}{2}) \cdot w + \log_2(3) \leq p_w$. Therefore,

$$2^{-p_w} \leq 2^{-\log_2(\frac{3}{2}) \cdot w - \log_2(3)} = \frac{1}{3} \left(\frac{2}{3}\right)^w. \quad \square$$

2.3.3 The Universal Graph $G_{w,p}$. For integers $w \geq 0$ and $p \geq p_w$, define the graph $G_{w,p}$ recursively. Let $G_{w,p_w} := K_{\lceil \delta(w) \rceil}$ and for $p \geq p_w + 1$, let $G_{w,p}$ be the graph formed by adding all the possible edges between a complete graph K_{w+1} and two disjoint copies of $G_{w,p-1}$, as shown in Figure 2.10.

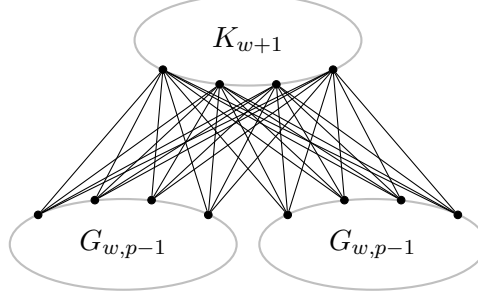


Figure 2.10: Constructing $G_{w,p}$.

Theorem 2.19. *For all integers $w \geq 0$ and $p \geq p_w$, $G_{w,p}$ contains every tree with at most $\gamma(w,p)$ edges.*

Proof. Proceed by induction on $p \geq p_w$ with w fixed. If $p = p_w$, then Fact 2.16 says $\gamma(w, p_w) < \delta(w)$. It follows that any tree on at most $\gamma(w, p_w)$ edges has at most $\lceil \delta(w) \rceil$ vertices, and is therefore contained in $G_{w,p_w} = K_{\lceil \delta(w) \rceil}$. Now assume $p \geq p_w + 1$ and the claim holds for all lower values of p . Let T be a tree on at most $\gamma(w, p)$ edges. Fact 2.15 says $\gamma(w, p) \geq \alpha(w, p) + 3$, so we can add vertices and edges to T if necessary (without creating cycles and maintaining connectivity) so that $\alpha(w, p) + 1 \leq |E(T)| \leq \gamma(w, p)$. By the definition of p_w , we have $\alpha(w, p) \geq \beta(w) = 3(\frac{3}{2})^w$. By Lemma 2.12, there exists a set $W \subseteq V(T)$ with $|W| \leq w + 1$ such that some set $\mathcal{C}$ of W -subtrees of T satisfies:

$$\alpha(w, p) < \sum_{T' \in \mathcal{C}} |E(T')| \leq \left(1 + \left(\frac{2}{3}\right)^w\right) \alpha(w, p) = \gamma(w, p-1),$$

where the last equality follows from (2.5). Let F_1 be the union of the W -subtrees in $\mathcal{C}$, and let F_2 be the union of the W -subtrees not in $\mathcal{C}$. Since the subtrees in $\mathcal{C}$ are edge-disjoint, the above inequality implies $\alpha(w, p) < |E(F_1)| \leq \gamma(w, p-1)$. Furthermore, since $\{E(F_1), E(F_2)\}$ is a partition of $E(T)$,

$$\begin{aligned} |E(F_2)| &= |E(T)| - |E(F_1)| \leq \gamma(w, p) - \alpha(w, p) = \left(2 + \left(\frac{2}{3}\right)^w\right) \alpha(w, p) - \alpha(w, p) \\ &= \gamma(w, p-1), \end{aligned}$$

where the last two equalities follow from (2.5). For each $i \in \{1, 2\}$, let T_i be the tree obtained from $F_i - W$ by adding edges if necessary to join up components. Observe that $|E(T_i)| \leq |E(F_i)| \leq \gamma(w, p-1)$. By the induction hypothesis, T_i is contained in $G_{w,p-1}$. Map T_i to the i -th copy of $G_{w,p-1}$ in $G_{w,p}$. Map W to the K_{w+1} in $G_{w,p}$. Since the K_{w+1} is complete to each copy of $G_{w,p-1}$ in $G_{w,p}$, this shows that T is contained in $G_{w,p}$. The claim holds by induction. $\square$

We now estimate $|V(G_{w,p})|$ and $|E(G_{w,p})|$. Let $m := p - p_w$.

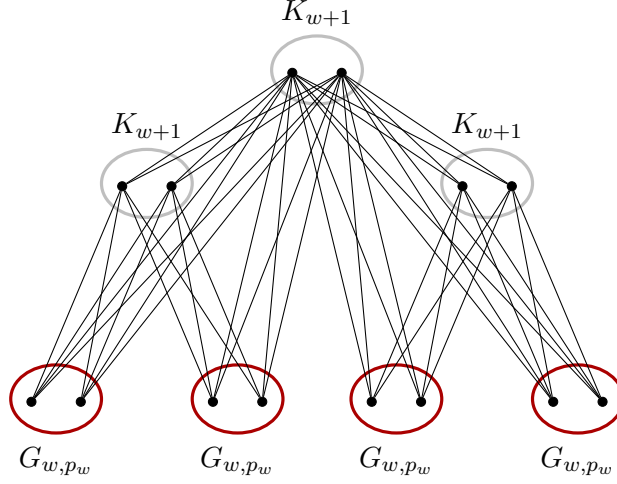


Figure 2.11: $G_{w,p}$ when $m = 2$.

Lemma 2.20. *For all sufficiently large integers $w \geq 0$, if $p \geq p_w$, then $|V(G_{w,p})| \leq 5 \cdot 2^p$.*

Proof. Since $G_{w,p}$ contains $2^m - 1$ copies of K_{w+1} and 2^m copies of G_{w,p_w} (see Figure 2.11),

$$|V(G_{w,p})| \leq (2^m - 1) \cdot |V(K_{w+1})| + 2^m \cdot |V(G_{w,p_w})| < 2^m(w + 1) + 2^m \lceil \delta(w) \rceil.$$

When w is sufficiently large, $w + 1 \leq (\frac{3}{2})^w$. By Fact 2.17, $\lceil \delta(w) \rceil \leq 12(\frac{3}{2})^w$. Therefore,

$$|V(G_{w,p})| \leq 2^m \cdot \left(\frac{3}{2}\right)^w + 2^m \cdot 12 \left(\frac{3}{2}\right)^w = 2^m \cdot 13 \left(\frac{3}{2}\right)^w.$$

It follows from Fact 2.18 that $2^m \leq 2^{p-1}(\frac{2}{3})^w$. Thus, $|V(G_{w,p})| \leq \frac{13}{3} \cdot 2^p \leq 5 \cdot 2^p$ as desired. $\square$

For the remainder of this section, let $w := \lceil \log_{3/2} p \rceil$ and consider p large. Note that $p \approx (\frac{3}{2})^w$ and $p_w \leq w + \log_{3/2}(9)$ by Fact 2.14, thus $p \geq p_w$ for all large enough p .

Lemma 2.21. $|E(G_{w,p})| \leq 5 \cdot 2^p p \log_{3/2} p + \mathcal{O}(2^p p)$.

Proof. Each of the $2^m - 1$ copies of K_{w+1} in $G_{w,p}$ contains $\binom{w+1}{2}$ edges. Each of the 2^m copies of G_{w,p_w} in $G_{w,p}$ contains $\binom{\lceil \delta(w) \rceil}{2} \leq 12^2 (\frac{3}{2})^{2w}$ edges by Fact 2.17. The contribution of the ‘downward’ edges from each K_{w+1} (see Figure 2.11) is the final term in:

$$|E(G_{w,p})| \leq (2^m - 1) \binom{w+1}{2} + 2^m \cdot 12^2 \left(\frac{3}{2}\right)^{2w} + \sum_{i=1}^m 2^i (w+1) |V(G_{w,p-i})|.$$

By Lemma 2.20, it follows that for each $i \in \{1, \dots, m\}$, $2^i |V(G_{w,p-i})| \leq 5 \cdot 2^p$. Therefore,

$$|E(G_{w,p})| < 2^m \binom{w+1}{2} + 2^m \cdot 12^2 \left(\frac{3}{2}\right)^{2w} + 5 \cdot 2^p m (w+1).$$

Since $m \leq p$ and $2^m \leq 2^{p\frac{1}{3}}(\frac{2}{3})^w$ by Fact 2.18,

$$|E(G_{w,p})| \leq 2^p \frac{1}{3} \left(\frac{2}{3}\right)^w \binom{w+1}{2} + 12^2 \cdot 2^p \frac{1}{3} \left(\frac{3}{2}\right)^w + 5 \cdot 2^p p(w+1) .$$

Since $w = \lceil \log_{3/2} p \rceil$, $5 \cdot 2^p p w$ is the dominating function in the above inequality. The lower order terms are $\mathcal{O}(2^p p)$. It follows that

$$|E(G_{w,p})| \leq 5 \cdot 2^p p w + \mathcal{O}(2^p p) \leq 5 \cdot 2^p p \log_{3/2} p + \mathcal{O}(2^p p)$$

as claimed. $\square$

For the remainder of this section, let $p := \lceil \log_2(2n) \rceil$ and consider n large.

Theorem 2.22. *The graph $G_{w,p}$ satisfies $|V(G_{w,p})| \leq 20n$ and $|E(G_{w,p})| \leq 20n(\log_2 n)(\log_{3/2} \log_2 n) + \mathcal{O}(n \log n)$. Moreover, $G_{w,p}$ contains every n -vertex tree.*

Proof. By Lemma 2.20, $|V(G_{w,p})| \leq 5 \cdot 2^{\lceil \log_2(2n) \rceil} \leq 20n$. By Lemma 2.21,

$$\begin{aligned} |E(G_{w,p})| &\leq 5 \cdot 2^{\lceil \log_2(2n) \rceil} \cdot \lceil \log_2(2n) \rceil \cdot \log_{3/2} \lceil \log_2(2n) \rceil + \mathcal{O}(2^{\lceil \log_2(2n) \rceil} \cdot \lceil \log_2(2n) \rceil) \\ &\leq 20n(\log_2 n)(\log_{3/2} \log_2 n) + \mathcal{O}(n \log n) . \end{aligned}$$

We now show $\gamma(w, p) \geq n$. We have

$$2 + \left(\frac{2}{3}\right)^w > 2 + \left(\frac{2}{3}\right)^w - \left(\frac{2}{3}\right)^{2w} = \left(2 - \left(\frac{2}{3}\right)^w\right) \left(1 + \left(\frac{2}{3}\right)^w\right) ,$$

implying $\frac{2 + (\frac{2}{3})^w}{1 + (\frac{2}{3})^w} \geq 2 - (\frac{2}{3})^w$. Furthermore,

$$\left(1 - \frac{1}{2} \left(\frac{2}{3}\right)^w\right)^p \geq 1 - \frac{p}{2} \left(\frac{2}{3}\right)^w \geq \frac{1}{2} ,$$

where the first inequality follows from Bernoulli's inequality, and the second inequality follows since $w = \lceil \log_{3/2} p \rceil$. Multiplying both extremes of the above inequality by 2^p implies $(2 - (\frac{2}{3})^w)^p \geq 2^{p-1}$. Hence,

$$\gamma(w, p) = \left(\frac{2 + (\frac{2}{3})^w}{1 + (\frac{2}{3})^w}\right)^p \geq \left(2 - \left(\frac{2}{3}\right)^w\right)^p \geq 2^{p-1} \geq n .$$

Therefore, Theorem 2.19 implies that $G_{w,p}$ contains every n -vertex tree. $\square$

Hence, Theorem 2.22 immediately implies the desired bound for $s(n)$.

Corollary 2.23. $s(n) \leq \mathcal{O}(n(\log n)(\log \log n))$.

Chapter 3

Universal Graphs for Tree-Like Graphs

Recall that $s_k(n)$ is the minimum number of edges in a graph that is universal for n -vertex treewidth- k graphs. The central problem of interest in my Honours project was to understand the asymptotic behaviour of $s_k(n)$. For this reason, the first goal of this chapter is to formally introduce treewidth and tree-decompositions. In Section 3.1, we present proofs of some of the elementary properties of treewidth, and give intuition as to how treewidth captures ‘tree-likeness’ of graphs. We then use these properties of treewidth to explain how to apply the methods involving perfectly balanced separators (shown in Section 2.1) to obtain the initial bound $s_k(n) \leq \mathcal{O}(kn \log^2 n)$.

The second goal of this chapter is to present original proofs of the new best known bounds for $s_k(n)$. In Section 3.2, we prove the following lower and upper bounds on $s_k(n)$:

$$\Omega(kn \log n) \leq s_k(n) \leq \mathcal{O}(kn(\log n)(\log \log n)) .$$

The content of Sections 2.2, 2.3 and 3.2 were assembled into an arXiv submission. This led to many interesting/fruitful developments over the course of this project (see the ‘Postscript’ at the end of this chapter).

Section 3.1 formally introduces treewidth. But, before we delve into the technical details, we first motivate the discussion of treewidth. Informally, the treewidth of a graph G measures how ‘tree-like’ G is. As mentioned before, the graphs with treewidth at most 1 are precisely the forests.

A cornerstone of structural graph theory is the study of graph minors and minor-closed classes. A graph class $\mathcal{G}$ is *minor-closed* if $\mathcal{G}$ is closed under taking minors. The concept of treewidth is central in the Graph Minors Project of Robertson and Seymour [66], in which the following monumental result (among others) is proved:

Theorem 3.1 (Graph Minor Theorem, Robertson and Seymour [66]). *In any infinite set of graphs, there exists two distinct graphs such that one is a minor of the other.*



Neil Robertson



Paul Seymour

Every minor-closed class $\mathcal{G}$ can be uniquely characterised by a set of minimally excluded minors. For example, recall that a graph is planar if it can be drawn in the plane without edge-crossings; the class of planar graphs is minor-closed and the set of minimally excluded minors is $\{K_5, K_{3,3}\}$ (see Figure 1.6).

Theorem 3.2 (Wagner [80]). *A graph G is planar if and only if G is K_5 -minor-free and $K_{3,3}$ -minor-free.*

The set of minimally excluded minors of any minor-closed class $\mathcal{G}$ forms an antichain with respect to the minor relation. Thus, a consequence of the Graph Minor Theorem is that the set of minimally excluded minors of $\mathcal{G}$ is always finite (like in the case of planar graphs). Therefore, to check whether a graph G is contained in some minor-closed class $\mathcal{G}$, one only needs to go through a finite set of minimally excluded minors of $\mathcal{G}$ and verify that G contains none of them as a minor. Robertson and Seymour [66] showed that there exists an algorithm to test whether a fixed graph H is a minor of G that runs in $\mathcal{O}(|V(G)|^3)$ time, therefore implying:

Theorem 3.3 (Robertson and Seymour [66]). *For any minor-closed class $\mathcal{G}$, there exists an algorithm that can decide whether a given graph G is in $\mathcal{G}$ in $\mathcal{O}(|V(G)|^3)$ time.*

Note that Theorem 3.3 asserts the existence of an algorithm without describing it. This is because no one knows the set of minimally excluded minors for a general minor-closed class $\mathcal{G}$.

As an example, the class of knotless graphs is minor-closed. These are the graphs G that can be embedded into Euclidean 3-space such that no cycle of G forms a nontrivial knot. Theorem 3.3 implies that there exists an algorithm that can decide whether an n -vertex graph is knotless in $\mathcal{O}(n^3)$ time. Before Theorem 3.3 was proved, it was not even known if a decision algorithm for knotless graphs existed, let alone one that runs in polynomial time. We note that the time complexity in Theorem 3.3 was recently improved to $\mathcal{O}(|V(G)|^2)$ by Kawarabayashi et al. [57].

Returning to treewidth, for every integer $k \geq 0$, the set of graphs with treewidth at most k is a minor-closed class (we show this in Section 3.1). The graphs with treewidth at most 1 are the forests; the set of minimally excluded minors is $\{K_3\}$. For the graphs with treewidth at most 2, the set of minimally excluded minors is $\{K_4\}$. One would hope that the obvious pattern continues for graphs with treewidth at most 3 (and beyond). It does not.

Theorem 3.4 (Arnborg, Corneil, and Proskurowski [9]). *A graph G has treewidth at most 3 if and only if G is K_5 -minor-free, $K_{2,2,2}$ -minor-free, V_8 -minor-free and $C_5 \square K_2$ -minor-free.*

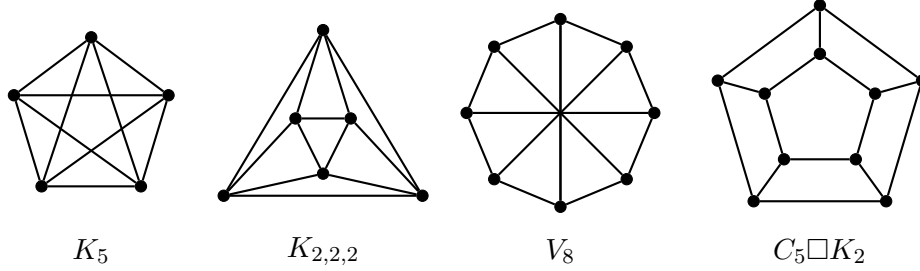


Figure 3.1: Minimally excluded minors for graphs with treewidth at most 3.

Although the forbidden-minor characterisation of graphs with treewidth at most k only gets more and more complicated for large k , algorithmically, graphs of bounded treewidth are simple. To be precise, a class of graphs $\mathcal{G}$ (not necessarily minor-closed) is said to have *bounded treewidth* if $\text{tw}(G) \leq k$ for all graphs G in $\mathcal{G}$, for some absolute constant k . Then Courcelle's Theorem says that a large class of graph properties can be efficiently tested on bounded treewidth graphs.

Theorem 3.5 (Courcelle [26]). *For every graph property $\mathcal{P}$ that can be expressed in MSO_2 , for every integer $k \geq 1$, there is an algorithm that can decide whether a graph G with treewidth at most k has property $\mathcal{P}$ in $\mathcal{O}(|V(G)|)$ time.*

MSO_2 refers to monadic second-order logic; briefly, this gives a formal way to express graph properties, that allows for quantification over the vertex set and edge set of a graph [27]. Examples of graph properties that can be expressed in MSO_2 include 3-colourability¹ and Hamiltonicity² [31]. Note that both the problem of 3-colourability and Hamiltonicity of a graph are NP-complete.

Now that we have established the motivation for studying treewidth, we move to a formal introduction of treewidth and tree-decompositions.

3.1 Treewidth

Tree-decompositions and treewidth are defined similarly to path-decompositions and path-width in Chapter 2. A *tree-decomposition* of a graph G is a collection of subsets $(B_x : x \in V(T))$ of $V(G)$ (called *bags*) indexed by the vertices of a tree T , such that each of the following hold:

- (*vertex-property*) for all $v \in V(G)$, $T[\{x \in V(T) : v \in B_x\}]$ is a nonempty subtree of T , and

¹A graph G is *3-colourable* if the numbers in $\{1, 2, 3\}$ can be assigned to the vertices of G such that no two adjacent vertices receive the same number.

²A graph G is *Hamiltonian* if G contains a spanning cycle as a subgraph.

- (*edge-property*) for all $uv \in E(G)$, there exists $x \in V(T)$ such that $\{u, v\} \subseteq B_x$.

The *width* of the tree-decomposition $(B_x : x \in V(T))$ equals $\max_{x \in V(T)} |B_x| - 1$. The *treewidth* of G , denoted by $\text{tw}(G)$, is the minimum width of a tree-decomposition of G . Note that a path-decomposition of G is a tree-decomposition of G . This shows that treewidth is well-defined and for every graph G , $\text{tw}(G) \leq \text{pw}(G)$.

For a graph G and a tree-decomposition $(B_x : x \in V(T))$, the vertex-property says that each vertex of G is mapped to a subtree of T . This can be seen in Figure 3.2. Furthermore, each edge of G appears in some bag by the edge-property.

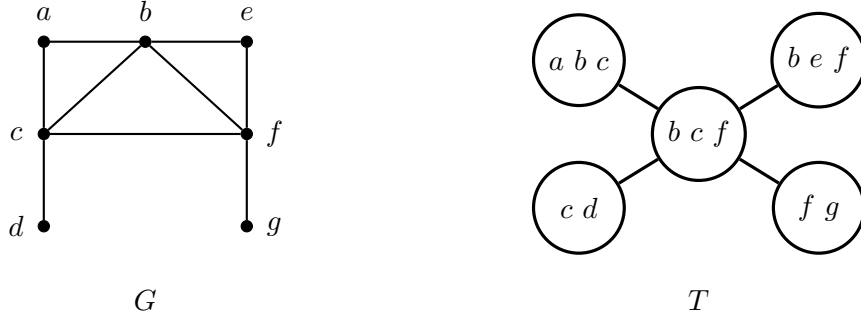


Figure 3.2: A graph G (left) and a tree-decomposition of G (right).

We now prove some of the basic properties of treewidth and tree-decompositions.

Just as vertices of a graph G are mapped to nonempty subtrees of the underlying tree of a tree-decomposition, the same is true for connected subgraphs of G .

Lemma 3.6. *For any tree-decomposition $(B_x : x \in V(T))$ of a graph G , if H is a connected subgraph of G , then $T[\{x \in V(T) : B_x \cap V(H) \neq \emptyset\}]$ is a nonempty subtree of T .*

Proof. Proceed by induction on n in the range $\{1, \dots, |V(G)|\}$ with the hypothesis that for every n -vertex connected subgraph H of G , $T[\{x \in V(T) : B_x \cap V(H) \neq \emptyset\}]$ is a nonempty subtree of T . The $n = 1$ case is immediate by the vertex-property of tree-decompositions. Assume now that the result holds for some $n \geq 1$. Suppose H is an $(n + 1)$ -vertex connected subgraph of G . Let $v \in V(H)$ be a leaf of a spanning tree of H . So $H' := H - v$ is an n -vertex connected subgraph of G . By the induction hypothesis, $A := \{x \in V(T) : B_x \cap V(H') \neq \emptyset\}$ and $B := \{x \in V(T) : v \in B_x\}$ induce subtrees of T . Observe that $A \cup B = \{x \in V(T) : B_x \cap V(H) \neq \emptyset\}$. Let u be a neighbour of v in H . By the edge-property in T , there exists a vertex $z \in V(T)$ such that $\{u, v\} \subseteq B_z$, implying $z \in A \cap B$. Since $T[A]$ and $T[B]$ are intersecting subtrees of T , $T[A \cup B]$ is a subtree of T . The result follows by induction. $\square$

The next lemma implies the following fundamental property of treewidth used in graph minor theory: for every positive integer k , the set of all graphs with treewidth at most k is a minor-closed class.

Lemma 3.7. *For every graph G and every minor H of G , $\text{tw}(H) \leq \text{tw}(G)$.*

Proof. Let $k := \text{tw}(G)$. We show that the treewidth does not increase after deleting a vertex, deleting an edge, and contracting an edge. Let $(B_x : x \in V(T))$ be a tree-decomposition of G with minimum width. Then for any vertex v of G , $(B_x \setminus \{v\} : x \in V(T))$ is a tree-decomposition of $G - v$ with width at most k . Furthermore, for any edge e of G , $(B_x : x \in V(T))$ is a tree-decomposition of $G - e$ with width at most k .

Let H be the graph obtained from G by contracting an edge uw of G to a new vertex z . For all $x \in V(T)$, let $B'_x := (B_x \setminus \{u, w\}) \cup \{z\}$ if u or w is in B_x , and let $B'_x := B_x$ otherwise. Note that $|B'_x| \leq |B_x|$ for all x . We show that $(B'_x : x \in V(T))$ is a tree-decomposition of H . The vertex z maps to $T[\{x \in V(T) : B_x \cap \{u, w\} \neq \emptyset\}]$, which is a nonempty subtree of T by Lemma 3.6. Every vertex y in $V(H) \setminus \{z\}$ maps to $T[\{x \in V(T) : y \in B_x\}]$. So the vertex-property holds. Furthermore, by the way the bags B'_x are defined, every edge of H is in some bag. This shows the edge-property. Hence, $(B'_x : x \in V(T))$ is a tree-decomposition of H with width at most k . $\square$

The study of relationships between treewidth and other graph parameters makes up a large body of the research on treewidth; such relationships are often given in the form of numerical bounds [51]. The most natural approach to upper bound the treewidth is to give a tree-decomposition with the desired width. On the other hand, it is not immediately clear how one should prove a lower bound for the treewidth. The simplest nontrivial lower bound for the treewidth is the minimum degree.

Lemma 3.8. *For any nonempty graph G , the minimum degree of a vertex of G is at most $\text{tw}(G)$.*

Proof. Let $(B_x : x \in V(T))$ be a tree-decomposition of G with $|V(T)|$ as small as possible subject to having minimum width. If $|V(T)| = 1$, then $\text{tw}(G) \geq |V(G)| - 1$, which is at least the minimum degree of a vertex in G . Now assume $|V(T)| \geq 2$. Let y be a leaf in T and z be its neighbour. If $B_y \subseteq B_z$, then $(B_x : x \in V(T - y))$ is a tree-decomposition of G with minimum width and $|V(T - y)| < |V(T)|$, contradicting the minimal choice of T . Thus, there exist a vertex $v \in B_y \setminus B_z$. By the edge-property of tree-decompositions, we deduce that $N_G(v) \subseteq B_y$. Therefore, $\deg_G(v) \leq |B_y \setminus \{v\}| \leq \text{tw}(G)$, implying the desired result. $\square$

For the purposes of this thesis, the bound in Lemma 3.8 is the only lower bound for treewidth that is needed. We now compute the treewidth of the complete graphs and the complete bipartite graphs. Note that in both of these computations, the treewidth equals the minimum degree.

Lemma 3.9. *For all integers $n \geq 1$, $\text{tw}(K_n) = \text{pw}(K_n) = n - 1$.*

Proof. The minimum degree of a vertex in K_n is $n - 1$, so Lemma 3.8 implies that $\text{tw}(K_n) \geq n - 1$. The path-decomposition of K_n consisting of a single bag $V(K_n)$ has width $n - 1$. Therefore, $\text{tw}(K_n) \leq \text{pw}(K_n) \leq |V(K_n)| - 1 = n - 1$. $\square$

Lemma 3.10. *For all integers $n, m \geq 1$, $\text{tw}(K_{n,m}) = \text{pw}(K_{n,m}) = \min\{n, m\}$.*

Proof. Assume $n \leq m$ without loss of generality. The minimum degree of a vertex in a complete bipartite graph $K_{n,m}$ is n . By Lemma 3.8, $\text{tw}(K_{n,m}) \geq n$. We now describe a path-decomposition of $K_{n,m}$ with width n . Label the vertices of $K_{n,m}$ by $a_1, \dots, a_n, b_1, \dots, b_m$ such that $\{\{a_1, \dots, a_n\}, \{b_1, \dots, b_m\}\}$ is the bipartition of $K_{n,m}$. For all $i \in \{1, \dots, m\}$, let $B_i := \{a_1, \dots, a_n\} \cup \{b_i\}$. Then a_i lies in every B_j , and b_i lies only in B_i . Furthermore, every edge in $K_{n,m}$ is of the form $a_i b_j$. It follows that $(B_1, \dots, B_m)$ is a path-decomposition of G . Since each bag contains exactly $n + 1$ vertices, the width of this path-decomposition is n , implying $\text{tw}(G) \leq \text{pw}(G) \leq n$. $\square$

The following section discusses how the ‘tree-likeness’ of a graph is captured by the treewidth.

3.1.1 Tree-Like Properties of Tree-Decompositions. We consider some of the ways the tree-like properties of any graph can be understood through its treewidth and its tree-decompositions. To start, any good measure of tree-likeness should be able to identify trees (or more generally, forests). This is indeed the case for treewidth.

Theorem 3.11. *A graph G has treewidth at most 1 if and only if G is a forest.*

Proof. ($\Rightarrow$) Proceed by contraposition. If G is not a forest, then G contains a cycle as a subgraph. Therefore, G contains a K_3 -minor. By Lemmas 3.7 and 3.9, $\text{tw}(G) \geq \text{tw}(K_3) = 2$. ($\Leftarrow$) By Lemma 3.7 it follows that the treewidth of a graph is at most the treewidth of its components, so we may assume that G is a tree. Root G at some vertex $r \in V(G)$ and let $B_r := \{r\}$. For each vertex $v \neq r$ with parent u , let $B_v := \{v, u\}$. It follows that $(B_x : x \in V(G))$ is a tree-decomposition of G with width at most 1, so $\text{tw}(G) \leq 1$. $\square$

Furthermore, for any graph G , many of the tree-like properties of the underlying tree in a tree-decomposition of G can be meaningfully transferred to G . Recall the following elementary tree facts.

Fact 3.12. *Each of the following hold:*

- (1) *a tree on at least two vertices contains a leaf,*
- (2) *every non-leaf vertex in a tree is a cut-vertex,*
- (3) *every edge in a tree is a bridge, and*
- (4) *a nonempty tree T has exactly $|V(T)| - 1$ edges.*

Lemma 3.8 guarantees that every nonempty treewidth- k graph has a vertex of degree at most k . This essentially extends (1) to treewidth- k graphs (although, the $k \leq 1$ case is strictly weaker than (1)).

A bag of a tree-decomposition of a graph G is a separator in G ; this extends (2).

Theorem 3.13. *For any tree-decomposition $(B_x : x \in V(T))$ of a graph G , for any vertex z of T , for any two distinct components T_1, T_2 of $T - z$, B_z is an $A_1 A_2$ -separator in G , where $A_i = \bigcup_{x \in V(T_i)} B_x$ for each $i \in \{1, 2\}$.*

Proof. Let P be an A_1A_2 -path in G . By Lemma 3.6, $T' := T[\{x : V(P) \cap B_x \neq \emptyset\}]$ is a nonempty subtree of T . Since T' contains a $V(T_1)V(T_2)$ -path, and T_1 and T_2 are components of $T - z$, $z \in V(T')$. By the definition of T' , $V(P) \cap B_z \neq \emptyset$, implying B_z is an A_1A_2 -separator in G . $\square$

For a tree-decomposition $(B_x : x \in V(T))$ of a graph G , an *adhesion* set is the intersection $B_x \cap B_y$ for any edge xy of T . Adhesion sets are separators in G ; this extends (3).

Theorem 3.14. *For any tree-decomposition $(B_x : x \in V(T))$ of a graph G , for any edge yz of T , for the two components T_1, T_2 of $T - yz$, $B_y \cap B_z$ is an A_1A_2 -separator in G , where $A_i = \bigcup_{x \in V(T_i)} B_x$ for each $i \in \{1, 2\}$.*

Proof. If there is no A_1A_2 -path in G , then we are done. So assume that there is an A_1A_2 -path P . By Lemma 3.6, $T' := T[\{x : V(P) \cap B_x \neq \emptyset\}]$ is a nonempty subtree of T . Since T' contains a $V(T_1)V(T_2)$ -path, and T_1 and T_2 are components of $T - yz$, we deduce that $yz \in E(T')$. By the definition of T' , $V(P) \cap B_y \neq \emptyset$ and $V(P) \cap B_z \neq \emptyset$. If $B_y \cap B_z = \emptyset$, then there is no A_1A_2 -path in G , contradicting our choice of P . Therefore, $B_y \cap B_z \neq \emptyset$, implying $V(P) \cap B_y \cap B_z \neq \emptyset$. Hence, $B_y \cap B_z$ is an A_1A_2 -separator in G . $\square$

A slight weakening of (4) says that a nonempty forest F has at most $|V(F)| - 1$ edges. This statement can be extended to graphs with treewidth at most k as follows:

Theorem 3.15. *For all integers $k \geq 1$ and $n \geq k + 1$, every n -vertex graph with treewidth at most k has at most $kn - \binom{k+1}{2}$ edges.*

Proof. Proceed by induction on n with k fixed. The $n = k + 1$ case holds since any graph on $k + 1$ vertices has at most $\binom{k+1}{2} = kn - \binom{k+1}{2}$ edges. Assume now that $n \geq k + 2$ and the result holds for $n - 1$. Let G be an n -vertex graph with treewidth at most k . By Lemma 3.8, G contains a vertex v with degree at most k . Then $H := G - v$ is an $(n - 1)$ -vertex graph with treewidth at most k . By the induction hypothesis, $|E(H)| \leq k(n - 1) - \binom{k+1}{2}$. Hence, $|E(G)| \leq |E(H)| + k \leq kn - \binom{k+1}{2}$. The result follows by induction. $\square$

We now consider a subtle property that goes beyond those in Fact 3.12, and essentially characterises treewidth. Recall that Lemma 2.8 shows that every tree T contains a so-called balanced separating vertex; a vertex v of T such that each component of $T - v$ has at most $\frac{1}{2}|V(T)|$ vertices. Every tree-decomposition $(B_x : x \in V(T))$ of a graph G has a bag B_x such that each component of $G - B_x$ has at most $\frac{1}{2}|V(G)|$ vertices. In fact, we can say even more.

Theorem 3.16 (Robertson and Seymour [68]). *For any tree-decomposition $(B_x : x \in V(T))$ of a graph G , for any set $S \subseteq V(G)$, there is a bag B_x such that each component of $G - B_x$ contains at most half of the vertices of S .*

Proof. For each edge $xy \in E(T)$, let $V_{x:y} := \bigcup_z B_z \setminus B_x$ where the union is taken over all the vertices z of the component of $T - x$ containing y . Note that $V_{x:y}$ is not necessarily

equal to $V_{y:x}$. Orient xy from x to y if $|V_{x:y} \cap S| \leq \frac{1}{2}|S|$. Every edge of T is oriented. Since T is a tree, T contains no cycles. Thus, there is a vertex $x \in V(T)$ such that each of its edges is oriented outward from x . For every component C of $G - B_x$, $V(C) \subseteq V_{x:y}$ for some edge $xy \in E(T)$. Thus, by the way the edges of T are oriented, $|V(C) \cap S| \leq \frac{1}{2}|S|$ as claimed. $\square$

Theorem 3.16 can be used to obtain lower bounds for the treewidth. For a graph G and a subset S of $V(G)$, let $f(S)$ be the minimum size of a subset X of $V(G)$ such that each component of $G - X$ contains at most half of the vertices of S . Note that f is well-defined since $G - V(G)$ has no vertices. Then Theorem 3.16 implies that for any tree-decomposition of G , for any subset S of $V(G)$, there exists a bag of size at least $f(S)$. Therefore, $\text{tw}(G) \geq \max_{S \subseteq V(G)} f(S) - 1$. Define the *balance number* of G by $\text{bal}(G) := \max_{S \subseteq V(G)} f(S)$. It turns out that, up to a constant factor, the treewidth of a graph equals its balance number.

Theorem 3.17. *For any graph G , $\text{tw}(G) \leq \mathcal{O}(\text{bal}(G))$. In particular, $\text{tw}(G) = \Theta(\text{bal}(G))$.*

Theorem 3.17 is implied by an induction proof of the following stronger statement in Lemma 3.18.

Lemma 3.18. *Let G be a graph with the property that for every set $S \subseteq V(G)$ with $|S| = 2k + 1$ there is a set $X \subseteq V(G)$ with $|X| \leq k$ such that each component of $G - X$ contains at most k vertices of S . Then for every set $R \subseteq V(G)$ with $|R| \leq 2k + 1$, there is a tree-decomposition $(B_x : x \in V(T))$ of G with width at most $3k$, such that R is contained in some bag of T .*

Proof of Theorem 3.17 assuming Lemma 3.18. Let $k := \text{bal}(G)$. By the definition of $\text{bal}(G)$, for every subset S of $V(G)$ with $|S| = 2k + 1$, there exists a subset X of $V(G)$ such that $|X| \leq k$ and each component of $G - X$ contains at most $\lfloor \frac{1}{2}|S| \rfloor = k$ vertices of S . The hypothesis of Lemma 3.18 is satisfied, hence, $\text{tw}(G) \leq 3k = 3 \text{bal}(G)$. $\square$

Proof of Lemma 3.18. Proceed by induction on $|V(G)|$ with k fixed. If $|V(G)| \leq 3k + 1$, the tree-decomposition consisting of a single bag $V(G)$ suffices. Assume now that $|V(G)| \geq 3k + 2$ and the result holds for all lower values. Note that the assumption about separators in G holds for every subgraph of G . Let $R \subseteq V(G)$ with $|R| \leq 2k + 1$. We may assume $|R| = 2k + 1$ (by adding more vertices to R if necessary). Then there exists a set $X \subseteq V(G)$ with $|X| \leq k$ such that each component of $G - X$ has at most k vertices of R . Let $C_1, \dots, C_p$ be the components of $G - X$. For each $i \in \{1, \dots, p\}$, let $R_i := (R \cap V(C_i)) \cup X$ and $G_i := G[V(C_i) \cup X]$. Then $|R_i| \leq |R \cap V(C_i)| + |X| \leq 2k$. Observe that $p \geq 2$: if not, then $R \subseteq R_1$, which contradicts that $|R| = 2k + 1$. Therefore, $|V(G_i)| < |V(G)|$. By induction, for each $i \in \{1, \dots, p\}$, there is a tree-decomposition $(B_x : x \in V(T_i))$ of G_i with width at most $3k$ such that for some $z_i \in V(T_i)$, $R_i \subseteq B_{z_i}$. Let T be the tree obtained from the disjoint union $\bigsqcup_{i=1}^p T_i$ by adding a new vertex z adjacent to the vertices $z_1, \dots, z_p$. Let $B_z := R \cup X$. It follows that $(B_x : x \in V(T))$ is a tree-decomposition of G . Since

$|B_z| \leq |R| + |X| \leq 3k + 1$, the width of this tree-decomposition is at most $3k$. Furthermore, the bag B_z contains R . The result follows by induction. $\square$

Theorem 3.17 says that tree-decompositions and balanced separators are intrinsically linked concepts. We note that the result of Lemma 3.18 was recently qualitatively strengthened.

Theorem 3.19 (Dvořák and Norin [37]). *Let G be a graph with the property that for subgraph H of G there is a set $X \subseteq V(H)$ with $|X| \leq k$ such that each component of $H - X$ has at most $\frac{1}{2}|V(H)|$ vertices. Then $\text{tw}(G) \leq 15k$.*

For surveys on treewidth, see [15, 51, 65].

3.1.2 Perfectly Balanced Separators in Treewidth- k Graphs. In this section, we apply the universal graph construction involving perfectly balanced separators from Chapter 2 to the class of graphs with treewidth at most k . Recall that for any subgraph-closed class $\mathcal{G}$, a uniform bound $f(n)$ on the pathwidth of the n -vertex graphs in $\mathcal{G}$ implies the existence of an n -vertex graph with $\mathcal{O}(nf(n) \log n)$ edges that is universal for the n -vertex graphs in $\mathcal{G}$ (see Corollary 2.6).

Let $\mathcal{G}_k$ be the set of all graphs with treewidth at most k . By Lemma 3.7, $\mathcal{G}_k$ is minor-closed, so it is subgraph-closed. We apply Corollary 2.6 to the class $\mathcal{G}_k$ by showing that every n -vertex graph in $\mathcal{G}_k$ has pathwidth at most $(k + 1) \log_{3/2} n$, thus implying $s_k(n) \leq \mathcal{O}(kn \log^2 n)$. We require the notion of normal tree-decompositions.

A tree-decomposition with width k is *normal* if every bag has size $k + 1$, and the intersection of any two adjacent bags has size k . One can always take a normal tree-decomposition of a graph [51]; the proof of this result is the exact same as the analogous result for normal path-decompositions (see Lemma 2.4).

Lemma 3.20. *If G has a tree-decomposition with width k , then G has a normal tree-decomposition with width k .*

To show the $\mathcal{O}(kn \log^2 n)$ bound for $s_k(n)$, we only require the following property of normal tree-decompositions. That being said, normal tree-decompositions are a key tool used later in Section 3.2.

Lemma 3.21. *If $(B_x : x \in V(T))$ is a normal tree-decomposition of a graph G , then $|V(T)| \leq |V(G)|$.*

Proof. Root T at some vertex $r \in V(T)$. By normality, for each vertex $x \in V(T) \setminus \{r\}$, B_x contains a vertex that is not in the bag of the parent of x (and hence not in B_r by the vertex-property), call it $f(x)$. Note that $B_r \neq \emptyset$, so we can pick a vertex $f(r)$ in B_r . By the vertex-property in T , it follows that f is an injection of $V(T)$ into $V(G)$, implying $|V(T)| \leq |V(G)|$. $\square$

By applying Theorem 2.10 to the underlying tree of a normal tree-decomposition, we obtain the desired bound on the pathwidth of an n -vertex treewidth- k graph.

Lemma 3.22 (Scheffler [72]). *For every n -vertex graph G , $\text{pw}(G) \leq (\text{tw}(G) + 1) \log_{3/2} n$.*

Proof. By Lemma 3.21, G has a tree-decomposition $(B_x : x \in V(T))$ with at most n bags. By Theorem 2.10, T has a path-decomposition $(C_1, \dots, C_t)$ with width at most $\log_{3/2} n$. For each $i \in \{1, \dots, t\}$, let $D_i := \bigcup_{x \in C_i} B_x$. Then it follows that $(D_1, \dots, D_t)$ is a path-decomposition of G with width at most $(\text{tw}(G) + 1) \log_{3/2} n$. $\square$

Hence, applying Corollary 2.6 with $f(n) = (k + 1) \log_{3/2} n$ and $\mathcal{G} = \mathcal{G}_k$ implies:

Theorem 3.23. $s_k(n) \leq \mathcal{O}(kn \log^2 n)$ when $n \gg k$.

In the next section, we prove the lower bound $s_k(n) \geq \Omega(kn \log n)$; this is only a $\log n$ factor away from matching the upper bound in Theorem 3.23. Then, by extending the methods shown in Section 2.3, we prove the upper bound: $s_k(n) \leq \mathcal{O}(kn(\log n)(\log \log n))$.

3.2 New Bounds for $s_k(n)$

The previously best known upper bound for $s_k(n)$ is $\mathcal{O}(kn \log^2 n)$ as explained by Joret [53]. This section presents original proofs of the new bounds:

$$\Omega(kn \log n) \leq s_k(n) \leq \mathcal{O}(kn(\log n)(\log \log n)) .$$

This is an original result and these bounds for $s_k(n)$ were the best known at the time they were proved (see the ‘Postscript’ for details on the current best known bounds for $s_k(n)$). The proof of the upper bound is similar to the proof for trees. In particular, the proof for trees is in terms of the number of edges, whereas the proof for treewidth- k graphs solely uses the number of vertices. In the proof for trees, the separator in the base case has one vertex, whereas in the $k = 1$ case of the treewidth- k proof, the analogous separator has two vertices.

3.2.1 Lower Bound. For the remainder of this chapter, we say that a graph U ‘contains’ a graph G if G is isomorphic to a subgraph of U . We first show the lower bound.

Theorem 3.24. $s_k(n) \geq \Omega(kn \log n)$ when $n \gg k$.

Proof. Let $t(n) := \sqrt{\frac{n}{k} + \frac{1}{4k^2}} - \frac{1}{2k}$. Assume n is sufficiently large (relative to k) so that $2 \leq \frac{1}{2}\sqrt{\frac{n}{k}} + 1 \leq t(n) \leq \frac{n}{k+1}$. The function $t(n)$ is chosen so $\frac{n}{j} - kj - 1$ is a nonnegative decreasing function of j in the interval $[1, t(n)]$.

Let U be a graph that contains every graph with n vertices and treewidth at most k . We show that $|E(U)| \geq \Omega(kn \log n)$.

For each $j \in \{1, \dots, \lfloor t(n) \rfloor\}$, let S_j be the complete bipartite graph $K_{k, \lfloor \frac{n}{j} \rfloor - k}$; the vertices in the part of size k are called *inner* vertices. We write $j \cdot S_j$ for the disjoint union of j copies of S_j . Since $|V(j \cdot S_j)| \leq n$ and $\text{tw}(j \cdot S_j) \leq k$, U contains $j \cdot S_j$. Let H_j be a copy of $j \cdot S_j$ in U . Note that H_j has kj inner vertices.

Mark some of the edges of U as follows. Let X_1 be the set of inner vertices of H_1 . Mark every edge of H_1 with an end vertex in X_1 ; there are $k(\lfloor \frac{n}{1} \rfloor - k)$ such edges. Apply the following algorithm: for $j = 2, \dots, \lfloor t(n) \rfloor$, let X_j be a set of k inner vertices of H_j such that $(X_1 \cup \dots \cup X_{j-1}) \cap X_j = \emptyset$. Mark every edge of H_j with an end vertex in X_j .

At the end of step j , for each $x \in X_j$, the number of new marked edges in U incident to x is at least $\lfloor \frac{n}{j} \rfloor - k$ minus the number of vertices in $X_1 \cup \dots \cup X_{j-1}$ that are in the same component of H_j as x . Since $|X_1 \cup \dots \cup X_{j-1}| = k(j-1)$ and $|X_j| = k$, it follows that at the end of step j , the number of new marked edges in U is at least $k(\lfloor \frac{n}{j} \rfloor - k) - k^2(j-1) \geq k(\frac{n}{j} - kj - 1)$. Thus, at the end of the algorithm, the total number of marked edges in U is at least

$$\begin{aligned} \sum_{j=1}^{\lfloor t(n) \rfloor} k \left(\frac{n}{j} - kj - 1 \right) &\geq k \int_1^{\frac{1}{2}\sqrt{\frac{n}{k}}} \left(\frac{n}{j} - kj - 1 \right) dj = k \left[n \log j - \frac{k}{2} j^2 - j \right]_{j=1}^{\frac{1}{2}\sqrt{\frac{n}{k}}} \\ &\geq kn \log \left(\frac{1}{2} \sqrt{\frac{n}{k}} \right) - k \left[\frac{k}{2} j^2 + j \right]_{j=1}^{\frac{1}{2}\sqrt{\frac{n}{k}}} \\ &\geq \frac{1}{2} kn \log n - \mathcal{O}(kn \log k) - \mathcal{O}(kn) \\ &\geq \Omega(kn \log n). \quad \square \end{aligned}$$

3.2.2 Normal Tree-Decompositions. Recall that a tree-decomposition with width k is normal if every bag has size $k+1$, and the intersection of any two adjacent bags has size k . In this section we prove Lemma 3.25; a result relating to normal tree-decompositions which is used in Section 3.2.3. We first introduce some new notation.

Given a tree-decomposition $(B_x : x \in V(T))$ of a graph G and a vertex z of T , for any subtree T' of T define $G(T', z) := (\bigcup_{x \in V(T')} B_x) \setminus B_z$. Then for any real number $\alpha \geq 1$, T' is said to be (α, z) -light if $|G(T', z)| \leq \alpha$, and (α, z) -heavy if $|G(T', z)| > \alpha$.

Recall that a z -subtree of T is a maximal subtree of T in which z only appears as a leaf (see Section 2.3). Note that if $\mathcal{F}$ is the set of all z -subtrees of T , then $\{G(F, z) : F \in \mathcal{F}\}$ partitions $V(G) \setminus B_z$.

Lemma 3.25. *For every integer $k \geq 1$, for every real number $\alpha \geq 1$, for every graph G with $|V(G)| > \alpha + k + 1$ and treewidth k , for every normal tree-decomposition $(B_x : x \in V(T))$ of G with width k , there exists a vertex $z \in V(T)$ such that for some set $\mathcal{C}$ of z -subtrees of T ,*

$$\alpha < \sum_{T' \in \mathcal{C}} |G(T', z)| \leq 2\alpha.$$

Proof. Let y be a leaf in T . Since $|G(T, y)| = |V(G)| - (k+1) > \alpha$, T is (α, y) -heavy. We may assume T is $(2\alpha, y)$ -heavy, otherwise $\{T\}$ is a desired set of y -subtrees of T . Let z be the neighbour of y in T and let $v \in B_y \setminus B_z$. Note that v does not appear in a bag of $T - y$. Therefore,

$$\left| \left(\bigcup_{x \in V(T)} B_x \right) \setminus \{v\} \right| = |G(T, y)| + k > 2\alpha + k \geq \alpha + k + 1.$$

Suppose that every z -subtree of $T - y$ is (α, z) -light. Then let $\{T_1, \dots, T_j\}$ be a minimal set of z -subtrees of $T - y$ such that $|G(T_1, z)| + \dots + |G(T_j, z)| > \alpha$ (this is well-defined since the union of all the z -subtrees of $T - y$ is (α, z) -heavy). If $|G(T_1, z)| + \dots + |G(T_j, z)| > 2\alpha$, then the minimality of $\{T_1, \dots, T_j\}$ implies

$$|G(T_j, z)| = |G(T_1, z)| + \dots + |G(T_j, z)| - (|G(T_1, z)| + \dots + |G(T_{j-1}, z)|) > 2\alpha - \alpha = \alpha,$$

contradicting that T_j is (α, z) -light. Thus, $\{T_1, \dots, T_j\}$ is a desired set of z -subtrees of T . Now assume some z -subtree of $T - y$ is (α, z) -heavy. Let $P = (v_0, v_1, \dots, v_t)$ be a maximal path in T such that $v_0 = y$, $v_1 = z$ and for each $i \geq 1$ there exists an (α, v_i) -heavy v_i -subtree of $T - v_{i-1}$ that contains v_i . Let T' be an (α, v_t) -heavy v_t -subtree of $T - v_{t-1}$ that contains v_t . We may assume T' is $(2\alpha, v_t)$ -heavy, otherwise $\{T'\}$ is a desired set of v_t -subtrees of T . Let v_{t+1} be the neighbour of v_t in T' , and let $u \in B_{v_t} \setminus B_{v_{t+1}}$. Note that u does not appear in a bag of $T' - v_t$. Therefore,

$$\left| \left(\bigcup_{x \in V(T')} B_x \right) \setminus \{u\} \right| = |G(T', v_t)| + k > 2\alpha + k \geq \alpha + k + 1.$$

By the maximality of P , every v_{t+1} -subtree of $T' - v_t$ is (α, v_{t+1}) -light. By the same argument given above, a minimal set of v_{t+1} -subtrees of $T' - v_t$ whose union is (α, v_{t+1}) -heavy will suffice. This proves the lemma. $\square$

3.2.3 W -Components. In this section, we prove a result analogous to Lemma 2.12 but for graphs with bounded treewidth. To do this, we first introduce a modified notion of W -subtrees from Section 2.3.1 that generalises to arbitrary graphs. For a graph G and a set $W \subseteq V(G)$, a **W -component** of G is a connected component of $G - W$.

Lemma 3.26. *For all integers $k \geq 1$ and $w \geq 0$, for every real number $\alpha \geq (2k+4)\left(\frac{3}{2}\right)^w$, for every graph G with $|V(G)| > \alpha + k + 1$ and $\text{tw}(G) \leq k$, there exists $W \subseteq V(G)$ with $|W| \leq (k+1)(w+1)$ such that for some set $\mathcal{S}$ of W -components of G ,*

$$\alpha < \sum_{C \in \mathcal{S}} |V(C)| \leq \left(1 + \left(\frac{2}{3}\right)^w\right) \alpha.$$

The proof of this result requires the following lemma analogous to Lemma 2.13.

Lemma 3.27. *For every integer $k \geq 1$, for every real number $\alpha \geq 1$, for every graph G with $|V(G)| > \alpha + k + 1$ and $\text{tw}(G) \leq k$, there exists $W \subseteq V(G)$ with $|W| = k + 1$ such that for some set $\mathcal{S}$ of W -components of G ,*

$$\alpha < \sum_{C \in \mathcal{S}} |V(C)| \leq 2\alpha.$$

Proof. Let G' be the graph obtained from G by adding edges if necessary such that $\text{tw}(G') = k$. Let $(B_x : x \in V(T))$ be a normal tree-decomposition of G' with width k . By Lemma 3.25, there is a vertex $z \in V(T)$ such that for some set $\mathcal{C}$ of z -subtrees of T , $\alpha < \sum_{T' \in \mathcal{C}} |G(T', z)| \leq 2\alpha$. Let $W := B_z$ and let $\mathcal{S}$ be the set of components of $G[\bigcup_{T' \in \mathcal{C}} G(T', z)]$. The proof is completed by observing that $|W| = k + 1$ and $\alpha < \sum_{C \in \mathcal{S}} |V(C)| \leq 2\alpha$. $\square$

Proof of Lemma 3.26. Proceed by induction on w with k fixed. The $w = 0$ case is a direct application of Lemma 3.27. Assume the result holds for some $w \geq 0$. Let $\alpha \geq (2k + 4)(\frac{2}{3})^{w+1}$ and let G be a graph with $|V(G)| > \alpha + k + 1$ and $\text{tw}(G) \leq k$. Since $\alpha \geq (2k + 4)(\frac{2}{3})^w$, by the induction hypothesis there exists $W \subseteq V(G)$ with $|W| \leq (k + 1)(w + 1)$ such that for some set $\mathcal{S}(W)$ of W -components of G , $\alpha < \sum_{C \in \mathcal{S}(W)} |V(C)| \leq (1 + (\frac{2}{3})^w)\alpha$. We are done if $\sum_{C \in \mathcal{S}(W)} |V(C)| \leq (1 + (\frac{2}{3})^{w+1})\alpha$, so assume the contrary. Then

$$\left(1 + \left(\frac{2}{3}\right)^{w+1}\right)\alpha < \sum_{C \in \mathcal{S}(W)} |V(C)| \leq \left(1 + \left(\frac{2}{3}\right)^w\right)\alpha. \quad (3.1)$$

Define the graph $H := \bigcup_{C \in \mathcal{S}(W)} C$ and let $\beta := (\frac{2}{3})^{w+1}\frac{\alpha}{2} - k - 1 \geq 1$. Then $\text{tw}(H) \leq \text{tw}(G) \leq k$. Since the graphs in $\mathcal{S}(W)$ are vertex-disjoint,

$$|V(H)| = \sum_{C \in \mathcal{S}(W)} |V(C)| > \alpha + \left(\frac{2}{3}\right)^{w+1}\alpha = \alpha + 2(\beta + k + 1) > \beta + k + 1.$$

Since $\beta \geq 1$, Lemma 3.27 implies that there exists $X \subseteq V(H)$ with $|X| = k + 1$ such that for some set $\mathcal{S}(X)$ of X -components of H ,

$$\left(\frac{2}{3}\right)^{w+1}\frac{\alpha}{2} - k - 1 = \beta < \sum_{C \in \mathcal{S}(X)} |V(C)| \leq 2\beta = \left(\frac{2}{3}\right)^{w+1}\alpha - 2k - 2. \quad (3.2)$$

Then (3.1) and (3.2) together imply

$$\begin{aligned} \alpha < \alpha + (k + 1) &= \left(1 + \left(\frac{2}{3}\right)^{w+1}\right)\alpha - (k + 1) - \left(\left(\frac{2}{3}\right)^{w+1}\alpha - 2k - 2\right) \\ &< \sum_{C \in \mathcal{S}(W)} |V(C)| - (k + 1) - \sum_{C \in \mathcal{S}(X)} |V(C)| \\ &\leq \left(1 + \left(\frac{2}{3}\right)^w\right)\alpha - (k + 1) - \left(\left(\frac{2}{3}\right)^{w+1}\frac{\alpha}{2} - k - 1\right) \\ &= \alpha + \left(\frac{2}{3}\right)^w\alpha - \left(\frac{2}{3}\right)^{w+1}\frac{\alpha}{2} \\ &= \left(1 + \left(\frac{2}{3}\right)^{w+1}\right)\alpha. \end{aligned} \quad (3.3)$$

Let $\overline{\mathcal{S}(X)}$ be the set of all X -components of H not in $\mathcal{S}(X)$. Since $\{V(C) : C \in \mathcal{S}(W)\}$

and $\{X\} \cup \{V(C) : C \in \mathcal{S}(X)\} \cup \{V(C) : C \in \overline{\mathcal{S}(X)}\}$ are both partitions of $V(H)$,

$$|V(H)| = \sum_{C \in \mathcal{S}(W)} |V(C)| = (k+1) + \sum_{C \in \mathcal{S}(X)} |V(C)| + \sum_{C \in \overline{\mathcal{S}(X)}} |V(C)|.$$

Therefore, (3.3) implies

$$\alpha < \sum_{C \in \overline{\mathcal{S}(X)}} |V(C)| \leq \left(1 + \left(\frac{2}{3}\right)^{w+1}\right) \alpha.$$

Thus, $\overline{\mathcal{S}(X)}$ is a desired set of $(W \cup X)$ -components of G . The result follows by induction. $\square$

3.2.4 More Useful Functions. This section proves a number of elementary facts that are direct extensions of the analogous observations in Section 2.3.2. For integers $w, p \geq 0$ define $\alpha(w, p)$ and $\gamma(w, p)$ as in Section 2.3.2. For integers $w, p \geq 0$ and $k \geq 1$, define

$$\beta_k(w) := (2k+4) \left(\frac{3}{2}\right)^w \quad \text{and} \quad \delta_k(w) := \left(2 + \left(\frac{2}{3}\right)^w\right) \beta_k(w).$$

While the $k = 1$ case corresponds to trees, $\beta_1(w)$ and $\delta_1(w)$ are slightly larger than $\beta(w)$ and $\delta(w)$ respectively. This is because of the $(k+1)(w+1)$ term in Lemma 3.26.

For integers $w \geq 0$ and $k \geq 1$, observe that for all sufficiently large integers p , $\alpha(w, p) \geq \frac{1}{2} \left(\frac{3}{2}\right)^{p-1} \geq \beta_k(w)$. Define $p_{w,k}$ to be the smallest integer such that $\alpha(w, p_{w,k} + 1) \geq \beta_k(w)$. Since $\alpha(w, p)$ is an increasing function of p , if $p \geq p_{w,k} + 1$ then $\alpha(w, p) \geq \beta_k(w)$, and if $p \leq p_{w,k}$ then $\alpha(w, p) < \beta_k(w)$.

For each of the following facts, assume k is a fixed positive integer.

Fact 3.28. *For every integer $w \geq 0$, $\log_2(\frac{3}{2}) \cdot w + \log_2(2k+4) \leq p_{w,k} \leq w + \log_{3/2}(6k+12)$.*

Proof. Observe that $p_{w,k} = \lceil x \rceil$ where $\alpha(w, x+1) = \beta_k(w)$. Rearranging $\alpha(w, x+1) = \beta_k(w)$ for x gives

$$x = \frac{\log \left(\left(1 + \left(\frac{2}{3}\right)^w\right) \cdot (2k+4) \left(\frac{3}{2}\right)^w \right)}{\log \left(\frac{2 + \left(\frac{2}{3}\right)^w}{1 + \left(\frac{2}{3}\right)^w} \right)}.$$

Let $a = \log \left(\left(1 + \left(\frac{2}{3}\right)^w\right) \cdot (2k+4) \left(\frac{3}{2}\right)^w \right)$ and $b = \log \left(\frac{2 + \left(\frac{2}{3}\right)^w}{1 + \left(\frac{2}{3}\right)^w} \right)$. Then, $a > \log \left((2k+4) \left(\frac{3}{2}\right)^w \right)$ and $b \leq \log(2)$, implying $\log_2(\frac{3}{2}) \cdot w + \log_2(2k+4) \leq \frac{a}{b} \leq p_{w,k}$. Furthermore,

$$\begin{aligned} a &\leq \log \left(\left(1 + \left(\frac{2}{3}\right)^0\right) \cdot (2k+4) \left(\frac{3}{2}\right)^w \right) = \log \left((4k+8) \left(\frac{3}{2}\right)^w \right) \quad \text{and} \\ b &\geq \log \left(\frac{2 + \left(\frac{2}{3}\right)^0}{1 + \left(\frac{2}{3}\right)^0} \right) = \log \left(\frac{3}{2} \right), \end{aligned}$$

implying $\frac{a}{b} \leq w + \log_{3/2}(4k+8)$. Thus, $p_{w,k} \leq w + \log_{3/2}(4k+8) + 1 = w + \log_{3/2}(6k+12)$. $\square$

Fact 3.29. For all integers $w \geq 0$ and $p \geq p_{w,k} + 1$, $\gamma(w, p) \geq \alpha(w, p) + k + 4$.

Proof. By the definition of $p_{w,k}$, $\alpha(w, p) \geq \beta_k(w)$ for all $w \geq 0$ and $p \geq p_{w,k} + 1$. By (2.5),

$$\begin{aligned} \gamma(w, p) - \alpha(w, p) &= \left(2 + \left(\frac{2}{3}\right)^w\right)\alpha(w, p) - \alpha(w, p) = \left(1 + \left(\frac{2}{3}\right)^w\right)\alpha(w, p) \\ &\geq \left(1 + \left(\frac{2}{3}\right)^w\right)\beta_k(w) \\ &> \beta_k(w). \end{aligned}$$

Since $\beta_k(w) \geq 2k + 4$, $\gamma(w, p) - \alpha(w, p) \geq k + 4$. □

Fact 3.30. For every integer $w \geq 0$, $\gamma(w, p_{w,k}) < \delta_k(w)$.

Proof. By the definition of $p_{w,k}$, $\alpha(w, p_{w,k}) < \beta_k(w)$. By (2.5),

$$\gamma(w, p_{w,k}) = \left(2 + \left(\frac{2}{3}\right)^w\right)\alpha(w, p_{w,k}) < \left(2 + \left(\frac{2}{3}\right)^w\right)\beta_k(w) = \delta_k(w). \quad \square$$

Fact 3.31. For every integer $w \geq 0$, $\lceil \delta_k(w) \rceil \leq (8k + 16)\left(\frac{3}{2}\right)^w$.

Proof. By the definition of $\delta_k(w)$ and $\beta_k(w)$,

$$\lceil \delta_k(w) \rceil = \left\lceil \left(2 + \left(\frac{2}{3}\right)^w\right)\beta_k(w) \right\rceil \leq \lceil 3\beta_k(w) \rceil \leq 4\beta_k(w) = (8k + 16)\left(\frac{3}{2}\right)^w. \quad \square$$

Fact 3.32. For every integer $w \geq 0$, $2^{-p_{w,k}} \leq \frac{1}{2k+4}\left(\frac{2}{3}\right)^w$.

Proof. By Fact 3.28, $\log_2\left(\frac{3}{2}\right) \cdot w + \log_2(2k + 4) \leq p_{w,k}$. Therefore,

$$2^{-p_{w,k}} \leq 2^{-\log_2\left(\frac{3}{2}\right) \cdot w - \log_2(2k+4)} = \frac{1}{2k+4}\left(\frac{2}{3}\right)^w. \quad \square$$

3.2.5 The Universal Graph $G_{w,p}^k$. For integers $w \geq 0$, $k \geq 1$ and $p \geq p_{w,k}$, define the graph $G_{w,p}^k$ recursively. Let $G_{w,p_{w,k}}^k := K_{\lceil \delta_k(w) \rceil}$ and for $p \geq p_{w,k} + 1$, let $G_{w,p}^k$ be the graph formed by adding all the possible edges between a complete graph $K_{(k+1)(w+1)}$ and two disjoint copies of $G_{w,p-1}^k$, as shown in Figure 3.3.

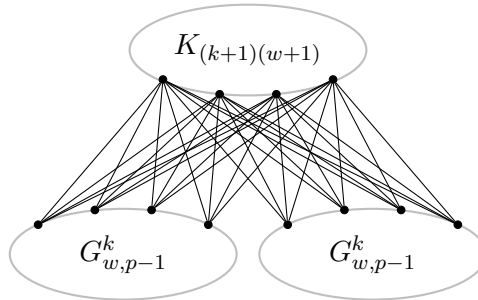


Figure 3.3: Constructing $G_{w,p}^k$.

Theorem 3.33. *For all integers $w \geq 0$, $k \geq 1$ and $p \geq p_{w,k}$, $G_{w,p}^k$ contains every graph with at most $\gamma(w, p)$ vertices and treewidth at most k .*

Proof. Proceed by induction on $p \geq p_{w,k}$ with w and k fixed. If $p = p_{w,k}$, then Fact 3.30 says $\gamma(w, p_{w,k}) < \delta_k(w)$. Every graph on at most $\gamma(w, p_{w,k})$ vertices is contained in $G_{w,p_{w,k}}^k = K_{\lceil \delta_k(w) \rceil}$, so the base case holds. Now assume $p \geq p_{w,k} + 1$ and the claim holds for all lower values of p . Let G be a graph on at most $\gamma(w, p)$ vertices with $\text{tw}(G) \leq k$. Fact 3.29 says $\gamma(w, p) \geq \alpha(w, p) + k + 4$, so we can add vertices and edges to G if necessary (without increasing the treewidth) so that $\alpha(w, p) + k + 1 < |V(G)| \leq \gamma(w, p)$. By the definition of $p_{w,k}$, we have $\alpha(w, p) \geq \beta_k(w) = (2k + 4)(\frac{3}{2})^w$. By Lemma 3.26, there exists a set $W \subseteq V(G)$ with $|W| \leq (k + 1)(w + 1)$ such that some set $\mathcal{S}$ of W -components of G satisfies:

$$\alpha(w, p) < \sum_{C \in \mathcal{S}} |V(C)| \leq \left(1 + \left(\frac{2}{3}\right)^w\right) \alpha(w, p) = \gamma(w, p - 1),$$

where the last equality follows from (2.5). Let H_1 be the union of the W -components in $\mathcal{S}$, and let H_2 be the union of the W -components not in $\mathcal{S}$. Since the graphs in $\mathcal{S}$ are vertex-disjoint, the above inequality implies $\alpha(w, p) < |V(H_1)| \leq \gamma(w, p - 1)$. Furthermore, since $\{W, V(H_1), V(H_2)\}$ is a partition of $V(G)$,

$$\begin{aligned} |V(H_2)| &= |V(G)| - |W| - |V(H_1)| \leq |V(G)| - |V(H_1)| \leq \gamma(w, p) - \alpha(w, p) \\ &= \left(2 + \left(\frac{2}{3}\right)^w\right) \alpha(w, p) - \alpha(w, p) \\ &= \gamma(w, p - 1), \end{aligned}$$

where the last two equalities follow from (2.5). Note that for each $i \in \{1, 2\}$, $\text{tw}(H_i) \leq \text{tw}(G) \leq k$. By the induction hypothesis, H_i is contained in $G_{w,p-1}^k$. Map H_i to the i -th copy of $G_{w,p-1}^k$ in $G_{w,p}^k$. Map W to the $K_{(k+1)(w+1)}$ in $G_{w,p}^k$. Since the $K_{(k+1)(w+1)}$ is complete to each copy of $G_{w,p-1}^k$ in $G_{w,p}^k$, this shows that G is contained in $G_{w,p}^k$. The claim holds by induction. $\square$

We now estimate $|V(G_{w,p}^k)|$ and $|E(G_{w,p}^k)|$. Let $m := p - p_{w,k}$.

Lemma 3.34. *For every integer $k \geq 1$, for every integer w sufficiently large relative to k , if $p \geq p_{w,k}$, then $|V(G_{w,p}^k)| \leq 5 \cdot 2^p$.*

Proof. Since $G_{w,p}^k$ contains $2^m - 1$ copies of $K_{(k+1)(w+1)}$ and 2^m copies of $G_{w,p_{w,k}}^k$ (see Figure 3.4),

$$|V(G_{w,p}^k)| \leq (2^m - 1) \cdot |V(K_{(k+1)(w+1)})| + 2^m \cdot |V(G_{w,p_{w,k}}^k)| < 2^m(k + 1)(w + 1) + 2^m \lceil \delta_k(w) \rceil.$$

Fact 3.31 says that $\lceil \delta_k(w) \rceil \leq (8k + 16)(\frac{3}{2})^w$. Furthermore, when w is sufficiently large relative to k , $(k + 1)(w + 1) \leq (\frac{3}{2})^w$. Therefore,

$$|V(G_{w,p}^k)| \leq 2^m \cdot \left(\frac{3}{2}\right)^w + 2^m \cdot (8k+16) \left(\frac{3}{2}\right)^w = 2^m \cdot (8k+17) \left(\frac{3}{2}\right)^w.$$

By Fact 3.32, $2^m \leq 2^p \frac{1}{2k+4} \left(\frac{2}{3}\right)^w$. Thus, $|V(G_{w,p}^k)| \leq \frac{8k+17}{2k+4} \cdot 2^p < 5 \cdot 2^p$ as desired. $\square$

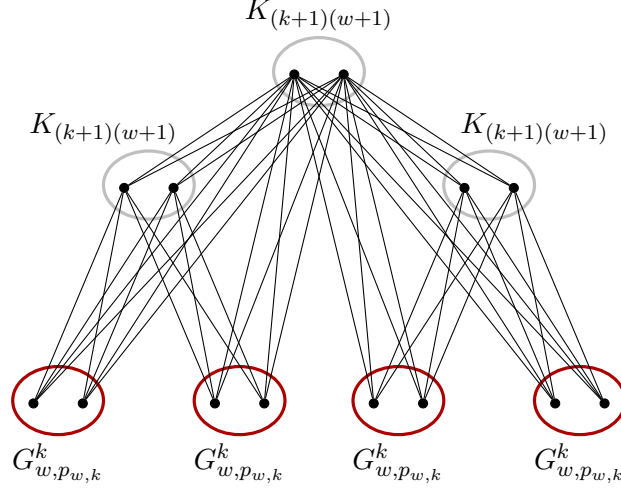


Figure 3.4: $G_{w,p}^k$ when $m = 2$.

For the remainder of this section, let $w := \lceil \log_{3/2} p \rceil$ and consider p large. Note that $p \approx \left(\frac{3}{2}\right)^w$ and $p_{w,k} \leq w + \log_{3/2}(6k+12)$ by Fact 3.28, so $p \geq p_{w,k}$ for all large enough p .

Lemma 3.35. $|E(G_{w,p}^k)| \leq 5k2^p p \log_{3/2} p + \mathcal{O}(k2^p p)$.

Proof. Each of the $2^m - 1$ copies of $K_{(k+1)(w+1)}$ in $G_{w,p}^k$ contains $\binom{(k+1)(w+1)}{2}$ edges. Each of the 2^m copies of $G_{w,p_w,k}^k$ in $G_{w,p}^k$ contains $\binom{\lceil \delta_k(w) \rceil}{2} \leq (8k+16)^2 \left(\frac{3}{2}\right)^{2w}$ edges by Fact 3.31. The contribution of the ‘downward’ edges from each $K_{(k+1)(w+1)}$ (see Figure 3.4) is the final term in:

$$\begin{aligned} |E(G_{w,p}^k)| &\leq (2^m - 1) \binom{(k+1)(w+1)}{2} + 2^m \cdot (8k+16)^2 \left(\frac{3}{2}\right)^{2w} \\ &\quad + \sum_{i=1}^m 2^i (k+1)(w+1) |V(G_{w,p-i}^k)|. \end{aligned}$$

By Lemma 3.34 it follows that, for each $i \in \{1, \dots, m\}$, $2^i |V(G_{w,p-i}^k)| \leq 5 \cdot 2^p$. Therefore,

$$|E(G_{w,p}^k)| < 2^m \binom{(k+1)(w+1)}{2} + 2^m \cdot (8k+16)^2 \left(\frac{3}{2}\right)^{2w} + 5 \cdot 2^p m (k+1)(w+1).$$

Since $m \leq p$ and $2^m \leq 2^p \frac{1}{2k+4} \left(\frac{2}{3}\right)^w$ by Fact 3.32,

$$|E(G_{w,p}^k)| \leq \frac{2^p}{2k+4} \left(\frac{2}{3}\right)^w \binom{(k+1)(w+1)}{2} + 2^p \frac{(8k+16)^2}{2k+4} \left(\frac{3}{2}\right)^w + 5 \cdot 2^p p (k+1)(w+1).$$

Since $w = \lceil \log_{3/2} p \rceil$, $5k2^p pw$ is the dominating function in the above inequality. The lower order terms are $\mathcal{O}(k2^p p)$. It follows that

$$|E(G_{w,p}^k)| \leq 5k2^p pw + \mathcal{O}(k2^p p) \leq 5k2^p p \log_{3/2} p + \mathcal{O}(k2^p p)$$

as claimed. $\square$

For the remainder of this section, let $p := \lceil \log_2(2n) \rceil$ and consider n large relative to k .

Theorem 3.36. *The graph $G_{w,p}^k$ satisfies $|V(G_{w,p}^k)| \leq 20n$ and*

$$|E(G_{w,p}^k)| \leq 20kn(\log_2 n)(\log_{3/2} \log_2 n) + \mathcal{O}(kn \log n).$$

Moreover, $G_{w,p}^k$ contains every graph with n vertices and treewidth at most k .

Proof. By Lemma 3.34, $|V(G_{w,p}^k)| \leq 5 \cdot 2^{\lceil \log_2(2n) \rceil} \leq 20n$. By Lemma 3.35,

$$\begin{aligned} |E(G_{w,p}^k)| &\leq 5k2^{\lceil \log_2(2n) \rceil} \cdot \lceil \log_2(2n) \rceil \cdot \log_{3/2} \lceil \log_2(2n) \rceil + \mathcal{O}(k2^{\lceil \log_2(2n) \rceil} \cdot \lceil \log_2(2n) \rceil) \\ &\leq 20kn(\log_2 n)(\log_{3/2} \log_2 n) + \mathcal{O}(kn \log n). \end{aligned}$$

Furthermore, $\gamma(w, p) \geq n$ by the same calculation in Theorem 2.22. Thus, Theorem 3.33 implies that $G_{w,p}^k$ contains every graph with n vertices and treewidth at most k . $\square$

Hence, Theorem 3.36 immediately implies the desired bound for $s_k(n)$.

Corollary 3.37. $s_k(n) \leq \mathcal{O}(kn(\log n)(\log \log n))$.

3.3 Postscript

This section gives a timeline of some key events over the course of my Honours project, with the goal of clarifying all of the contributions I made.

All of the preceding original results presented in this thesis had been found by July.

In early August, David Wood and I submitted the paper [55] to arXiv. This paper consists of the following:

- (1) an explanation of the mistake by Chung and Graham [24],
- (2) a proof of the bound $s(n) \leq \mathcal{O}(n(\log n)(\log \log n))$, and
- (3) proofs of the bounds $\Omega(kn \log n) \leq s_k(n) \leq \mathcal{O}(kn(\log n)(\log \log n))$, which were the best known at the time.

This is exactly the content of Sections 2.2, 2.3 and 3.2. When we submitted the paper to arXiv, we could not find any evidence of the mistake in [24] being known. Furthermore, the previously best known upper bound for $s(n)$ prior to the claimed $\mathcal{O}(n \log n)$ bound in [24] is $s(n) \leq \mathcal{O}(n(\log n)(\log \log n)^2)$ by Chung, Graham, and Pippenger [25]. So it seemed as though our arXiv paper proved the best known upper bound for $s(n)$.

Several people contacted us regarding our arXiv paper; each of them expressed doubts with Chung and Graham’s proof.

In September, Fabrizio Frati informed us that his paper ‘Universal geometric graphs’ with Hoffmann and Tóth [44] from 2023 also discusses the mistake in Chung and Graham’s proof and how to fix it to recover the $\mathcal{O}(n \log n)$ bound for $s(n)$.

Two of the people that responded to our arXiv paper, Jaehoon Kim and Minseo Kim, also submitted an arXiv paper [58] in which they explain Chung and Graham’s mistake, how to fix it, and improve the bound on $s(n)$ by a constant factor. We agreed to write a joint journal paper that merged our results. The joint paper (which is yet to be submitted to a journal) consists of the following:

- (1) an explanation of how to fix the mistake in [24] to show that there exists an n -vertex graph with $\frac{7}{2}n \log_2(n) + \mathcal{O}(n)$ edges that is universal for n -vertex trees,
- (2) a proof that there exists an n -vertex graph with $\frac{14}{5}n \log_2 n + \mathcal{O}(n)$ edges that is universal for the n -vertex trees (this is the first improvement to the bound in [24] in over four decades),
- (3) a proof that there exists a graph with $(3 + o(1))n$ vertices, $\mathcal{O}(kn \log n)$ edges and treewidth $\mathcal{O}(k \log n)$ that is universal for the n -vertex treewidth- k graphs, and
- (4) a lower bound construction showing that $s_k(n) \geq \Omega(kn \log n)$. In particular, $s_k(n) = \Theta(kn \log n)$.

Chapter 4

Universal Graphs for Planarity (and Beyond)

In Chapters 2 and 3, universality was discussed for trees and more generally, graphs of bounded treewidth. As explained in the ‘Postscript’ in Chapter 3, it is known how to construct a universal graph that is asymptotically optimal in the number of edges for bounded treewidth graphs. Naturally, the next problem to consider is universality for graph classes with unbounded treewidth.

Arguably, the class of planar graphs is the most renowned of them all, partly due to the famous 4-Colour Theorem of Appel and Haken [8] which states that the vertices of any planar graph can be coloured with 4 colours such that no two adjacent vertices receive the same colour.

One can show that the $n \times n$ grid (a planar graph) has treewidth n [51]. Therefore the class of planar graphs has unbounded treewidth.

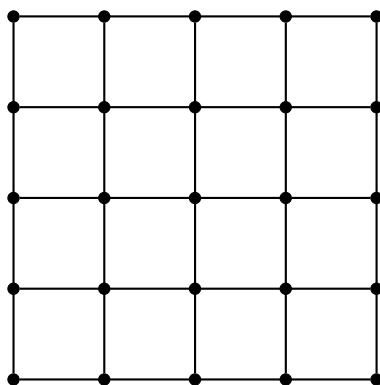


Figure 4.1: Drawing of the $n \times n$ grid ($n = 5$).

In particular, in the context of graph minor theory, the class of planar graphs is the simplest minor-closed class with unbounded treewidth. This statement is made precise by the following corollary of the Grid Minor Theorem by Robertson and Seymour [67] that we recall from Section 1.1.2: a minor-closed class $\mathcal{G}$ has bounded treewidth if and only if some planar graph is not in $\mathcal{G}$. Thus, the class of planar graphs is a natural starting point

when considering universality for classes with unbounded treewidth.

Recall that $r(n)$ is the minimum number of edges in a graph that is universal for n -vertex planar graphs. This chapter is mainly concerned with determining the asymptotic behaviour of $r(n)$.

In Section 4.1, we apply the methods of perfectly balanced separators (from Chapter 2) to show the initial bound: $r(n) \leq \mathcal{O}(n^{3/2})$. As an intermediate step, we prove a $\mathcal{O}(\sqrt{n})$ upper bound on the pathwidth of an n -vertex planar graph; the proof uses the Planar Separator Theorem of Lipton and Tarjan [59], which says that n -vertex planar graphs have balanced separators of size $\mathcal{O}(\sqrt{n})$. We present a proof of the Planar Separator Theorem in full.

The best known lower bound for $r(n)$ is $\Omega(n \log n)$, the same as for trees as explained in Chapter 2. For a long time, $\mathcal{O}(n^{3/2})$ was the best known upper bound for $r(n)$. In Section 4.1, we explain why a $\mathcal{O}(n^{3/2})$ upper bound for $r(n)$ is the best one can hope for when using arguments involving perfectly balanced separators. A different approach is needed to improve this bound.

The main goal of this chapter is to present a proof of the following breakthrough bound:

Theorem 4.1 (Esperet, Joret, and Morin [43]). $r(n) \leq n^{1+o(1)}$.

An essential tool in the proof of Theorem 4.1 is the seminal Planar Graph Product Structure Theorem, which says that every planar graph is isomorphic to a subgraph of the strong product of some bounded treewidth graph and a path. In Section 4.2, we introduce the notion of strong products of graphs and we present the original proof of the Planar Graph Product Structure Theorem in full.

The proof of Theorem 4.1 that we present in Section 4.3 gives near-linear bounds on the minimum number of edges in a universal graph for graph classes far beyond planar graphs. In fact, for any graph class $\mathcal{G}$ that admits a product structure theorem (like the planar graphs do), there is a universal graph with $n^{1+o(1)}$ edges for the n -vertex graphs in $\mathcal{G}$. This is made precise by considering the graph parameter ‘row-treewidth.’

4.1 Separators in Planar Graphs

In this section, we show that $r(n) \leq \mathcal{O}(n^{3/2})$ by using perfectly balanced separators. In light of Corollary 2.6, we proceed by first bounding the pathwidth of n -vertex planar graphs. To do so, we use the Planar Separator Theorem.

Theorem 4.2 (Planar Separator Theorem, Lipton and Tarjan [59]). *For every planar graph G with $n \geq 1$ vertices, there is a partition $\{A, S, B\}$ of $V(G)$ with $|A|, |B| < 2n/3$ and $|S| \leq 2\sqrt{2n}$ such that there is no AB -edge in G .*

We present a full proof of Theorem 4.2 shortly. For now, we assume Theorem 4.2 holds. Then, the pathwidth of an n -vertex planar graph is $\mathcal{O}(\sqrt{n})$.

Lemma 4.3. *If $c := \frac{2\sqrt{6}}{\sqrt{3}-\sqrt{2}} \approx 15.4$, then for every n -vertex planar graph G , $\text{pw}(G) \leq c\sqrt{n}$.*

Proof of Lemma 4.3 assuming Theorem 4.2. Note the following relation: $c = c\sqrt{2/3} + 2\sqrt{2}$. Proceed by induction on n . In the $n = 1$ case, $\text{pw}(G) = 0 \leq c$. Now assume that $n \geq 2$ and the result holds for all lower values. By Theorem 4.2, there is a partition $\{A, S, B\}$ of $V(G)$ with $|A|, |B| < 2n/3$ and $|S| \leq 2\sqrt{2n}$ such that there is no AB -edge in G . By the induction hypothesis, $\text{pw}(G[A])$ and $\text{pw}(G[B])$ are at most $c\sqrt{2n/3}$. Let $(C_1, \dots, C_g)$ and $(D_1, \dots, D_h)$ be path-decompositions of $G[A]$ and $G[B]$ respectively with minimum widths. Then $(C_1 \cup S, \dots, C_g \cup S, D_1 \cup S, \dots, D_h \cup S)$ is a path-decomposition of G with width at most

$$c\sqrt{2n/3} + |S| \leq c\sqrt{2n/3} + 2\sqrt{2n} = (c\sqrt{2/3} + 2\sqrt{2})\sqrt{n} = c\sqrt{n}.$$

The result follows by induction. $\square$

Let c be the constant in Lemma 4.3. Then applying Corollary 2.6 with the function $c\sqrt{n}$ implies $r(n) \leq \mathcal{O}(n^{3/2} \log n)$; this is worse than $\mathcal{O}(n^{3/2})$ by a $\log n$ factor. The $\log n$ factor can be removed by using the fact that $c\sqrt{n}$ grows relatively quickly and re-estimating the number of edges in the universal graph U_n shown in Section 2.1.

Let $\mathcal{G}$ be the class of planar graphs. Recall that $\text{pbs}_{\mathcal{G}}(n)$ is the minimum size of a perfectly balanced separator in every n -vertex graph in $\mathcal{G}$. Consider the universal graph U_n constructed in Section 2.1 by picking $f(n) = \text{pbs}_{\mathcal{G}}(n)$. Then the number of edges in the resulting universal graph U_n is bounded above by a function $e(n)$ satisfying the recurrence:

$$e(n) \geq 2e\left(\frac{n - f(n)}{2}\right) + (n - 1)f(n). \quad (4.1)$$

Note that $f(n) \leq c\sqrt{n} + 2$ by Theorem 2.5.

Lemma 4.4. *If $c' := \frac{(c+2)\sqrt{2}}{\sqrt{2}-1} \approx 59.4$, then $e(n) = c'n^{3/2}$ satisfies (4.1) for all $n \geq 2$.*

Proof. Note the following relation: $\frac{1}{\sqrt{2}} + \frac{c+2}{c'} = 1$. When $e(n) = c'n^{3/2}$, the right hand side of (4.1) is bounded above by

$$\begin{aligned} 2e\left(\frac{n - f(n)}{2}\right) + (n - 1)f(n) &\leq 2c'\left(\frac{n}{2}\right)^{3/2} + (n - 1)f(n) = c'n^{3/2}\left(\frac{1}{\sqrt{2}} + \frac{(n - 1)f(n)}{c'n^{3/2}}\right) \\ &\leq c'n^{3/2}\left(\frac{1}{\sqrt{2}} + \frac{c + 2}{c'}\right) \\ &= e(n), \end{aligned}$$

as desired. $\square$

Hence, the number of edges in U_n is $\mathcal{O}(n^{3/2})$, implying:

Theorem 4.5 (Babai et al. [10]). $r(n) \leq \mathcal{O}(n^{3/2})$.

It is natural to ask whether one can obtain a better bound for $\text{pbs}_{\mathcal{G}}(n)$; by Corollary 2.3, this would give an improved upper bound for $r(n)$. It turns out that if $\mathcal{G}$ is the class of

planar graphs, then $\text{pbs}_G(n) \geq \Omega(\sqrt{n})$, implying that the bound in Theorem 4.5 is the best possible bound for $r(n)$ obtainable via the construction in Section 2.1.

To see why $\text{pbs}_G(n) \geq \Omega(\sqrt{n})$, one only needs to consider the grids (see Figure 4.1). Let G_n be the $n \times n$ grid. Then any set of vertices X of G_n with $|X| < n/2$ avoids at least half of the rows and half of the columns of G_n , implying $G_n - X$ has a component with more than $n^2/2$ vertices. So X is not a perfectly balanced separator of G_n . This shows that $\text{pbs}_G(n^2) \geq n/2$.

4.1.1 Proof of Planar Separator Theorem. In this section, we give a full proof of Theorem 4.2; we follow the simpler proof by Alon, Seymour, and Thomas [4].

A *near-triangulation* is a drawing of a connected planar graph (in the plane) in which the outer face is bounded by a cycle, and every internal face is a triangle. We require the following variation of Menger's Theorem for near-triangulations; it is folk-lore. We give a proof for completeness. Recall that for any graph H and any two subsets X, Y of $V(H)$, $s(H, X, Y)$ is the size of a minimum XY -separator.

Lemma 4.6. *For any near-triangulation H , for any distinct vertices u, v on the outer face of H , let P_1 and P_2 be the two internally disjoint uv -paths such that the cycle $P_1 \cup P_2$ bounds the outer face of H . Then there exists a uv -path in H with $s(H, V(P_1), V(P_2))$ vertices.*

Proof. Let $X := V(P_1)$ and let $Y := V(P_2)$. Let $s := s(H, X, Y)$. Let D be the compact disk bounded by the cycle $P_1 \cup P_2$. Let S be an XY -separator in H of size s . Since $X \cap Y = \{u, v\}$, S contains u and v . By planarity there is an arc¹ γ in D starting at u and ending at v such that the intersection of γ with the drawing of H is S . Let $u = w_1, w_2, \dots, w_s = v$ be the vertices in S in the order they appear along γ . Observe that for each $i \in \{1, \dots, s-1\}$, w_i and w_{i+1} are joined by a subarc of γ whose interior is contained in an internal (triangle) face of H (that contains w_i and w_{i+1}). Therefore $w_i w_{i+1} \in E(H)$. Hence, $(w_1, w_2, \dots, w_s)$ is a uv -path in H as desired. $\square$

The following elementary fact is needed.

Fact 4.7. *For every positive integer k , $\sum_{i=0}^k \min\{i, k-i\} \geq \frac{1}{4}(k^2 - 1)$.*

Proof. Observe that

$$\begin{aligned} \sum_{i=0}^k \min\{i, k-i\} &= \sum_{i=0}^k \min\{2i, k\} - \sum_{i=0}^k i = \sum_{i=0}^{\lfloor k/2 \rfloor} 2i + \lceil k/2 \rceil k - \sum_{i=0}^k i \\ &= (\lfloor k/2 \rfloor + 1)\lfloor k/2 \rfloor + \lceil k/2 \rceil k - \frac{(k+1)k}{2}. \end{aligned} \tag{4.2}$$

¹That is, a continuous injection of the unit-interval $I = [0, 1]$ into the disk D .

If k is even, (4.2) implies

$$\sum_{i=0}^k \min\{i, k-i\} = (k/2+1)(k/2) + (k/2)k - \frac{(k+1)k}{2} = k^2/4.$$

If k is odd, (4.2) implies

$$\sum_{i=0}^k \min\{i, k-i\} = \left(\frac{k-1}{2} + 1\right) \frac{k-1}{2} + \frac{k+1}{2}k - \frac{(k+1)k}{2} = k^2/4 - 1/4.$$

In both cases, $\sum_{i=0}^k \min\{i, k-i\} \geq \frac{1}{4}(k^2 - 1)$. □

We now prove the Planar Separator Theorem.

Proof of Theorem 4.2. We may assume that $n \geq 3$ and that G is an edge-maximal planar graph. Consider a drawing of G in the plane with no edge-crossings. Then every face of G is a triangle. Let $k := \lfloor \sqrt{2n} \rfloor$. For every cycle C of G , let $\{A(C), V(C), B(C)\}$ be the partition of $V(G)$ where $A(C)$ consists of the vertices drawn inside the bounded face of C , and $B(C)$ consists of the vertices drawn inside the unbounded face of C . Note that there is no $A(C)B(C)$ -edge in G . Furthermore, observe that if C is the triangle bounding the outer face of G , then C satisfies each of the following:

- (1) $|V(C)| \leq 2k$, and
- (2) $|B(C)| < 2n/3$.

Hence, we can pick a cycle C of G satisfying (1) and (2) such that $|A(C)| - |B(C)|$ is as small as possible. Assume for a contradiction, that $|A(C)| \geq 2n/3$.

Let $H := G - B(C)$. Then H is a near-triangulation and C is the cycle that bounds the outer face of H . We claim that:

- (a) $\text{dist}_C(u, v) = \text{dist}_H(u, v)$ for all $u, v \in V(C)$.

Since $C \subseteq H$, $\text{dist}_C(u, v) \geq \text{dist}_H(u, v)$. For a contradiction, suppose that for some $u, v \in V(C)$, $\text{dist}_C(u, v) > \text{dist}_H(u, v)$. Pick $u, v \in V(C)$ such that $\text{dist}_C(u, v) > \text{dist}_H(u, v)$, and $\text{dist}_H(u, v)$ is as small as possible. Let P be a shortest uv -path in H . If P has an interior vertex w belonging to C , then

$$\text{dist}_C(u, w) + \text{dist}_C(w, v) \geq \text{dist}_C(u, v) > \text{dist}_H(u, v) = \text{dist}_H(u, w) + \text{dist}_H(w, v),$$

where the first inequality follows from the triangle inequality, and the last equality follows since P is a shortest path in H . We deduce that $\text{dist}_C(u, w) > \text{dist}_H(u, w)$ or $\text{dist}_C(w, v) > \text{dist}_H(w, v)$; both cases contradict our choice of the vertices u and v . Hence, such a vertex w does not exist, implying $C \cup P$ is equal to the union of three cycles C_1, C_2 and C . Assume without loss of generality that $|A(C_1)| \geq |A(C_2)|$. Since $|A(C_1)| + |V(C_1)| + |B(C_1)| = n$,

$$2(n - |B(C_1)|) = 2(|A(C_1)| + |V(C_1)|) > |A(C_1)| + |A(C_2)| + |V(C_1)| > |A(C)| \geq 2n/3.$$

The extremes of the above inequality imply $|B(C_1)| < 2n/3$. Furthermore, since $|E(P)| < \text{dist}_C(u, v)$, $|V(C_1)| < |V(C)| \leq 2k$. Note that $A(C_1) \subseteq A(C)$ and $B(C) \subseteq B(C_1)$. By our choice of C , it follows that

$$|A(C_1)| - |B(C_1)| = |A(C)| - |B(C)|. \quad (4.3)$$

Observe that $|B(C_1)| > |B(C)|$:

$$0 = (|V(C)| - |V(C_1)|) + (|A(C)| - |A(C_1)|) + (|B(C)| - |B(C_1)|) > |B(C)| - |B(C_1)|.$$

Therefore, (4.3) implies $|A(C_1)| > |A(C)|$, contradicting that $A(C_1) \subseteq A(C)$. This proves (a).

We claim that $|V(C)| = 2k$. For a contradiction, assume $|V(C)| < 2k$. Let $uv \in E(C)$. Since $|A(C)| \geq 2n/3 > 0$, there is a vertex $w \in A(C)$ such that (u, w, v) is a path in H . Then $C' := (u, w, v) \cup (C - uv)$ is a cycle in G such that $|V(C')| = |V(C)| + 1 \leq 2k$, $|B(C')| = |B(C)| < 2n/3$ and $|A(C')| = |A(C)| - 1$, which contradicts our choice of C . This proves that $|V(C)| = 2k$. Let $v_0, \dots, v_{2k-1}$ be the vertices of C in cyclic order; for convenience, let $v_{2k} = v_0$.

Let $X := \{v_0, \dots, v_k\}$ and let $Y := \{v_k, \dots, v_{2k}\}$. If $s(H, X, Y) \leq k$, then by Lemma 4.6 there is a $v_0 v_k$ -path in H on at most k vertices, contradicting (a). Thus, $s(H, X, Y) \geq k+1$. By Theorem 1.18, there are $k+1$ disjoint XY -paths in H . Let $P_0, \dots, P_k$ be disjoint XY -paths in H . We may demand that for each $i \in \{0, \dots, k\}$, the ends of P_i are v_i and v_{2k-i} . By (a),

$$|V(P_i)| \geq 1 + \text{dist}_H(v_i, v_{2k-i}) = 1 + \text{dist}_C(v_i, v_{2k-i}) = 1 + \min\{2i, 2(k-i)\}.$$

Therefore, by Fact 4.7,

$$n \geq \sum_{i=0}^k |V(P_i)| \geq k+1 + 2 \sum_{i=0}^k \min\{i, k-i\} \geq k+1 + \frac{1}{2}(k^2 - 1) = \frac{1}{2}(k+1)^2.$$

The extremes of the above inequality imply $\sqrt{2n} \geq k+1$, contradicting that $k = \lfloor \sqrt{2n} \rfloor$. Hence, our assumption that $|A(C)| \geq 2n/3$, is false. Thus, $|A(C)| < 2n/3$, implying that $\{A(C), V(C), B(C)\}$ a desired partition of $V(G)$. $\square$

We note that Theorem 4.2 can be extended to any proper minor-closed class (that is, a minor-closed class that does not contain every graph).

Theorem 4.8 (Kawarabayashi and Reed [56]). *For all integers $n, t \geq 1$ with $n \gg t$, for every n -vertex K_t -minor-free graph G , there is a partition $\{A, S, B\}$ of $V(G)$ with $|A|, |B| < 2n/3$ and $|S| \leq \mathcal{O}(t\sqrt{n})$ such that there is no AB -edge in G .*

Thus, one can generalise the universal graph construction for planar graphs to any proper minor-closed class, implying:

Theorem 4.9. *For all integers $n, t \geq 1$ with $n \gg t$, there exists an n -vertex graph with $\mathcal{O}(tn^{3/2})$ edges that is universal for the n -vertex K_t -minor-free graphs.*

For the remainder of this chapter, we build towards a construction of a sparse universal graph for planarity. In fact, the proof that we follow shows the existence of sparse universal graphs for any graph class with ‘bounded row-treewidth’ (defined in Section 4.2.2). The family of graph classes with bounded row-treewidth reaches far beyond the class of planar graphs. This requires the discussion of the strong product of graphs.

4.2 Strong Products

The *strong product* of two graphs G_1 and G_2 , denoted by $G_1 \boxtimes G_2$, is the graph with vertex set $V(G_1 \boxtimes G_2) := V(G_1) \times V(G_2)$ and with the following edges. Two vertices (x_1, y_1) and (x_2, y_2) of $G_1 \boxtimes G_2$ are adjacent if one of the following holds:

- (1) $x_1x_2 \in E(G_1)$ and $y_1 = y_2$,
- (2) $x_1 = x_2$ and $y_1y_2 \in E(G_2)$, or
- (3) $x_1x_2 \in E(G_1)$ and $y_1y_2 \in E(G_2)$.

For vertices $v_1 \in V(G_1)$ and $v_2 \in V(G_2)$, the set $V(G_1) \times \{v_2\}$ is a *row* of $G_1 \boxtimes G_2$, and the set $\{v_1\} \times V(G_2)$ is a *column* of $G_1 \boxtimes G_2$.

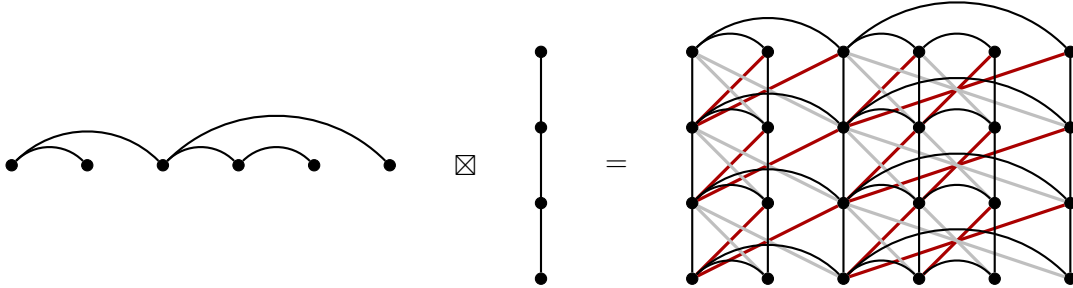


Figure 4.2: Example of strong graph product.

In this section, we present the full proof of the Planar Graph Product Structure Theorem from the original paper [33].

Theorem 4.10 (Planar Graph Product Structure Theorem, Dujmović, Joret, Micek, Morin, Ueckerdt, and Wood [33]). *Every planar graph is a subgraph of $H \boxtimes P$ for some planar graph H with $\text{tw}(H) \leq 8$ and for some path P .*

4.2.1 Proof of Planar Graph Product Structure Theorem. A *partition* $\mathcal{P}$ of a graph G is a partition of $V(G)$ into nonempty subsets (called *parts*). The *quotient* of $\mathcal{P}$, denoted by $G/\mathcal{P}$, is the auxiliary graph with vertex set $\mathcal{P}$ and for all distinct parts A and B of $\mathcal{P}$, A and B are adjacent if and only if there is an AB -edge in G . Note that if every part of $\mathcal{P}$ induces a connected subgraph of G , then $G/\mathcal{P}$ is isomorphic to the minor of G obtained by contracting each part of $\mathcal{P}$ to a vertex.

For a tree T rooted at some vertex r , a *vertical path* in T is a nonempty subpath of an xr -path in T for some $x \in V(T)$. Theorem 4.10 is implied by the following result.

Theorem 4.11. *For any connected planar graph G , for any rooted spanning tree T of G , there is a partition $\mathcal{P}$ of G into vertical paths in T such that $\text{tw}(G/\mathcal{P}) \leq 8$.*

Proof of Theorem 4.10 assuming Theorem 4.11. It is enough to prove the result for connected planar graphs. Let G be an n -vertex connected planar graph. Pick any vertex r of G . By Lemma 1.20, there is a BFS spanning tree T of G rooted at r . By Theorem 4.11, there is a partition $\mathcal{P}$ of G into vertical paths in T such that $H := G/\mathcal{P}$ has treewidth at most 8. Since the parts of $\mathcal{P}$ induce connected subgraphs of G , H is a minor of G . Since G is planar and planarity is a minor-closed property, H is planar.

Consider the path graph $P = (0, \dots, n-1)$. Define the map $\psi : V(G) \rightarrow V(H \boxtimes P)$ as follows. For each part A of $\mathcal{P}$, ψ maps a vertex a of A to the ordered pair $(A, \text{dist}_T(r, a))$. Since $\mathcal{P}$ is a partition of G , and each part of $\mathcal{P}$ gives a vertical path in T , it follows that ψ is injective.

Since T is a BFS spanning tree of G , vertical paths in T are induced subgraphs of G . In particular, $G[A]$ is a path for all $A \in \mathcal{P}$, implying ψ maps adjacent vertices of $G[A]$ to adjacent vertices of $H \boxtimes P$.

Now let A and B be distinct parts of $\mathcal{P}$. Suppose there exists vertices $u \in A$ and $v \in B$ such that $uv \in E(G)$. Then A and B are adjacent in H . Since T is a BFS spanning tree of G , the numbers $\text{dist}_T(r, u)$ and $\text{dist}_T(r, v)$ are either equal or consecutive. In the former case, $(A, \text{dist}_T(r, u))(B, \text{dist}_T(r, v))$ is a Type (1) edge of $H \boxtimes P$. In the latter case, $(A, \text{dist}_T(r, u))(B, \text{dist}_T(r, v))$ is a Type (3) edge of $H \boxtimes P$. Hence, $\psi(u)\psi(v) \in E(H \boxtimes P)$. This shows that ψ embeds G into $H \boxtimes P$. $\square$

We note that the proof of Theorem 4.11 builds on top of the following earlier result by Pilipczuk and Siebertz [63].

Theorem 4.12. *For any planar graph G , there is a partition $\mathcal{P}$ of G into shortest paths in G such that $\text{tw}(G/\mathcal{P}) \leq 8$.*

Note that Theorem 4.11 is a strengthening of Theorem 4.12.

Recall that a near-triangulation is a drawing of a connected planar graph in which the outer face is bounded by a cycle, and every internal face is a triangle. We require the following variation of a well-known result by Sperner [74] for near-triangulations; we follow the proof in [1]. If C is a cycle, then we write $C = [R_1, \dots, R_\ell]$ if $R_1, \dots, R_\ell$ are nonempty pairwise-disjoint paths in C such that for each i , the end vertices of R_i are labelled x_i and y_i , and $y_i x_{i+1} \in E(C)$ (where $x_{\ell+1} = x_1$). We say that C is the *concatenation* of the paths $R_1, \dots, R_\ell$. Note that $V(C) = V(R_1) \cup \dots \cup V(R_\ell)$.

Theorem 4.13 (Sperner [74]). *Let G be a near-triangulation and let R_1, R_2, R_3 be pairwise-disjoint paths such that $C = [R_1, R_2, R_3]$ is the cycle that bounds the outer face of G . If the vertices of G are assigned numbers from the set $\{1, 2, 3\}$, where each vertex of R_i is numbered i , then there exists an internal face of G whose vertices are numbered 1, 2, 3.*

Proof. An edge xy of G is *good* if x is numbered 1 and y is numbered 2. Let G^* be the auxiliary graph defined in the following way. The vertex set of G^* is the set of faces of G . Two distinct faces E and F of G are adjacent in G^* if and only if E and F share a good edge of G .

Note that the outer face of G has degree 1 in G^* . Furthermore, since every internal face of G has at most 2 good edges, the degree of every vertex in G^* is at most 2. Moreover, for an internal face F of G , the degree of F in G^* equals 1 if and only if the vertices of F are numbered 1, 2, 3.

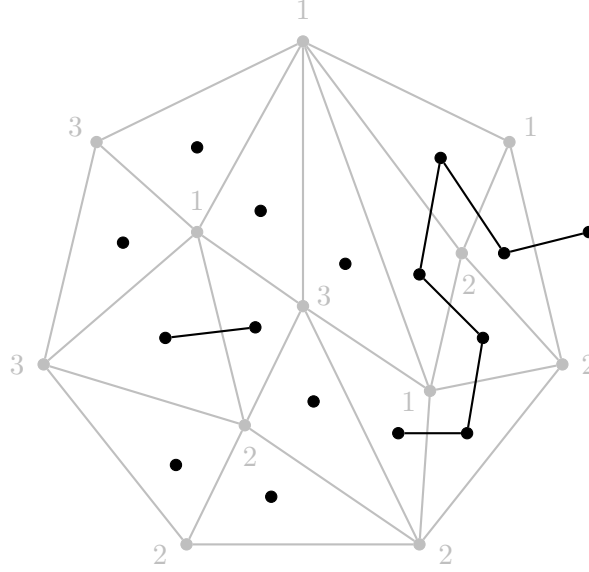


Figure 4.3: Drawing of G (light grey) and G^* (black).

Since the sum of the degrees in a graph is even, there must be an even number of odd-degree vertices. In particular, since the outer face of G has odd-degree in G^* , there is some internal face of G with odd-degree. This is the desired internal face of G . $\square$

Theorem 4.11 is proved by showing the following stronger statement.

Lemma 4.14. *For any plane triangulation G^+ , for any spanning tree T of G^+ that is rooted at some vertex r on the outer face of G^+ , for any set of pairwise-disjoint vertical paths $P_1, \dots, P_k$ in T with $k \in \{1, \dots, 6\}$ such that $C = [P_1, \dots, P_k]$ is a cycle in G^+ , if G is the near-triangulation that is equal to the subgraph of G^+ induced by the vertices drawn on and inside C , then each of the following hold:*

- (1) *there is a partition $\mathcal{P}$ of G into vertical paths in T ,*
- (2) *$V(P_1), \dots, V(P_k) \in \mathcal{P}$, and*
- (3) *$H = G/\mathcal{P}$ has a tree-decomposition with width at most 8 in which some bag contains every vertex of H that corresponds to one of $P_1, \dots, P_k$.*

Proof of Theorem 4.11 assuming Lemma 4.14. The result holds trivially if $|V(G)| < 3$, so we may assume $|V(G)| \geq 3$. Let r be the root vertex of T . Let G^+ be a plane triangulation that contains G as a spanning subgraph and r is on the outer face of G^+ . Since the three

vertices on the outer face of G^+ form three single-vertex vertical paths in T , we can apply Lemma 4.14 to G^+ , implying there is partition $\mathcal{P}$ of G^+ into vertical paths in T such that $\text{tw}(G^+/\mathcal{P}) \leq 8$. Note that $\mathcal{P}$ is a partition of G into vertical paths in T such that $G/\mathcal{P}$ is isomorphic to a subgraph of $G^+/\mathcal{P}$. Therefore, $\text{tw}(G/\mathcal{P}) \leq 8$. $\square$

Proof of Lemma 4.14. Proceed by induction on $|V(G)|$. If $|V(G)| = 3$, then G is isomorphic to K_3 , implying $\{P_1, \dots, P_k\}$ is a partition of G into vertical paths in T . Since $k \leq 3$, $H = G/\mathcal{P}$ has at most 3 vertices. Therefore, the tree-decomposition of H consisting of a single bag suffices. Assume now that $|V(G)| > 3$.

We will apply Theorem 4.13 to G . First, we partition C into three consecutive paths R_1, R_2, R_3 . There are three cases depending on the value of k :

- if $k = 1$, then $|V(P_1)| \geq 3$. Let u and v be the end vertices of P_1 and let $P'_1 := P_1 - u - v$ such that $C = [u, P'_1, v]$. Define $R_1 := u$, $R_2 := P'_1$ and $R_3 := v$,
- if $k = 2$, then without loss of generality $|V(P_1)| \geq 2$. Let u be the end vertex of P_1 and let $P'_1 := P_1 - u$ such that $C = [u, P'_1, P_2]$. Define $R_1 := u$, $R_2 := P'_1$ and $R_3 := P_2$, and
- if $k \in \{3, 4, 5, 6\}$, then for each $i \in \{1, 2, 3\}$, define R_i to be the subpath of C obtained by grouping the consecutive paths $P_{\lfloor (i-1)k/3 \rfloor + 1}, \dots, P_{\lfloor ik/3 \rfloor}$. Note that R_i consists of either 1 or 2 of $P_1, \dots, P_k$.

Assign numbers from $\{1, 2, 3\}$ to the vertices of G in the following way. Assign i to each vertex of R_i . For each vertex u drawn inside C , let P_u be the unique (vertical) ur -path in T . Since r is on the outer face of G^+ , P_u contains a vertex of C . Assign i to u if the first vertex of P_u that is contained in C is in R_i . By Theorem 4.13, there is an internal face u_1, u_2, u_3 of G such that u_i is numbered i .

For each $i \in \{1, 2, 3\}$, let Q_i be the subpath of P_{u_i} that starts at u_i and ends at the first vertex v_i contained in R_i . By the way the vertices of G are numbered, it follows that Q_1, Q_2, Q_3 are nonempty pairwise-disjoint vertical paths in T . Let $Q'_i := Q_i - v_i$. Note that Q'_i could be empty. For each $i \in \{1, 2, 3\}$, let L_i be the $v_i v_{i+1}$ -path in $C - V(R_{i+2})$ (where $v_4 = v_1$, $R_4 = R_1$ and $R_5 = R_2$). Define the subgraphs C_1, C_2, C_3 of G as follows. For each $i \in \{1, 2, 3\}$,

- if $|V(Q'_i) \cup V(L_i) \cup V(Q'_{i+1})| = 2$ (where $Q'_4 = Q'_1$), then Q'_i and Q'_{i+1} are empty. Let C_i be the subgraph of G consisting of the edge $u_i u_{i+1}$ (where $u_4 = u_1$); C_i is said to be *degenerate*,
- if $|V(Q'_i) \cup V(L_i) \cup V(Q'_{i+1})| \geq 3$, then $C_i := [Q'_i, L_i, Q'_{i+1}]$ is a cycle in G (where we remove Q'_i and Q'_{i+1} from the notation if they are empty).

Observe that the subgraph of G consisting of C , the triangle u_1, u_2, u_3 , and $Q_1 \cup Q_2 \cup Q_3$, has an outer face bounded by C , an internal face triangle face bounded by u_1, u_2, u_3 , and at most three more internal faces bounded by the non-degenerate cycles in $\{C_1, C_2, C_3\}$ (see Figure 4.4).

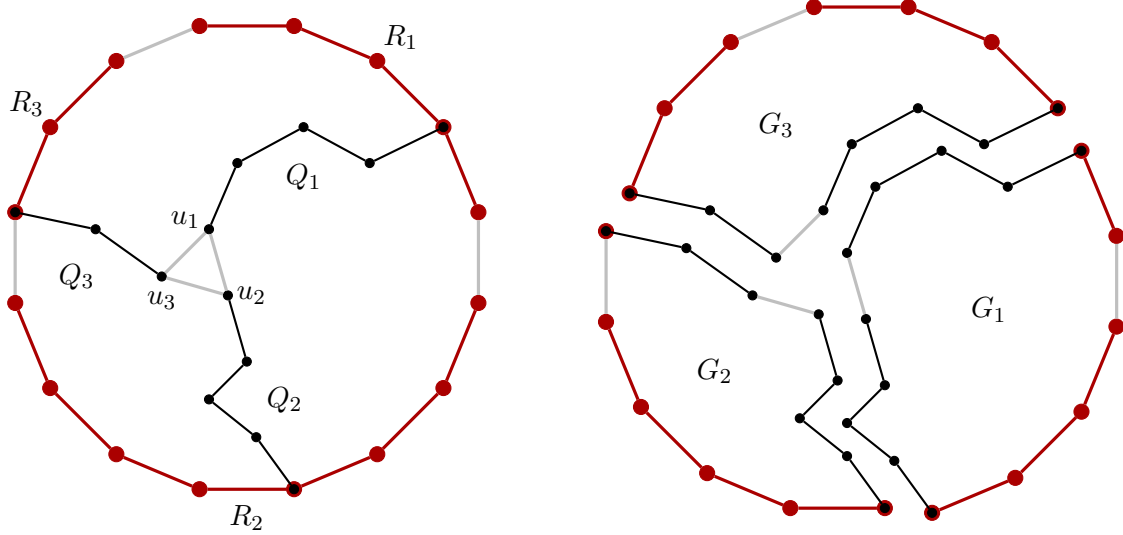


Figure 4.4: Drawing of G (left) and the subgraphs G_1, G_2, G_3 (right).

For each non-degenerate C_i , let G_i be the near-triangulation that is equal to the subgraph of G^+ induced by the vertices drawn on and inside C_i . Observe that C_i is the concatenation of at most six vertical paths in T , namely, Q'_i, Q'_{i+1} (if they are nonempty), and at most four more from L_i (each of which is a subpath of one of $P_1, \dots, P_k$). Furthermore, note that G_i does not contain u_{i+2} (where $u_4 = u_1$ and $u_5 = u_2$), in particular, $|V(G_i)| < |V(G)|$. Therefore, by applying the induction hypothesis to G_i and C_i , we obtain a partition $\mathcal{P}_i$ of G_i into vertical paths in T such that $H_i := G_i/\mathcal{P}_i$ has a tree-decomposition $(B_x^i : x \in V(T_i))$ with width at most 8 for which there is some bag $B_{w_i}^i$ containing each of the vertices of H_i that correspond to the (at most) six vertical paths that concatenate to C_i . Let $\mathcal{J}_i$ be the parts of $\mathcal{P}_i$ that do not intersect $V(C_i)$. Then $\mathcal{J}_i$ is a partition of $G_i - V(C_i)$ into vertical paths in T , and each remaining part of $\mathcal{P}_i$ is a subset of one of $V(P_1), \dots, V(P_k)$. We now combine these partitions to form a desired partition $\mathcal{P}$ of G . Let $\mathcal{P}$ be the union of the following sets: $\{V(P_1), \dots, V(P_k)\}$, the set of all nonempty $V(Q'_i)$, and for each non-degenerate C_i , the set $\mathcal{J}_i$. Then $\mathcal{P}$ is a partition of G into vertical paths in T . Note that there are unique maps $f_i : \mathcal{P}_i \rightarrow \mathcal{P}$ such that $A \subseteq f_i(A)$.

Let $H := G/\mathcal{P}$. It remains to give a tree-decomposition of H with width at most 8 in which some bag contains every vertex of H that corresponds to one of $P_1, \dots, P_k$. Let T' be the tree formed by taking the disjoint union of the trees T_i , and adding a new vertex w that is adjacent to each w_i . Let B_w be the union of $\{V(P_1), \dots, V(P_k)\}$ and the set of all nonempty $V(Q'_i)$. Then $|B_w| \leq k+3 \leq 9$. For each $x \in V(T_i)$, let $B_x := \{f_i(A) : A \in B_x^i\}$. Then $|B_x| \leq |B_x^i| \leq 9$. We check that $(B_x : x \in V(T'))$ is a tree-decomposition of H .

Let A be a vertex of H . Then A is a part of $\mathcal{P}$. If A is in $\mathcal{J}_i$, then A only appears in the bags of T_i . Since $(B_x^i : x \in V(T_i))$ is a tree-decomposition, A is mapped to a nonempty subtree of T_i , which is a nonempty subtree of T' . If A is one of $V(P_1), \dots, V(P_k)$, $V(Q'_1), V(Q'_2), V(Q'_3)$, then $A \in B_w$. Furthermore, if the preimage $f_i^{-1}(A)$ is nonempty, then $T_i[\{x \in V(T_i) : f_i^{-1}(A) \cap B_x\}]$ is a nonempty subtree of T_i that contains w_i . Since w is adjacent to each w_i , it follows that A is mapped to a nonempty subtree of T' . This

shows the vertex-property in T' .

Let A and B be adjacent vertices of H . We may assume that $\{A, B\} \not\subseteq B_w$. Say $A \notin B_w$. Then A is in $\mathcal{J}_i$ for some i . Since $V(C_i)$ separates $V(G_i)$ and $V(G) \setminus V(G_i)$ in G , there exists a part X of $\mathcal{P}_i$ that is adjacent to A in H_i , and $f_i(X) = B$. Since $(B_x^i : x \in V(T_i))$ is a tree-decomposition, there is a vertex $x \in V(T_i)$ such that $\{A, X\} \subseteq B_x^i$, implying $\{A, B\} = \{f_i(A), f_i(X)\} \subseteq B_x$. This shows the edge-property in T' . Hence, $(B_x : x \in V(T'))$ is a tree-decomposition of H with width at most 8 as desired. $\square$

This completes the proof of Theorem 4.10. We note that the bound on the treewidth of H in the Planar Graph Product Structure Theorem can be reduced further.

Theorem 4.15 (Dujmović et al. [33]). *Every planar graph is a subgraph of $H \boxtimes P \boxtimes K_3$ for some planar graph H with $\text{tw}(H) \leq 3$ and for some path P .*

4.2.2 Row-Treewidth. The Planar Graph Product Structure Theorem says that every planar graph is a subgraph of $H \boxtimes P$ where H is a graph with treewidth at most 8 and P is some path. There are other graph classes that admit product structure theorems, for example, graphs embeddable on a fixed surface.

Theorem 4.16 (Distel et al. [30]). *Every graph with Euler genus g is a subgraph of $H \boxtimes P \boxtimes K_{\max\{2g, 3\}}$ for some graph H with treewidth at most 3 and for some path P .*

Even some non-minor-closed classes admit product structure theorems. A graph G is *k -planar* if it can be drawn in the plane with at most k crossings per edge. The class of k -planar graphs is not minor-closed when $k \geq 1$ [32].

Theorem 4.17 (Dujmović et al. [33]). *Every k -planar graph is a subgraph of $H \boxtimes P$ for some graph H with treewidth $\mathcal{O}(k^6)$ and for some path P .*

In light of these results, it is natural to make the following definition. For a graph G , the *row-treewidth* of G , denoted by $\text{rtw}(G)$, is the minimum treewidth of a graph H such that G is isomorphic to a subgraph of $H \boxtimes P$ for some path P . Note, we do not require H to be planar.

In general, $\text{rtw}(G) \leq \text{tw}(G)$ for every graph G . Theorem 4.10 says that the row-treewidth of every planar graph is at most 8. Recall that the class of planar graphs has unbounded treewidth. One might ask: is the row-treewidth always much less than the treewidth? The answer is no.

Theorem 4.18 (Bose et al. [18]). *For every integer $k \geq 1$, there exists a graph G such that $\text{rtw}(G) = \text{tw}(G) = k$.*

A graph class $\mathcal{G}$ has *bounded row-treewidth* if $\text{rtw}(G) \leq t$ for every graph G in $\mathcal{G}$, for some absolute constant t . As stated in Section 1.1.2, planar graphs can be used to characterise the minor-closed graph classes with bounded treewidth. Minor-closed classes with bounded row-treewidth can be characterised similarly. A graph G is *apex* if there exists a vertex v of G such that $G - v$ is planar.

Theorem 4.19 (Dujmović et al. [34]). *A minor-closed class $\mathcal{G}$ has bounded row-treewidth if and only if some apex graph is not in $\mathcal{G}$.*

For a survey on graph product structure theory see [36]. In the next section, we present a proof that any graph class (not necessarily minor-closed) with bounded row-treewidth has sparse universal graphs.

4.3 Sparse Universal Graphs for Classes with Bounded Row-Treewidth

Let $\mathcal{Q}_t$ be the set of all graphs with row-treewidth at most t . Let $r_t(n)$ be the minimum number of edges in a graph that is universal for n -vertex graphs in $\mathcal{Q}_t$. By Theorem 4.10, $\mathcal{Q}_8$ contains every planar graph, implying $r(n) \leq r_8(n)$.

The remainder of this chapter presents a proof of the bound: $r_t(n) \leq t^2 n^{1+o(1)}$ when $n \gg t$. This bound is implied by the next theorem.

Theorem 4.20 (Esperet, Joret, and Morin [43]). *For all integers $n, t \geq 1$ with $n \gg t$, there exists a graph with $t^2 n^{1+o(1)}$ edges and $tn^{1+o(1)}$ vertices that is universal for the class of n -vertex graphs in $\mathcal{Q}_t$.*

We follow the proof from the original paper [43]. At the end we show a trick that is used to reduce the number of vertices in the universal graph given by Theorem 4.20 from $tn^{1+o(1)}$ to $(1 + o(1))n$.



Louis Esperet



Gwenaël Joret



Pat Morin

We begin by introducing some machinery related to binary search trees.

4.3.1 Bulk Tree Sequences. A *binary tree* T is a rooted tree in which every vertex except the root r is the *left* or *right* child of its parent, and every vertex has at most one left and one right child. For any vertex v of T , let $P = (r = x_0, \dots, x_n = v)$ be the path in T from r to v . Then the *depth* of v in T , denoted by $\text{depth}_T(v)$, is the length of P . The *height* of T , denoted by $\text{height}(T)$, is the maximum depth of a vertex in T . A vertex u of P is a *T -ancestor* of v ; we write $u \preceq_T v$. The *signature* of v in T , denoted by $\sigma_T(v)$, is the binary string $\alpha_1 \dots \alpha_n$ such that for each $i \in \{1, \dots, n\}$,

$$\alpha_i = \begin{cases} 0 & \text{if } x_i \text{ is the left child of } x_{i-1}, \\ 1 & \text{if } x_i \text{ is the right child of } x_{i-1}. \end{cases}$$

Note that $\sigma_T(r)$ is the empty string.

A *binary search tree* T is a binary tree with $V(T) \subseteq \mathbb{R}$ such that T satisfies the *BST property*: for every vertex v of T , $u < v$ for every vertex u in the left subtree of v , and $v < w$ for every vertex w in the right subtree of v .

Given a weighted set of finitely many real numbers S , the following lemma shows that there is a binary search tree on the vertex set S such that the vertices with (relatively) larger weight are closer to the root vertex.

Lemma 4.21. *For every finite subset S of $\mathbb{R}$ and every function $f : S \rightarrow \mathbb{R}^+$, there exists a binary search tree T such that $|V(T)| = S$ and for each $v \in S$,*

$$\text{depth}_T(v) \leq \log_2 \left(\frac{\sum_{s \in S} f(s)}{f(v)} \right).$$

Proof. Proceed by induction on $|S|$. The $|S| = 0$ case holds vacuously, so assume that $|S| \geq 1$ and the result holds for all lower values. Put $W := \sum_{s \in S} f(s)$. Let $r \in S$ be maximum such that $\sum_{\substack{s \in S, \\ s < r}} f(s) \leq W/2$. Let $S_1 := \{s \in S : s < r\}$ and let $S_2 := \{s \in S : r < s\}$. If $|S_2| \geq 1$, then by the maximality of r ,

$$\sum_{s \in S_2} f(s) = W - \sum_{\substack{s \in S, \\ s \leq r}} f(s) < W - W/2 = W/2.$$

If $|S_2| = 0$, then again $\sum_{s \in S_2} f(s) < W/2$. Since $|S_1|, |S_2| < |S|$, by the induction hypothesis, for each $i \in \{1, 2\}$ we obtain a binary search tree T_i with $V(T_i) = S_i$ such that for each $v \in S_i$,

$$\text{depth}_{T_i}(v) \leq \log_2 \left(\frac{\sum_{s \in S_i} f(s)}{f(v)} \right) \leq \log_2 \left(\frac{W}{2f(v)} \right).$$

Then a binary search tree T is formed by choosing the root r , and T_1 and T_2 respectively to be the left and right subtrees of r in T .

Then $\text{depth}_T(r) = 0 \leq \log_2 \left(\frac{W}{f(r)} \right)$. Furthermore, for each $i \in \{1, 2\}$ and each vertex $v \in S_i$,

$$\text{depth}_T(v) = 1 + \text{depth}_{T_i}(v) \leq 1 + \log_2 \left(\frac{W}{2f(v)} \right) = \log_2 \left(\frac{W}{f(v)} \right).$$

Hence, the result follows by induction. $\square$

The following technical lemma shows the existence of so-called bulk tree sequences. We skip the proof as it is quite long and cumbersome. Instead, we cite the previous paper by the same authors in which the lemma is proved.

Recall that $\{0, 1\}^*$ is the set of all finite binary strings (including the empty string).

Lemma 4.22 (Dujmović et al. [35]). *There exists a function $\lambda : \mathbb{N} \rightarrow \mathbb{N}$ with $\lambda(n) \leq \mathcal{O}(\sqrt{(\log n)(\log \log n)})$ and a function $\Phi : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ such that for all positive integers m and h , for any collection of finite subsets $S_1, \dots, S_h$ of $\mathbb{R}$ with $|S_1| +$*

$\dots + |S_h| = m$, there exists a sequence of binary search trees $T_1, \dots, T_h$ (called a *bulk tree sequence*) such that each of the following hold:

- (1) $S_h \subseteq V(T_h)$ and for each $i \in \{1, \dots, h-1\}$, $S_i \cup S_{i+1} \subseteq V(T_i)$,
- (2) $|V(T_1)| + \dots + |V(T_h)| \leq 4m$,
- (3) for each $i \in \{1, \dots, h\}$, $\text{height}(T_i) \leq \log_2 |V(T_i)| + \lambda(m)$, and
- (4) for each $i \in \{1, \dots, h-1\}$ and each $s \in V(T_i) \cap V(T_{i+1})$, there exists $\tau \in \{0, 1\}^*$ such that $|\tau| \leq \lambda(m)$ and $\Phi(\sigma_{T_i}(s), \tau) = \sigma_{T_{i+1}}(s)$.

Roughly speaking, the bulk tree sequence $T_1, \dots, T_h$ is intended to efficiently ‘encode’ the rows of the strong product $H \boxtimes P$.

For a binary search tree T , two vertices x and y of T (with $x < y$ say) are *consecutive* if there is no vertex z of T such that $x < z < y$. The *length* of a binary string σ , denoted by $|\sigma|$, is the number of bits.

For each $\sigma \in \{0, 1\}^*$ and each integer $h \geq |\sigma|$, define the set $\Lambda_\sigma(h)$ of binary strings as follows. A binary string τ is contained in $\Lambda_\sigma(h)$ if and only if at least one of the following holds:

- (a) σ equals $\tau 0$ followed by at most $|\sigma| - 1$ ones, or
- (b) τ equals $\sigma 1$ followed by at most $h - |\sigma| - 1$ zeros.

Lemma 4.23. *For every $\sigma \in \{0, 1\}^*$ and every integer $h \geq |\sigma|$, each of the following hold:*

- (1) *each binary string in $\Lambda_\sigma(h)$ has length at most h ,*
- (2) *$|\Lambda_\sigma(h)| \leq h$, and*
- (3) *for any binary search tree T of height at most h , for any two consecutive vertices $x < y$ of T , we have $\sigma_T(y) \in \Lambda_{\sigma_T(x)}(h)$.*

Proof. In (a) and (b), the length of τ is at most h , so (1) holds. Furthermore, $|\Lambda_\sigma(h)| \leq |\sigma| + h - |\sigma| = h$, implying (2) holds. It remains to check (3) for a given binary search tree T of height at most h , and two consecutive vertices $x < y$ of T .

Suppose x has no right child in T . By the BST property of T , x is contained in the left subtree of y . Since x is the greatest (real number) vertex in the left subtree of y , the signature of x equals $\sigma_T(y)0$ followed by $\text{depth}_T(x) - \text{depth}_T(y) - 1 \leq |\sigma_T(x)| - 1$ ones. Thus, $\sigma_T(y) \in \Lambda_{\sigma_T(x)}(h)$ by (a).

Now suppose x has a right child in T . By the BST property of T , y is contained in the right subtree of x . Since y is the smallest (real number) vertex in the right subtree of x , the signature of y equals $\sigma_T(x)1$ followed by at most $h - \text{depth}_T(x) - 1 = h - |\sigma_T(x)| - 1$ zeros. Thus, $\sigma_T(y) \in \Lambda_{\sigma_T(x)}(h)$ by (b), which proves the lemma. $\square$

4.3.2 Intervals in Complete Binary Search Trees. For integers $d \geq 0$, let B_d be the unique binary search tree of height d and vertex set $\{1, \dots, 2^{d+1} - 1\}$. Define the graph C_d by $V(C_d) := V(B_d)$ and $E(C_d) := \{uv : u \preceq_{B_d} v \text{ and } u \neq v\}$ (see Figure 4.5).

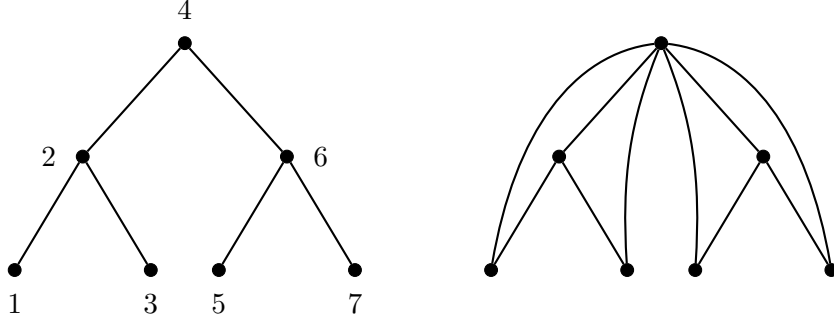


Figure 4.5: B_2 (left) and C_2 (right).

The *interval* of a vertex v of B_d is the set $I_d(v) := \{u \in V(B_d) : v \preceq_{B_d} u\}$. The adjacencies in C_d can be characterised by the intersections of intervals.

Lemma 4.24. *For any two distinct vertices u and v of C_d , each of the following are equivalent:*

- (1) $uv \in E(C_d)$,
- (2) $I_d(u) \cap I_d(v) \neq \emptyset$, and
- (3) $I_d(u) \subseteq I_d(v)$ or $I_d(v) \subseteq I_d(u)$.

Proof. (1 $\Rightarrow$ 2) If $uv \in E(C_d)$, then either $v \preceq_{B_d} u$ or $u \preceq_{B_d} v$. Assume without loss of generality that $v \preceq_{B_d} u$. Then $u \in I_d(v)$, which implies $I_d(u) \cap I_d(v) \neq \emptyset$. (2 $\Rightarrow$ 3) If $I_d(u) \cap I_d(v) \neq \emptyset$, then there exists a vertex $w \in V(T)$ such that $v \preceq_{B_d} w$ and $u \preceq_{B_d} w$. Let P be the path in T from the root to w . So $u, v \in V(P)$. Assume without loss of generality that u comes after v in P . Then $v \preceq_{B_d} u$, implying $I_d(u) \subseteq I_d(v)$. (3 $\Rightarrow$ 1) Suppose $I_d(u) \subseteq I_d(v)$. Then v is a T -ancestor of u , which implies $uv \in E(C_d)$. Similarly, if $I_d(v) \subseteq I_d(u)$, then $uv \in E(C_d)$. $\square$

We note that in the following lemma, although the vertex set of the binary search tree T is contained in $V(B_d)$, the structure of T may be completely different from B_d .

Lemma 4.25. *For any binary search tree T with $V(T) \subseteq V(B_d)$, for any vertex v of T , the minimum T -depth in $V(T) \cap I_d(v)$ is achieved by a unique vertex.*

Proof. Since v is a vertex of T , $V(T) \cap I_d(v) \neq \emptyset$. For a contradiction, assume there are two vertices u and w such that the minimum T -depth in $V(T) \cap I_d(v)$ equals $\text{depth}_T(u) = \text{depth}_T(w)$. Then u and w have a common T -ancestor. Let z be a common T -ancestor of u and w with $\text{depth}_T(z)$ as large as possible. Note that $z \neq u$ and $z \neq w$. By the BST property in T , either

$$u < z < w \quad \text{or} \quad w < z < u.$$

By the minimality of u and w , $z \notin I_d(v)$. By the BST property in B_d , it follows that z is either greater than every vertex in $I_d(v)$, or less than every vertex in $I_d(v)$. This contradicts the above inequalities, since $u, w \in I_d(v)$. $\square$

For any vertex v of T , let $m_T(v)$ be the unique vertex of $V(T) \cap I_d(v)$ with minimum T -depth (that is shown to exist in Lemma 4.25).

For two binary strings σ and τ , we write $\sigma \preceq \tau$ when σ is a prefix of τ . We write $\sigma \sim \tau$ if $\sigma \preceq \tau$ or $\tau \preceq \sigma$. Note that $\sim$ is not transitive.

Lemma 4.26. *For any binary search tree T with $V(T) \subseteq V(B_d)$, for any edge uv of $C_d[V(T)]$,*

$$m_T(u) \preceq_T m_T(v) \quad \text{or} \quad m_T(v) \preceq_T m_T(u),$$

and hence $\sigma_T(m_T(u)) \sim \sigma_T(m_T(v))$.

Proof. Let $x := m_T(u)$ and let $y := m_T(v)$. By Lemma 4.24, we may assume without loss of generality that $I_d(u) \subseteq I_d(v)$, since $uv \in E(C_d)$. For a contradiction, assume that neither $x \preceq_T y$ nor $y \preceq_T x$ holds. Then x and y have a common T -ancestor. Let z be a common T -ancestor of x and y with $\text{depth}_T(z)$ as large as possible. Note that $z \neq x$ and $z \neq y$. By the BST property in T , either

$$x < z < y \quad \text{or} \quad y < z < x.$$

Since y has minimum T -depth in $V(T) \cap I_d(v)$, $z \notin I_d(v)$. By the BST property in B_d , it follows that z is either greater than every vertex in $I_d(v)$, or less than every vertex in $I_d(v)$. This contradicts the above inequalities, since $x, y \in I_d(v)$. $\square$

4.3.3 Subgraphs of $C_{\lceil \log_2 n \rceil} \boxtimes P_n$. We construct a universal graph for the n -vertex subgraphs of $C_{\lceil \log_2 n \rceil} \boxtimes P_n$. We require the following results about strong products.

Lemma 4.27. *For any two graphs G_1 and G_2 , for any subgraph H of $G_1 \boxtimes G_2$ there exists induced subgraphs H_1 of G_1 and H_2 of G_2 such that $H \subseteq H_1 \boxtimes H_2$, and each row and column of $H_1 \boxtimes H_2$ contains vertices of H . In particular, $|V(H_1)|, |V(H_2)| \leq |V(H)|$.*

Proof. Let H_1 be a minimal induced subgraph of G_1 such that $H \subseteq H_1 \boxtimes G_2$ (this is well-defined since G_1 is an induced subgraph of itself such that $H \subseteq G_1 \boxtimes G_2$). If a column $\{v\} \times G_2$ of $H_1 \boxtimes G_2$ does not contain any vertices of H , then $H_1 - v$ is an induced subgraph of G_1 such that $H \subseteq (H_1 - v) \boxtimes G_2$, which contradicts the minimality of H_1 . Therefore, each column of $H_1 \boxtimes G_2$ contains vertices of H . Let H_2 be a minimal induced subgraph of G_2 such that $H \subseteq H_1 \boxtimes H_2$. By a similar argument, each row of $H_1 \boxtimes H_2$ contains vertices of H . Thus, each row and column of $H_1 \boxtimes H_2$ contains vertices of H , implying $|V(H_1)|, |V(H_2)| \leq |V(H)|$. $\square$

Lemma 4.28. *If a graph U is universal for the class of n -vertex subgraphs of a graph G_1 , then for any graph G_2 , $U \boxtimes G_2$ is universal for the class of n -vertex subgraphs of $G_1 \boxtimes G_2$.*

Proof. Note that U contains every subgraph of G_1 with at most n vertices. Let H be an n -vertex subgraph of $G_1 \boxtimes G_2$. By Lemma 4.27, there exists a subgraph H_1 of G_1 such that

$H \subseteq H_1 \boxtimes G_2$ and $|V(H_1)| \leq n$. Since U contains H_1 , $U \boxtimes G_2$ contains $H_1 \boxtimes G_2$, implying $U \boxtimes G_2$ contains H . Hence, $U \boxtimes G_2$ contains every n -vertex subgraph of $G_1 \boxtimes G_2$. $\square$

Let n be a positive integer and let $d := \lceil \log_2 n \rceil$. Let $\lambda = \lambda(n)$ and Φ be the functions from Lemma 4.22. Define the graph G_n on the vertex set

$$V(G_n) = \left\{ (x, y) : x, y \in \{0, 1\}^* \text{ and } |x| + |y| \leq d + \lambda + 2 \right\} \times \{0, \dots, d\},$$

and with the following edges. For convenience, we define the edges of G_n as directed edges; the graph G_n itself is undirected. For any two distinct vertices (x_1, y_1, z_1) and (x_2, y_2, z_2) , add a directed edge from (x_1, y_1, z_1) to (x_2, y_2, z_2) if one of the following holds:

- (1) $y_1 = y_2$ and $x_2 \preceq x_1$,
- (2) $y_1 \neq y_2$ and
 - (a) $y_2 \in \Lambda_{y_1}(d + 2)$ (where Λ is the set from Lemma 4.23),
 - (b) there exists $x_3 \in \{0, 1\}^*$ such that $|x_3| \leq d + \lambda + 2 - |y_1|$ and $x_1 \sim x_3$, and
 - (c) there exists $\tau \in \{0, 1\}^*$ such that $|\tau| \leq \lambda$ and $\Phi(x_3, \tau) = x_2$.

Theorem 4.29. G_n is universal for the n -vertex subgraphs of $C_d \boxtimes P_n$.

Proof. Let G be an n -vertex subgraph of $C_d \boxtimes P_n$. Let h be the number of rows of $C_d \boxtimes P_n$ that contain vertices of G . Then $C_d \boxtimes P_h$ has a subgraph H isomorphic to G such that each row of $C_d \boxtimes P_h$ contains vertices of H . We first define a function $\psi : V(H) \rightarrow V(G_n)$ and then verify it is an embedding. Let $S := \{1, \dots, h\}$. For each $i \in S$, let $S_i := \{v \in V(C_d) : (v, i) \in V(H)\}$. Then $S_1, \dots, S_h$ are subsets of $\mathbb{R}$ with $|S_1| + \dots + |S_h| = n$; let $T_1, \dots, T_h$ be the bulk tree sequence obtained by applying Lemma 4.22. Let T be a binary search tree obtained by applying Lemma 4.21 to the set S and the function $f : S \rightarrow \mathbb{R}^+$ given by $f(i) = |V(T_i)|$. Define $c : V(C_d) \rightarrow \{0, \dots, d\}$ by $c(v) = \text{depth}_{B_d}(v)$. By the way the edges in C_d are defined, c is a proper colouring² of C_d . Then for each vertex (v, i) of H , define

$$\psi(v, i) := (\sigma_{T_i}(m_{T_i}(v)), \sigma_T(i), c(v)).$$

Observe that ψ does indeed map into $V(G_n)$: let $(v, i) \in V(H)$ and let $(x, y, z) := \psi(v, i)$. By (3) from Lemma 4.22,

$$|x| = |\sigma_{T_i}(m_{T_i}(v))| \leq \text{height}(T_i) \leq \log_2 |V(T_i)| + \lambda. \quad (4.4)$$

By Lemma 4.21,

$$\begin{aligned} |y| = \text{depth}_T(i) &\leq \log_2 \left(\sum_{s \in S} f(s) \right) - \log_2(f(i)) = \log_2 \left(\sum_{s \in S} |V(T_s)| \right) - \log_2 |V(T_i)| \\ &\leq \log_2(4n) - \log_2 |V(T_i)| \\ &\leq d + 2 - \log_2 |V(T_i)|, \end{aligned} \quad (4.5)$$

²That is, for any edge uv of C_d , $c(u) \neq c(v)$.

where the second last inequality follows by (2) from Lemma 4.22 and the last inequality follows since $d = \lceil \log_2 n \rceil$. Then (4.4) and (4.5) together imply that

$$|x| + |y| \leq \log_2 |V(T_i)| + \lambda + \log_2(4n) - \log_2 |V(T_i)| \leq d + \lambda + 2.$$

Moreover, since $z \in \{0, \dots, d\}$, $(x, y, z) \in V(G_n)$. So ψ maps into $V(G_n)$.

We check that ψ is injective. Suppose (u, i) and (v, j) are distinct vertices of H . If $i \neq j$, then $\sigma_T(i) \neq \sigma_T(j)$ and so $\psi(u, i) \neq \psi(v, j)$. Now assume that $i = j$ (so $u \neq v$). Then u and v are vertices of T_i . If $m_{T_i}(u) \neq m_{T_i}(v)$, then $\sigma_{T_i}(m_{T_i}(u)) \neq \sigma_{T_i}(m_{T_i}(v))$ and so $\psi(u, i) \neq \psi(v, j)$. Thus, we may assume $m_{T_i}(u) = m_{T_i}(v)$. This implies that $I_d(u) \cap I_d(v) \neq \emptyset$. Since u and v are distinct vertices, Lemma 4.24 implies $uv \in E(C_d)$. Therefore, $c(u) \neq c(v)$ and so $\psi(u, i) \neq \psi(v, j)$. Hence, ψ is injective.

It remains to verify that ψ is a homomorphism. Suppose $(u, i)(v, j)$ is an edge of H . We may assume without loss of generality that $i \leq j$. Let $(x_1, y_1, z_1) := \psi(u, i)$ and let $(x_2, y_2, z_2) := \psi(v, j)$. By the way the edges in $C_d \boxtimes P_h$ are defined, either $j = i$ or $j = i + 1$. In the former case, $y_1 = \sigma_T(i) = y_2$. Furthermore, u and v are (distinct) vertices of T_i that are adjacent in C_d . By Lemma 4.26, $x_1 = \sigma_{T_i}(m_{T_i}(u)) \sim \sigma_{T_i}(m_{T_i}(v)) = x_2$. Thus, $\psi(u, i)\psi(v, j)$ is a Type (1) edge of G_n . Now assume $j = i + 1$. We show that $\psi(u, i)\psi(v, j)$ is a Type (2) edge of G_n .

By Lemma 4.21,

$$\begin{aligned} \text{height}(T) &= \max_{k \in V(T)} \text{depth}_T(k) \leq \max_{k \in V(T)} \left(\log_2 \left(\sum_{s \in S} |V(T_s)| \right) - \log_2 |V(T_k)| \right) \\ &\leq \log_2 \left(\sum_{s \in S} |V(T_s)| \right) \\ &\leq \log_2(4n), \end{aligned}$$

where the last inequality follows by (2) from Lemma 4.22. Since $d = \lceil \log_2 n \rceil$, the above inequality implies $\text{height}(T) \leq d + 2$. Since i and $j = i + 1$ are consecutive vertices of T , by Lemma 4.23, $y_2 \in \Lambda_{y_1}(d + 2)$. This shows condition (2a) for Type (2) edges.

By (1) from Lemma 4.22, $v \in V(T_i)$. Then $x_3 := \sigma_{T_i}(m_{T_i}(v))$ is well-defined. It follows from a similar derivation as (4.4) and (4.5) respectively, that

$$|x_3| \leq \log_2 |V(T_i)| + \lambda \quad \text{and} \quad |y_1| \leq d + 2 - \log_2 |V(T_i)|.$$

Therefore, $|x_3| \leq d + \lambda + 2 - |y_1|$. Since $(u, i)(v, j)$ is an edge of H , either $u = v$ or $uv \in E(C_d)$. In the former case, $x_2 = \sigma_{T_i}(m_{T_i}(u)) = \sigma_{T_i}(m_{T_i}(v)) = x_3$, which implies $x_1 \sim x_3$. In the latter case, observe that u and v are (distinct) vertices of T_i that are adjacent in C_d . By Lemma 4.26, $x_1 = \sigma_{T_i}(m_{T_i}(u)) \sim \sigma_{T_i}(m_{T_i}(v)) = x_3$. Thus, in either case, x_3 satisfies condition (2b) for Type (2) edges. By (4) from Lemma 4.22, there exists $\tau \in \{0, 1\}^*$ such that $|\tau| \leq \lambda$ and

$$\Phi(x_3, \tau) = \Phi(\sigma_{T_i}(m_{T_i}(v)), \tau) = \sigma_{T_{i+1}}(m_{T_i}(v)) = \sigma_{T_{i+1}}(m_{T_{i+1}}(v)) = x_2. \quad (4.6)$$

Therefore, x_2 and x_3 satisfy condition (2c) for Type (2) edges, implying $\psi(u, i)\psi(v, j)$ is a Type (2) edge of G_n . This shows that G is isomorphic to a subgraph of G_n . $\square$

There is one part in the proof of Theorem 4.29 that I do not understand, namely, the second last equality in (4.6). My supervisor and I have notified the authors of the paper about this anomaly. The authors of [43] are currently considering the issue.

We now estimate the number of vertices and edges in G_n .

Lemma 4.30. $|V(G_n)| \leq n2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})}$ and $|E(G_n)| \leq n2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})}$.

Proof. We count the number of vertices (x, y, z) in G_n . There are $d + 1$ choices for z . Furthermore, for each $r \in \{0, \dots, d + \lambda + 2\}$, the number of pairs (x, y) such that $x, y \in \{0, 1\}^*$ and $|x| + |y| = r$ is $(1 + r)2^r$. Thus,

$$\begin{aligned} |V(G_n)| &= \left(\sum_{r=0}^{d+\lambda+2} (1+r)2^r \right) (d+1) \leq (d+\lambda+3)2^{d+\lambda+3}(d+1) \leq 2^d(d+\lambda+3)^2 2^{\lambda+3} \\ &\leq n2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})}, \end{aligned}$$

as claimed. We bound the number of edges in G_n by considering the outgoing edges from a vertex (x_1, y_1, z_1) in G_n . The number of Type (1) edges equals the number of pairs (x_2, z_2) such that $x_2 \in \{0, 1\}^*$ with $x_2 \preceq x_1$ and $z_2 \in \{0, \dots, d\}$. There are at most $|x_1| + 1$ choices for x_2 , and there are $d + 1$ choices for z_2 . Thus, the number of Type (1) edges in G_n is at most

$$|V(G_n)|(|x_1| + 1)(d + 1) \leq n2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})}(d + \lambda + 3)^2 \leq n2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})}.$$

Now, for $r \in \{0, \dots, d + \lambda + 2\}$, we count the number of Type (2) outgoing edges from a vertex (x_1, y_1, z_1) with $|x_1| + |y_1| = r$. As before, there are exactly $(r + 1)2^r(d + 1)$ such vertices (x_1, y_1, z_1) . The number of Type (2) edges is at most the number of quad-tuples (x_3, τ, y_2, z_2) such that $x_3, \tau \in \{0, 1\}^*$ with $|x_3| \leq d + \lambda + 2$, $x_1 \sim x_3$, $|\tau| \leq \lambda$, $y_2 \in \Lambda_{y_1}(d + 2)$ and $z_2 \in \{0, \dots, d\}$.

If $x_1 \preceq x_3$, then there are at most $2^{d+\lambda+2-|y_1|-|x_1|} = 2^{d+\lambda+2-r}$ choices for x_3 . If $x_3 \preceq x_1$, then there are at most $|x_1| + 1 \leq d + \lambda + 3$ choices for x_3 . So if $x_1 \sim x_3$, then there are at most $d + \lambda + 3 + 2^{d+\lambda+2-r} \leq (d + \lambda + 3)2^{d+\lambda+2-r}$ choices for x_3 .

There are at most $2^{\lambda+1}$ choices for τ . By (2) from Lemma 4.23, there are at most $d + 3$ choices for y_2 . The number of choices for z_2 equals $d + 1$. Therefore, the number of Type (2) edges is at most

$$\begin{aligned} &\sum_{r=0}^{d+\lambda+2} \left[(1+r)2^r(d+1) \right] \left[(d+\lambda+3)2^{d+\lambda+2-r} \right] 2^{\lambda+1}(d+3)(d+1) \\ &= (d+\lambda+3)2^{d+2\lambda+3}(d+3)(d+1)^2 \sum_{r=0}^{d+\lambda+2} (1+r) \\ &\leq 2^d(d+\lambda+3)^6 2^{2\lambda+3}. \end{aligned}$$

Since $d = \lceil \log_2 n \rceil$ and $\lambda \leq \mathcal{O}(\sqrt{(\log n)(\log \log n)})$, $|E(G_n)| \leq n2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})}$. $\square$

4.3.4 The Universal Graph U_n^t . We construct the universal graph U_n^t for the class of n -vertex graphs in $\mathcal{Q}_t$.

For integers $m \geq 1$ and $\omega \geq 0$, observe that the strong product $C_m \boxtimes K_{\omega+2}$ can be formed by adding all the possible edges between a complete graph $K_{\omega+2}$ and two disjoint copies of $C_{m-1} \boxtimes K_{\omega+2}$. We require the following lemma.

Lemma 4.31. *For all integers $n \geq 1$ and $\omega \geq 0$, $C_{\lceil \log_2 n \rceil} \boxtimes K_{\omega+2}$ is universal for the class of n -vertex graphs with pathwidth at most ω .*

Proof. Proceed by induction on n with ω fixed. The $n = 1$ case holds trivially, so assume that $n \geq 2$ and the result holds for all lower values. Let G be an n -vertex graph with pathwidth at most ω . By Theorem 2.5, there is a partition $\{A, S, B\}$ of $V(G)$ with $|A|, |B| \leq n/2$ and $|S| \leq \omega + 2$ such that there is no AB -edge in G . Observe that $G[A]$ and $G[B]$ have pathwidth at most ω . By the induction hypothesis, $G[A]$ and $G[B]$ are contained in $C_{\lceil \log_2(n/2) \rceil} \boxtimes K_{\omega+2} = C_{\lceil \log_2 n \rceil - 1} \boxtimes K_{\omega+2}$. Map $G[A]$ and $G[B]$ respectively to the two disjoint copies of $C_{\lceil \log_2 n \rceil - 1} \boxtimes K_{\omega+2}$ in $C_{\lceil \log_2 n \rceil} \boxtimes K_{\omega+2}$. Map S to the $K_{\omega+2}$ in $C_{\lceil \log_2 n \rceil} \boxtimes K_{\omega+2}$ that is complete to the two copies of $C_{\lceil \log_2 n \rceil - 1} \boxtimes K_{\omega+2}$. This shows that G is contained in $C_{\lceil \log_2 n \rceil} \boxtimes K_{\omega+2}$. The result holds by induction. $\square$

For integers $n, t \geq 1$, let $\omega := \lfloor (t+1) \log_{3/2} n \rfloor$. By Lemma 3.22, every n -vertex graph with treewidth at most t has pathwidth at most ω . Define the graph $U_n^t := G_n \boxtimes K_{\omega+2}$. Recall that $d := \lceil \log_2 n \rceil$.

Theorem 4.32. *For all integers $n, t \geq 1$, U_n^t is universal for the class of n -vertex graphs in $\mathcal{Q}_t$.*

Proof. Let G be an n -vertex graph in $\mathcal{Q}_t$. By the definition of $\mathcal{Q}_t$, G is a subgraph of $H \boxtimes P$, where H is graph with treewidth at most t and P is a path. By Lemma 4.27, we may demand that H and P have at most n vertices. Therefore, $\text{pw}(H) \leq \omega$. By Lemma 4.31, H is isomorphic to a subgraph of $C_d \boxtimes K_{\omega+2}$. By Theorem 4.29, $H \boxtimes P$ is isomorphic to a subgraph of $G_n \boxtimes K_{\omega+2}$, implying G is isomorphic to a subgraph of U_n^t as desired. $\square$

We now estimate the number of vertices and edges in U_n^t .

Theorem 4.33. $|V(U_n^t)| \leq tn2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})}$ and $|E(U_n^t)| \leq t^2n2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})}$ when $n \gg t$.

Proof. By the definition of U_n^t , $|V(U_n^t)| = (\omega + 2)|V(G_n)|$. By Lemma 4.30,

$$|V(U_n^t)| \leq \omega n 2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})} \leq t(\log n)n 2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})} \leq tn 2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})},$$

as claimed. Furthermore, by basic properties of the strong product,

$$\begin{aligned}
|E(U_n^t)| &= |V(G_n)| \cdot |E(K_{\omega+1})| + |E(G_n)| \cdot |V(K_{\omega+1})| + |E(G_n)| \cdot |E(K_{\omega+1})| \\
&\leq t^2 (\log^2 n) n 2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})} \\
&\leq t^2 n 2^{\mathcal{O}(\sqrt{(\log n)(\log \log n)})},
\end{aligned}$$

as claimed. $\square$

Hence, Theorems 4.32 and 4.33 imply Theorem 4.20 as desired. Furthermore, we have:

Theorem 4.34. $r_t(n) \leq t^2 n^{1+o(1)}$ when $n \gg t$. In particular, $r(n) \leq n^{1+o(1)}$.

We note that Gawrychowski and Janczewski [47] showed that bulk tree sequences can be replaced with so-called ‘B-trees’ in the proof of Theorem 4.20 so that the resulting universal graph constructed for the n -vertex graphs in $\mathcal{Q}_t$ has $t^2 n 2^{\mathcal{O}(\sqrt{\log n})}$ edges. This improves the estimates in Theorem 4.33.

It is an open problem to asymptotically determine $r(n)$. We know

$$\Omega(n \log n) \leq r(n) \leq n 2^{\mathcal{O}(\sqrt{\log n})}.$$

4.3.5 Alon’s Trick. Although the universal graph U_n^t constructed in Section 4.3.4 has a near-linear number of edges, its number of vertices is far from linear, unlike the universal graphs for trees and bounded treewidth graphs shown in Chapters 2 and 3.

Nevertheless, there is a trick to reduce the number of vertices in the universal graph in Theorem 4.20 from $tn^{1+o(1)}$ to $(1+o(1))n$. Use the probabilistic method!

We require the following elementary facts.

Fact 4.35. $\binom{s}{t} \leq \left(\frac{se}{t}\right)^t$ for all integers $s \geq t \geq 1$.

Fact 4.36. For any constant $c > 0$, the function $f(x) = (c/x)^x$ is increasing over the interval $(0, c/e)$.

The next lemma is proved using the probabilistic method.

Lemma 4.37. For all sufficiently large integers n , for every integer $k \geq 1$, for every real number $\varepsilon \in (0, 1]$, for every integer $m \geq k(1 + \varepsilon)n$, there is a bipartite graph H with bipartition $\{A, B\}$ such that each of the following hold:

- (1) $m \leq |A| \leq m + k$ and $k|B| = |A|$,
- (2) every vertex of A has degree $\mathcal{O}(\frac{1}{\varepsilon} \log k)$, and
- (3) each n -vertex subset of A is saturated by a matching in H .

Proof. Let A and B be disjoint sets such that $|A| \geq m$, $|A|$ is the smallest integer at least m such that k divides $|A|$, and $|B| = |A|/k$. Then (1) holds. Let $d := 2 + \lceil \frac{1}{\varepsilon} \log_2(20e^2 k) \rceil$. Let H be a bipartite graph with bipartition $\{A, B\}$ such that for every vertex x of A , x is

joined to d randomly chosen vertices of B (uniformly with replacement). Then (2) holds. By Theorem 1.17, it follows that (3) is equivalent to the condition that:

$$\text{for every set } S \subseteq A \text{ with } |S| \leq n, |N_H(S)| \geq |S|. \quad (4.7)$$

For each set $S \subseteq A$ and $T \subseteq B$ with $n \geq |S| > |T| \geq 1$, let $E_{S,T}$ be the event that $N_H(S) \subseteq T$. Then $\mathbb{P}(E_{S,T}) = (|T|/|B|)^{|S|d}$. We show that with positive probability, none of the ‘bad’ events $E_{S,T}$ occur. By the union bound,

$$\begin{aligned} \mathbb{P}\left(\bigcup_{\substack{S \subseteq A, \\ |S| \leq n}} \bigcup_{\substack{T \subseteq B, \\ |T| < |S|}} E_{S,T}\right) &\leq \sum_{\substack{S \subseteq A, \\ |S| \leq n}} \sum_{\substack{T \subseteq B, \\ |T| < |S|}} \left(\frac{|T|}{|B|}\right)^{|S|d} \leq \sum_{i=1}^n \binom{|A|}{i} \sum_{j=1}^{i-1} \binom{|B|}{j} \left(\frac{i}{|B|}\right)^{id} \\ &\leq \sum_{i=1}^n \sum_{j=1}^{i-1} \left(\frac{|A|e}{i}\right)^i \left(\frac{|B|e}{j}\right)^j \left(\frac{i}{|B|}\right)^{id} \\ &\leq \sum_{i=1}^n \sum_{j=1}^{i-1} \left(\frac{|A|e}{i}\right)^i \left(\frac{|B|e}{i}\right)^i \left(\frac{i}{|B|}\right)^{id}, \end{aligned}$$

where the second last inequality follows from Fact 4.35, and the last inequality follows by applying Fact 4.36 to $(\frac{|B|e}{j})^j$. Since $|B| = |A|/k$ and $i \leq 2^i$,

$$\mathbb{P}\left(\bigcup_{\substack{S \subseteq A, \\ |S| \leq n}} \bigcup_{\substack{T \subseteq B, \\ |T| < |S|}} E_{S,T}\right) \leq \sum_{i=1}^n i \left(\frac{|A|e}{i}\right)^i \left(\frac{|A|e}{ik}\right)^i \left(\frac{ik}{|A|}\right)^{id} \leq \sum_{i=1}^n \left[2e^2 k \left(\frac{ik}{|A|}\right)^{d-2}\right]^i. \quad (4.8)$$

Since $\varepsilon \in (0, 1]$, $2^{-\varepsilon} \geq \frac{1}{1+\varepsilon} \geq \frac{nk}{|A|} \geq \frac{ik}{|A|}$. Therefore, $(\frac{ik}{|A|})^{d-2} \leq 2^{-\varepsilon(d-2)} \leq \frac{1}{20e^2k}$, where the last inequality follows from the definition of d . Hence, (4.8) implies

$$\mathbb{P}\left(\bigcup_{\substack{S \subseteq A, \\ |S| \leq n}} \bigcup_{\substack{T \subseteq B, \\ |T| < |S|}} E_{S,T}\right) \leq \sum_{i=1}^n \left[2e^2 k \frac{1}{20e^2k}\right]^i \leq \sum_{i \geq 1} \left(\frac{1}{10}\right)^i < 1.$$

Thus, H satisfies (4.7) with positive probability, implying the lemma holds. $\square$

We now show how to use Lemma 4.37 to reduce the number of vertices in the universal graph U_n^t described in the preceding section.

Theorem 4.38. *For all integers $n, t \geq 1$ with $n \gg t$, there exists a graph with $t^2 n^{1+o(1)}$ edges and $(1+o(1))n$ vertices that is universal for the class of n -vertex graphs in $\mathcal{Q}_t$.*

Proof. Let U_n^t be the universal graph in Theorem 4.32. Pick m such that $|V(U_n^t)| \leq m \leq tn^{1+o(1)}$. Let $\varepsilon := 1/\log n$ and let $k := \frac{m}{(1+\varepsilon)n} \leq tn^{o(1)} \log n \leq tn^{o(1)}$. Since $n \gg t$, $\log k = o(\log n)$. Apply Lemma 4.37 to obtain a bipartite graph H with bipartition $\{A, B\}$. By (1) from Lemma 4.37, $|V(U_n^t)| \leq m \leq |A|$, so we can assume without loss of generality that $V(U_n^t) \subseteq A$. Furthermore, $|B| = |A|/k \leq m/k + 1 = (1+\varepsilon)n + 1$.

Define the graph $\mathcal{U}_n^t$ in the following way. Let $V(\mathcal{U}_n^t) := B$ and for two distinct vertices

$u, v \in B$, $uv \in E(\mathcal{U}_n^t)$ if and only if there exists an edge $u'v' \in E(U_n^t)$ such that $uu' \in E(H)$ and $vv' \in E(H)$. By (2) from Lemma 4.37, $|E(\mathcal{U}_n^t)|$ is at most $\mathcal{O}(\frac{1}{\varepsilon^2} \log^2 k) |E(U_n^t)| \leq \mathcal{O}(\log^4 n) t^2 n^{1+o(1)} \leq t^2 n^{1+o(1)}$. Furthermore, $|V(\mathcal{U}_n^t)| \leq (1 + \varepsilon)n + 1 = (1 + 1/\log n + 1/n)n = (1 + o(1))n$. It remains to show that $\mathcal{U}_n^t$ is universal for the n -vertex graphs in $\mathcal{Q}_t$.

Let G be an n -vertex graphs in $\mathcal{Q}_t$. By (3) from Lemma 4.37, $V(G)$ is saturated by a matching in H ; let X be the vertices in B that are the end-vertices of the edges in this matching. By the way the edges in $\mathcal{U}_n^t$ are defined, $\mathcal{U}_n^t[X]$ contains a subgraph isomorphic to G as desired. $\square$

The authors of the paper [43] attribute this trick to Noga Alon.



Noga Alon

Chapter 5

Conclusion and Further Directions

This thesis began by considering the universal graph problem for the class of trees. The purpose of looking at trees (a relatively simple and highly structured class) was to gain an understanding of the basic techniques used to attack the universal graph problem, with the goal of eventually investigating universality for treewidth- k graphs. On route to achieving this goal, a mistake in the literature was found and explained in Section 2.2.

In Section 3.1, we introduced treewidth and presented a proof of the previously best known bound on $s_k(n)$, namely, $s_k(n) \leq \mathcal{O}(kn \log^2 n)$. Then the upper bound $s(n) \leq \mathcal{O}(n(\log n)(\log \log n))$ for trees in Section 2.3 was extended to treewidth- k graphs, giving the new best known upper bound $s_k(n) \leq \mathcal{O}(kn(\log n)(\log \log n))$. Furthermore, the lower bound $s(n) \geq \Omega(n \log n)$ for trees was extended to treewidth- k graphs, giving the new best known lower bound $s_k(n) \geq \Omega(kn \log n)$.

These original bounds for $s(n)$ and $s_k(n)$ lead directly to joint work in which the bounds for $s(n)$ and $s_k(n)$ are further improved as explained in the ‘Postscript’ in Chapter 3.

The last chapter of this thesis discussed a plethora of results (not original) relating to universality for planar graphs. A proof of the best known upper bound for $r(n)$ was presented in Section 4.3, namely, $r(n) \leq n^{1+o(1)}$. It is an open problem to asymptotically determine $r(n)$.

The following additional topics regarding universality in graph theory are of interest for further research:

- Sparse induced-universal graphs for treewidth- k graphs: It is known that n -vertex treewidth- k graphs have an adjacency labelling scheme with $\log_2 n + \mathcal{O}(k \log \log n)$ bits [46]. This implies that there is a graph with $n \log^{\mathcal{O}(k)}(n)$ vertices that is induced-universal for the n -vertex treewidth- k graphs. It is open whether the n -vertex treewidth- k graphs have an adjacency labelling scheme with $\log_2 n + \mathcal{O}(k)$ bits.
- Sparse universal graphs for any proper minor-closed class: It is open whether there exists a graph with $n^{1+o(1)}$ edges that is universal for n -vertex K_t -minor-free graphs.
- Faithful universal graphs: Given a graph class $\mathcal{G}$, a universal graph U for the n -vertex graphs in $\mathcal{G}$ is *faithful* if $U \in \mathcal{G}$. A recent paper by Bastide et al. [11] proved a number of results regarding faithful universal graphs for minor-closed classes.

- The induced-universal graph problem for the class of planar graphs: It was recently shown by Esperet et al. [43] that there is a graph with $n^{1+o(1)}$ vertices that is induced-universal for n -vertex planar graphs.
- Faithful minor-universal graphs for planarity: Recall that the $2n \times 2n$ grid is minor-universal for n -vertex planar graphs. It is open whether there exists a planar graph with $o(n^2)$ vertices that is minor-universal for n -vertex planar graphs.
- Universality for degree sequences: It is an open problem to asymptotically determine the minimum number of edges in a graph U that is universal for the class of graphs with a given degree sequence $(d_1, \dots, d_n)$. Note the obvious lower bound: $|E(U)| \geq \frac{1}{2}(d_1 + \dots + d_n)$.

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